

The Keakeya maximal conjecture in three dimensions

OpenAI

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Abstract

We prove the Keakeya maximal conjecture in three dimensions. For every $\varepsilon > 0$, the maximal average over unit tubes of radius δ maps $L^3(\mathbb{R}^3)$ to $L^3(S^2)$ with norm at most $C_\varepsilon \delta^{-\varepsilon}$.

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1 Introduction

For $a \in \mathbb{R}^3$, $\omega \in S^2$, and $0 < \delta < 1$, let

$$T_\delta(a, \omega) = \left\{ a + t\omega + u : |t| \leq \frac{1}{2}, \quad u \cdot \omega = 0, \quad |u| \leq \delta \right\}.$$

Thus $T_\delta(a, \omega)$ has volume $\pi\delta^2$. Define

$$K_\delta f(\omega) = \sup_{a \in \mathbb{R}^3} \frac{1}{\pi\delta^2} \int_{T_\delta(a, \omega)} |f(x)| \, dx.$$

The measure on S^2 below is its usual surface measure.

Theorem 1.1. *For every $\varepsilon > 0$ there exists a finite constant C_ε such that, for every $0 < \delta < 1$ and every $f \in L^3(\mathbb{R}^3)$,*

$$\|K_\delta f\|_{L^3(S^2)} \leq C_\varepsilon \delta^{-\varepsilon} \|f\|_{L^3(\mathbb{R}^3)}.$$

Theorem 1.1 resolves the three-dimensional Kakeya maximal conjecture affirmatively. Its essential quantitative feature is uniformity in the density of the part of each tube on which a function is large. We explain that feature before describing the proof.

1.1 The set problem and the maximal problem

A Kakeya set contains a unit line segment in every direction. The problem has its origins in the study of moving a needle inside a small planar region; the early convex formulation appears in Fujiwara and Kakeya [8, Section 10]. Besicovitch’s constructions showed that a set with segments in every direction can have measure zero, and that continuous needle reversal is possible in regions of arbitrarily small area [3, 6]. These are different requirements. The dimension conjecture asks whether directional containment nevertheless forces full Hausdorff dimension. Davies proved the planar case [6]; Córdoba’s planar maximal estimates provided a quantitative analytic counterpart [5].

The distinction relevant here is already visible for characteristic functions. Let $\mathcal{T}$ be a family of unit δ -tubes in a fixed bounded region, with δ -separated directions, and let a *shading* assign a measurable subset $Y(T) \subset T$ to each tube. If $|Y(T)| \geq \lambda|T|$, the maximal problem asks for the cubic density power in the union estimate

$$\left| \bigcup_{T \in \mathcal{T}} Y(T) \right| \gtrsim_{\varepsilon} \delta^{\varepsilon} \lambda^3 \sum_{T \in \mathcal{T}} |T|, \quad 0 < \lambda \leq 1. \quad (1.1)$$

The power loss in δ may be arbitrarily small, uniformly in λ . A set-dimension conclusion can follow from weaker dependence on λ . In particular, Wang and Zahl’s set theorem gives this form of estimate with a power $\lambda^{K(\varepsilon)}$; the need to replace it by λ^3 is stated explicitly after their theorem [20, Theorem 1.2]. The final reduction in Section 10 retains the density power through the passage from shadings to arbitrary L^3 functions.

The same overlapping tube configurations also enter Fourier restriction and summability problems. Fefferman’s ball-multiplier counterexample made this connection concrete [7]. Bourgain developed higher-dimensional maximal estimates and their Fourier-analytic applications [4], and Wolff’s hairbrush argument gave maximal estimates implying the Hausdorff-dimension bound $(n+2)/2$ in $\mathbb{R}^n$ [23]. In three dimensions this is $5/2$. Katz, Łaba, and Tao obtained a strict improvement for upper Minkowski dimension and analyzed three recurring structures in nearly extremal tube families: nearby tubes remain grouped across scales (stickiness), tubes meeting near a point have nearly coplanar directions (planiness), and their union organizes into thin planar pieces (graininess) [14]. Katz and Zahl subsequently obtained a strict Hausdorff-dimension improvement over $5/2$ [13].

The three-dimensional set problem was resolved through a sequence of works of Wang and Zahl. Their sticky Kakeya theorem was followed by the full Assouad-dimension theorem and then the Hausdorff- and Minkowski-dimension theorem for arbitrary Kakeya sets [20–22]. The corresponding preprints appeared in 2022, 2024, and 2025. The streamlined proof of Guth, Wang, and Zahl supplies the set estimate used in this paper [11, Theorem 1.1]. It applies under a convex-clustering bound and a shading-density threshold $\lambda \geq \delta^{\eta}$, with $\eta > 0$ chosen from the desired volume exponent. Convex clustering bounds how many complete tubes can lie in a convex body; it is a formulation that remains useful after the restrictions and changes of scale needed here. The task is to retain the sharper density dependence required by (1.1) through those operations. For a broader account of the maximal formulation and its relation to the set problem, see Zahl [24, Conjecture 1.3’’ and Section 3.3].

Two further geometric estimates enter the proof. The multilinear Kakeya theorem of Bennett, Carbery, and Tao [2, Theorem 1.15], sharpened to the endpoint by Guth [10, Theorem 1.3], controls three transverse tube families. The planar Furstenberg theorem of Ren and Wang [17, Theorem 4.1], in the equivalent shaded-tube formulation recorded by Wang and Wu [19, Theorem 0.6], controls planar incidence and projection configurations. We state these inputs and prove the weighted forms used here in Section 3.

1.2 Geometry of the proof

A density-sensitive extremal problem. Work in a bounded coordinate chart in which each tube follows an affine graph $M_i(t) = b_i + tu_i$, with $b_i, u_i \in \mathbb{R}^2$. At resolution $N = \delta^{-1}$, its shading is a set of marked time bins. We keep the original index i and its weight under restriction, even when two indices have the same affine graph. A full-time plank counts the indices whose complete traces it contains; marked bins instead determine the occupied cells and their multiplicity. This separation permits the set input to be used after rescaling without identifying different shadings on the same trace.

Total shading density alone does not describe what happens after a scale cut. A temporal deficit profile records, in logarithmic units, how the number of marked bins falls short of full branching inside each occupied block. We introduce a family of multiplicity inequalities whose temporal cost depends on this profile and whose clustering term depends on the two plank widths. At the parameter that gives the maximal theorem, the cost reduces to the total density penalty. The critical exponent is the least extra power of N needed to make the inequality hold for every configuration. A positive critical exponent would produce configurations attaining equality in the limit. We study their geometry, allowing restrictions that lose only a subpower fraction of the weighted incidence.

From equality to compatible plates. An equality configuration cannot contain a local packet with excessive mass: rescaling such a packet would improve the critical exponent. The resulting local bounds, together with multilinear Kakeya, force the directions on typical short blocks to lie close to a common plane. The planar Furstenberg estimate supplies density inside plates containing the reference segments. The plate Nikodym estimate then controls the support cost of filling their missing times. Equality therefore forces full temporal branching on an initial interval of logarithmic scales. On those scales, the weighted set theorem captures saturated plates: their masses attain the local bound. An anisotropic rescaling of aligned plates rules out equality with arbitrarily small total time deficit.

These local planes and plates are the forms of planiness and graininess needed by the present extremal argument. Their compatibility across scales must still be proved. We vary the two parameters in the critical inequality to select saturated shapes, then compare two ways of cutting an equality configuration. An outer cut removes initial scales; an inner cut retains a fine-scale model after aligning its plates. A minimum-delay argument forces a canonical temporal profile

$$F(s) = \beta(s - \tau)_+,$$

with a flat interval followed by constant deficit slope. Repeating the outer cut reproduces this profile and the same normalized packet geometry. This is the stationarity used below.

Four coordinates and two frames. A packet has a time interval and a nearly constant normal parameter ϑ . In that frame, write

$$X = M_i(t)_1 + \vartheta M_i(t)_2, \quad Y = M_i(t)_2, \quad U = u_{i,1} + \vartheta u_{i,2}, \quad V = u_{i,2}.$$

Thus X, Y describe position and U, V describe velocity; the first coordinate in each pair measures the normal component. Their admissible widths change at four prescribed rates, determined by the stationary time and angle scales. The masses of the packets prescribe how much information these four coordinates must carry. Conditional entropy, normalized by $\log N$, measures that information as a logarithmic cell count.

Now choose two marked events on the same original index. They give two frames for the same affine graph. If their time and normal differences are Δt and $\Delta \vartheta$, then

$$\begin{pmatrix} X' & Y' \\ U' & V' \end{pmatrix} = \begin{pmatrix} 1 & \Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} X & Y \\ U & V \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \Delta \vartheta & 1 \end{pmatrix}.$$

Changing time adds velocity to position; changing the normal mixes the two spatial components. The resulting scalar projections constrain the coordinate entropy rates through the planar Furstenberg theorem. When the two middle speeds coincide, the scale increments directly supply only their joint rate. Freezing the middle coordinates leaves the coefficient $\Delta t \Delta \vartheta$ multiplying V in X' ; this is the product slope for which the pinning estimate is needed.

The pinning argument adapts the radial-projection bootstrap of Shmerkin and Wang [18, Section 5.4] and of Orponen, Shmerkin, and Wang [16, Section 1.3]. On small cells, planar projection estimates improve the nonconcentration of rays joining parameter points; this improvement can then be iterated. Section 9 proves the finite-scale product-slope estimate used here, including control of both single-event marginals.

The two events share an index, so their matrix and slope are generally dependent. To apply a projection theorem, we compare their actual joint law with the product of its conditional matrix and label marginals. The actual law always lands in the second event's recorded support. If a coordinate-rate constraint failed, the product law would land there with polynomially small probability. Such a discrepancy requires a positive amount of mutual information. An averaged short-gap estimate shows that the available information is smaller. The resulting projection constraints contradict the entropy demand of the stationary packets, excluding a positive critical exponent.

1.3 Consequences and organization

The theorem also gives Nikodym maximal estimates in $\mathbb{R}^3$ and locally on three-dimensional manifolds of constant sectional curvature, as well as local curved Keakeya estimates for a fixed nondegenerate translation-invariant phase satisfying Bourgain's condition. These follow from the transfers of Gao, Liu, and Xi [9]; their full hypotheses and interpolation ranges appear in Section 11. An independent route to the same strong Keakeya maximal estimate through spherical Fourier extension appears in OpenAI [15, Corollary 1.3].

Section 2 defines the indexed model, explains the density reduction, and constructs the critical inequalities and equality sequences. Section 3 supplies the geometric inputs. Sections 4 and 5 establish the local geometry, canonical profile, and stationary packets. Section 6 proves angular nonconcentration and the conditional normal-position estimate.

Section 7 constructs the coordinate entropy rates, proves their lower demand, and obtains the velocity trace needed at the central offset. Section 8 derives the projection constraints from the two-frame experiment; Section 9 proves its auxiliary projection and pinning estimates. Section 10 closes the contradiction and passes from the density estimate to the full maximal theorem.

All limiting exponents arise from finite configurations. At each finite stage, the required scales, partitions, and conditional tests are imposed together before the resolution is increased. The regularization results in Section 2 give the losses and order of limits used in the later constructions.

2 Weighted configurations and critical exponents

The geometric problem is to bound the average multiplicity of shaded tubes in terms of their shading density and concentration in planks. We first express this problem for finite weighted, indexed

families of affine lines. Restrictions and changes of scale preserve inherited weights, but may produce repeated line parameters, so the class includes those repetitions. We then define a family of critical inequalities and construct sequences that attain their optimal exponents. These sequences will be the input to the local geometric arguments.

2.1 The indexed model and logarithmic comparisons

Let $N = 2^j$ tend to infinity. An index i consists of a positive weight ω_i , a matrix $M_i = (b_i, u_i)$ in a fixed bounded subset of $\mathbb{R}^{2 \times 2}$, and a nonempty subset S_i of the N dyadic time bins in $[0, 1]$. Different indices may have identical matrices and different shadings. The spatial trace is

$$M_i(t) = b_i + tu_i.$$

An event is a pair (i, J) with $J \in S_i$. Its location is $(t_J, M_i(t_J))$, where t_J is the bin center. Let E be the union of the space–time grid cells of side N^{-1} containing these locations. Write

$$n = \sum_i \omega_i, \quad \mathcal{I} = \sum_i \omega_i |S_i|, \quad k = \mathcal{I}/n, \quad m = \mathcal{I}/|E|. \quad (2.1)$$

Thus k is initially a weighted mean, and m is the raw average multiplicity. After regularization all $|S_i|$ will have the same exponent as k . Each event has mass ω_i ; probabilities on the event set always mean division by $\mathcal{I}$.

A full-time plank of widths a, b , where $1 \leq a \leq b \leq N$, is a set with cross-sections

$$\left\{ x \in \mathbb{R}^2 : |(x - b_0 - tu_0) \cdot e_1| \leq a/N, \quad |(x - b_0 - tu_0) \cdot e_2| \leq b/N \right\}, \quad 0 \leq t \leq 1,$$

for an orthonormal spatial frame (e_1, e_2) and affine center $b_0 + tu_0$. Constant factors in these widths are permitted. An index belongs to a plank only if its entire trace over the indicated time interval is contained in it. We use the same definition on shorter dyadic time blocks; there the admissible widths are at most the block length in finest-cell units. In particular, membership is stronger than having an event in the plank. The mass n_B of a plank is the sum of the inherited weights of its contained indices.

Fixed enlargements of cells, moving event locations a bounded number of fine cells, and replacing the bounded matrix patch by any other fixed-size patch do not affect the exponents below. For the last assertion, translate the matrix center and rescale space by a fixed constant. Both the old and new support grids admit bounded-overlap coverings by the other grid, and plank tests transfer after bounded enlargement and a fixed finite covering.

Convention 2.1 (Exponent notation). For positive quantities on a sequence with $N \rightarrow \infty$, write

$$A \leq_{\text{exp}} B \iff \limsup_{N \rightarrow \infty} \frac{\log(A/B)}{\log N} \leq 0, \quad A =_{\text{exp}} B \iff \frac{\log(A/B)}{\log N} \rightarrow 0.$$

A factor $N^{o(1)}$ has logarithm $o(\log N)$, uniformly over the objects being compared in that statement. Subsequence selection is allowed. Every contradiction will compare exponents separated by a fixed positive amount before the resolution tends to infinity. All these comparisons are homogeneous in the weights.

A refinement deletes events and then deletes empty indices; retained indices keep their original weights. It is *extensive* if its event mass is at least $N^{-o(1)}\mathcal{I}$. A restriction of loss exponent at most e retains at least $N^{-e+o(1)}\mathcal{I}$.

Definition 2.2 (Uniform temporal profile). An admissible profile is a nondecreasing 1-Lipschitz function $F : [0, 1] \rightarrow [0, 1]$ with $F(0) = 0$. A sequence of indexed configurations has uniform profile F if, uniformly over every retained index and every occupied dyadic time block containing K fine bins,

$$|S_i \cap I| = N^{\kappa - F(\kappa) + o(1)}, \quad \kappa = \log_N K. \quad (2.2)$$

The uniform error also applies over the scale range $0 \leq \kappa \leq 1$. In this case $k =_{\text{exp}} N^{1-\alpha}$, where $\alpha = F(1)$.

The variable κ measures block size, not physical time: increasing κ moves from fine bins to coarser time blocks. A block at this scale has N^κ available bins, so $F(\kappa)$ records the exponent lost from full occupation. Between scales $s < t$, the number of occupied s -blocks in an occupied t -block has exponent $(t - s) - (F(t) - F(s))$.

Complete shading has $F = 0$, while one marked bin per index gives $F(\kappa) = \kappa$. Total density alone does not determine the profile. For example, for $0 < \alpha < 1$ the admissible profiles

$$F_1(\kappa) = \alpha\kappa, \quad F_2(\kappa) = (\kappa - 1 + \alpha)_+$$

have the same endpoint deficit α ; here $r_+ = \max\{r, 0\}$. The first distributes the deficit at a constant rate across scales. The second is realized by marking one aligned dyadic interval of $N^{1-\alpha+o(1)}$ bins on every index: occupied blocks below that interval's size are full, and larger blocks contain the entire marked interval. This distinction matters when we change scales or fill occupied blocks.

It is enough to arrange (2.2) on meshes whose maximal gap tends to zero sufficiently slowly. Indeed the count in an occupied intermediate block is bounded below by that in one of its occupied finer mesh blocks and above by that in its coarser mesh ancestor. The two mesh exponents differ by at most their scale gap plus the errors at the mesh points.

2.2 The density bound and its deformation

Put $\lambda = k/N$. The multiplicity estimate we seek, for $p > 2$ arbitrarily close to two, is

$$m \leq_{\text{exp}} \lambda^{-p} \sup_B \frac{n_B}{ab}. \quad (2.3)$$

Here B ranges over full-time planks of widths a, b . To see the connection with the maximal theorem, consider unit-weight lines with pairwise c/N -separated slopes, where $c > 0$ is fixed. Comparing the two endpoint cross-sections of a plank confines those slopes to a rotated rectangle of side lengths $O(a/N)$ and $O(b/N)$. Slope separation therefore gives $n_B \leq Cab$. Since $m = nk/|E|$, the desired bound would give

$$|E| \geq_{\text{exp}} nN\lambda^{p+1}.$$

As p decreases to two, this approaches the cubic density bound required for the L^3 maximal estimate. The final passage from this discrete estimate to arbitrary functions is proved in Proposition 10.2.

We deform both the temporal penalty and the powers assigned to the two plank widths. The resulting critical exponent measures the additional power of resolution needed in the bound. One endpoint of the deformation will be elementary; derivatives of the optimal exponent at an interior point will later select temporal deficits and plank shapes.

Fix $0 < \eta < 1/2$. For $p > 2$, $z = -q \in [-1, 0]$ and $d = 2 - q$, set

$$\mathcal{D}(q, e) = \frac{1}{\eta} \int_{1/2-\eta}^{1/2} \min(q, re) dr, \quad P_{p,q}(e) = pe - \mathcal{D}(q, e), \quad 0 \leq e \leq 1. \quad (2.4)$$

For an admissible profile define

$$\Pi_F(t) = \int_0^t P_{p,q}(F'(u)) du, \quad 0 \leq t \leq 1. \quad (2.5)$$

The derivative exists almost everywhere because F is Lipschitz. When the profile and parameters are fixed we write Π for Π_F . Dependence on the fixed smoothing width η is suppressed. At $q = 0$, $\Pi_F(1) = pF(1)$, so only the total density enters. For $q > 0$ the discount $\mathcal{D}(q, F')$ also records how the deficit is distributed across scales. The averaging in r smooths this discount in q , as needed for the later derivative test.

For fixed q , the function $e \mapsto \min(q, re)$ is increasing, concave, and r -Lipschitz. Thus $P_{p,q}$ is increasing and convex, vanishes at zero, and has all secant slopes in $[p - 1/2, p]$. The bounds $0 \leq \mathcal{D}(q, e) \leq \min(q, e/2)$ also give

$$(p - 1/2)F(t) \leq \Pi_F(t) \leq pF(t), \quad \Pi_F(1) \geq pF(1) - q. \quad (2.6)$$

These elementary estimates suffice to define the critical problem and show that its admissible exponents form a nonempty set.

For $0 \leq A \leq d - 1$, define the plank parameter

$$\Delta_A = \sup_B \frac{n_B}{a^{d-1-A}b^{1+A}}, \quad (2.7)$$

where B runs over full-time planks. The letter A in this definition is only a candidate exponent. Every configuration is finite, and the supremum may equivalently be taken over a finite net with bounded thickening. The bounds below are independent of the size of that net.

Definition 2.3 (Critical exponent). For fixed p, z, η , let $h(p, z)$ be the least $A \in [0, d - 1]$ such that for every sequence with a uniform profile F ,

$$m \leq_{\text{exp}} N^{A+\Pi_F(1)} \Delta_A. \quad (2.8)$$

The definition ranges over all bounded-patch weighted indexed configurations. At $q = 0$ one has $d = 2$, and the candidate $A = 0$ gives exactly (2.3) for a uniform profile. There is then no dependence on η . Our task is to show that $h(p, 0) = 0$ for $p > 2$ arbitrarily close to two.

More precisely, Proposition 10.2 shows that $h(p, 0) = 0$ implies, for every $\varepsilon > 0$,

$$\|K_\delta f\|_{L^3(S^2)} \leq C_{p,\varepsilon} \delta^{-(p-2)/3-\varepsilon} \|f\|_{L^3(\mathbb{R}^3)}.$$

Lemma 2.4 (Existence and elementary comparisons). *Definition 2.3 is well-defined. Its admissible candidates form a closed upper interval in $[0, d - 1]$, and $h(p, -1) = 0$. For every $0 \leq A_1 \leq A_2 \leq d - 1$,*

$$N^{-(A_2-A_1)} \Delta_{A_1} \leq \Delta_{A_2} \leq \Delta_{A_1}, \quad N^{A_1} \Delta_{A_1} \leq N^{A_2} \Delta_{A_2} \leq N^{A_2-A_1} N^{A_1} \Delta_{A_1}. \quad (2.9)$$

All fixed geometric constants may be absorbed in exponent notation.

Proof. The comparisons follow by multiplying each plank denominator by $(b/a)^{A_2-A_1}$ and using $1 \leq b/a \leq N$. Thus a fixed plank's contribution to $N^A \Delta_A$ has logarithmic slope $\log_N(Na/b) \in [0, 1]$. Taking the supremum preserves these comparisons. Monotonicity of the bound and closedness of the set of valid candidates follow, the latter by letting $A_2 - A_1$ tend to zero after applying the bound on an arbitrary fixed sequence.

It remains to prove validity at $A = d - 1$. We bound multiplicity by the weighted count of pairs meeting in fine cells, then sum by slope separation. Let $\mu(Q)$ be the total weight of events in a fine space-time cell Q . Cauchy–Schwarz gives

$$m = \mathcal{I}/|E| \leq \frac{\sum_Q \mu(Q)^2}{\mathcal{I}}.$$

Fix one indexed line. Other lines with relative slope magnitude comparable to $\theta \geq N^{-1}$ that pass within bounded fine distance of it at some time are contained in a full-time plank of both widths $O(\theta N)$ around it. Their total mass is at most $O(\Delta_{d-1}(\theta N)^d)$. Two such lines can be within bounded fine distance for at most $O(1/\theta)$ time bins: their spatial difference is affine with velocity of magnitude comparable to θ . For relative slopes at most N^{-1} , use mass $O(\Delta_{d-1})$ and at most N bins. Fixed large slope ranges or patch boundaries require only fixed finite coverings. Summing over dyadic θ and then the reference line gives

$$\sum_Q \mu(Q)^2 \lesssim n \Delta_{d-1} N^d \left(1 + \sum_{\substack{\theta \text{ dyadic} \\ N^{-1} \leq \theta \lesssim 1}} \theta^{d-1} \right) \leq n \Delta_{d-1} N^{d+o(1)}.$$

This includes $d = 1$, where the sum costs a logarithm, and includes repeated indices. Consequently

$$m \leq_{\text{exp}} \Delta_{d-1} N^d / k =_{\text{exp}} \Delta_{d-1} N^{d-1+F(1)}.$$

Since $\Pi_F(1) \geq (p - 1/2)F(1) \geq F(1)$, (2.8) holds at $A = d - 1$. When $z = -1$, one has $d = 1$ and this is the only possible candidate, proving $h(p, -1) = 0$. $\square$

2.3 Finite cleaning and profile regularization

To apply the critical inequality after selecting events, we need uniform profiles and lower bounds on the mass retained in every tested cell. The following elementary deletion estimate preserves those lower bounds. Its partitions are allowed to have arbitrarily many cells; a bound on the number of cells is needed only for the probability pigeonholing that follows it.

Lemma 2.5 (Finite cleaning). *Let Ω be a finite set with positive measure μ , let $A \subseteq \Omega$ satisfy $\mu(A) \geq \rho\mu(\Omega)$, and let $\mathcal{P}_1, \dots, \mathcal{P}_L$ be partitions of Ω . If $0 < \vartheta_0 < \rho/L$, there is $A' \subseteq A$ with*

$$\mu(A') \geq (\rho - L\vartheta_0)\mu(\Omega)$$

such that every cell C of a tested partition meeting A' satisfies

$$\mu(A' \cap C) \geq \vartheta_0 \mu(C).$$

In particular one can take $\vartheta_0 = \rho/(2L)$ and retain at least $\rho\mu(\Omega)/2$.

Proof. Start with A . Whenever a tested cell has positive remaining mass less than $\vartheta_0 \mu(C)$, delete its remaining points. A cell used in such a deletion becomes empty and is never charged again. Each deletion costs less than $\vartheta_0 \mu(C)$, measured against the *original* measure. For each partition the sum of the original cell masses is $\mu(\Omega)$. Thus all deletions together cost at most $L\vartheta_0 \mu(\Omega)$. The process terminates because Ω is finite. Cascades between the partitions cause no additional charge. $\square$

A conditional probability test is specified by a partition into domains and by a partition of each domain into at most N^C cells, with C fixed for the test. Its value at an event is the probability of that event's cell conditional on its domain. The number of domains need not be polynomial. For example, the domains may be the individual indices. For any $A > C$, the events in conditional cells of probability below N^{-A} have total probability at most N^{C-A} , by summing within each domain and then averaging over the domains. A global partition with at most N^C cells is the special case of one domain.

For a fixed finite list of such tests, discard these negligible tails and round each remaining $-\log_N$ conditional probability to an interval of length ν . There are at most

$$\prod_{r=1}^L (2 + A_r/\nu)$$

possible label vectors. One vector carries at least the reciprocal of this number, up to the discarded mass. Clean both the domain and cell partitions using Lemma 2.5. On a surviving cell, both numerator and denominator of a conditional probability lie between their original values and ϑ_0 times those values. Its conditional exponent therefore changes by at most $|\log_N \vartheta_0|$. This proves the stated pigeonholing simultaneously for probabilities and conditional probabilities. Further finite partitions can be included solely for cleaning, without pigeonholing their cell masses.

Within an index, temporal counts can be used instead of conditional probabilities. They are integers between 1 and N and their logarithmic labels have a polynomially bounded range regardless of the number or weights of indices.

Lemma 2.6 (Regularization and inherited branching). *The following assertions hold.*

- (i) *Every sequence of indexed configurations has, after passage to a subsequence, an extensive refinement with a uniform temporal profile. A prescribed finite list of the preceding probability and conditional-count tests can be regularized at the same time. Countably many prescribed tests can be imposed by the finite-test diagonal described below.*
- (ii) *Suppose an original sequence has profile F , and a restriction followed by extensive regularization has profile G . Then*

$$G(t) - G(s) \geq F(t) - F(s), \quad 0 \leq s \leq t \leq 1. \quad (2.10)$$

If the restriction retains at least $N^{-e+o(1)}$ of the original event mass, then

$$0 \leq G(1) - F(1) \leq e. \quad (2.11)$$

In particular an extensive refinement preserves F .

- (iii) *The same branching comparison holds within a fixed interval of logarithmic scales when events are routed through charts, provided each derived temporal fiber is a subset of one original index and starting block, its weight is inherited, and each original index and starting block produces at most $N^{o(1)}$ chart fibers. Total event mass and total available starting-block weight are counted over that ensemble. If the routed ensemble retains at least $N^{-e+o(1)}$ of the original ensemble's event mass and is extensively regularized to a common profile, its total deficit increase on the interval is at most e in original logarithmic units. Without that retained-mass hypothesis only the increment comparison is asserted.*

Proof. Fix first a finite mesh $0 = \kappa_0 < \dots < \kappa_J = 1$ and an accuracy $\nu > 0$. At resolution N , round each interior mesh point to a multiple of $1/\log_2 N$, keeping the endpoints. For sufficiently large

N these rounded points remain ordered; they specify actual dyadic block sizes and differ from the prescribed points by at most $1/\log_2 N$. In this finite construction κ_r denotes the rounded value. The rounding error tends to zero before the mesh is refined. At every event label the original counts in its index's occupied ancestral blocks at these scales, rounding $\log_N$ of each count to intervals of length ν . There are at most $(2 + 1/\nu)^{J+1}$ labels. Include the other prescribed labels, retain a label class of maximal event mass, and clean all the partitions into pairs consisting of an index and a block. Those partitions may have many cells, which is permitted in Lemma 2.5. Let c_j be the lower endpoint, a multiple of ν , of the common rounded count interval at scale κ_j . Every surviving occupied block there has count between

$$\vartheta_0 N^{c_j} \quad \text{and} \quad N^{c_j + \nu}.$$

The same assertion at the root gives uniform counts per retained index. The weights cancel in these within-index comparisons.

Put $\lambda_N = \log_N(1/\vartheta_0)$ and $\epsilon_N = \nu + \lambda_N$. An occupied coarse block contains an occupied finer block. Conversely it has at most $N^{\kappa_k - \kappa_j}$ subblocks across the actual dyadic scale gap $[\kappa_j, \kappa_k]$. The preceding count bounds therefore give, for every $j < k$,

$$-\epsilon_N \leq c_k - c_j \leq \kappa_k - \kappa_j + \epsilon_N.$$

The error does not accumulate with the number of intervening scales. To replace these approximate count exponents by an admissible profile, first define a branching-count exponent and then its deficit:

$$b_N(s) = \min_{0 \leq j \leq J} \{c_j + (s - \kappa_j)_+\}, \quad F_N(s) = s - b_N(s).$$

Each function in the minimum is nondecreasing and 1-Lipschitz, and so is their minimum. Since $c_0 = 0$ and every $c_j \geq 0$, one has $0 \leq b_N(s) \leq s$ and $b_N(0) = 0$. Thus F_N is an admissible profile. At a tested scale the preceding increment bounds imply

$$c_j - \epsilon_N \leq b_N(\kappa_j) \leq c_j.$$

Taking an occupied tested descendant and a tested ancestor for an arbitrary occupied intermediate block I of K fine bins proves the finite, all-scale estimate

$$|\log_N |S'_i \cap I| - (s - F_N(s))| \leq \max_j (\kappa_{j+1} - \kappa_j) + 2\nu + \lambda_N, \quad s = \log_N K. \quad (2.12)$$

For clarity, the passage to arbitrarily many tests is sequential. At stage r take a fixed finite mesh of gap at most $1/r$, the first r prescribed tests and their required joint or domain partitions, and a fixed label accuracy at most $1/r$. Let C_r be the reciprocal of the retained mass fraction guaranteed by the finite construction, including cleaning, and let L_r be the number of cleaned partitions. Choose the resolutions for that stage so large that $\log(C_r L_r)/\log N \leq 1/r$ and all other fixed-stage errors are at most $1/r$. Let the stage grow sufficiently slowly along the original resolution sequence. The resulting refinements are extensive. Compactness of the finite profiles F_N gives a uniformly convergent subsequence with admissible limit F . Equation (2.12) gives uniformity on all scales. This proves (i).

For (ii), let $s < t$ be fixed. Uniformity shows that the number of occupied s -blocks in any occupied t -block on an original index has exponent

$$(t - s) - (F(t) - F(s)).$$

For the restricted and regularized index the corresponding exponent is $(t - s) - (G(t) - G(s))$. Its occupied subblocks are a subset of the original ones. Comparison proves (2.10); rational scales suffice, followed by continuity. Let n' be the remaining total index weight. Since $n' \leq n$, the assumed event-mass retention gives

$$n' N^{1-G(1)+o(1)} \geq n N^{1-F(1)-e+o(1)}.$$

This proves the upper bound in (2.11); the lower bound follows from (2.10) with $s = 0, t = 1$. If $e = 0$, all increments of $G - F$ are nonnegative and its total increment is zero, so $G = F$.

For (iii), apply the same child-block upper bounds within each derived fiber. The sum of its possible starting weights is at most $N^{o(1)}$ times the sum for the original index-starting-block pairs. The retained event mass supplies the same lower bound used in the preceding displayed formula. Thus the total branching loss is at most the event-loss exponent, and every subinterval has the inherited upper branching bound. After normalizing a logarithmic interval of length $L > 0$ to length one, both the logarithmic loss and the profile deficits are divided by L . No estimate uniform in a vanishing L is asserted before fixing that interval. $\square$

If earlier tests have already been homogenized, an extensive refinement can preserve their limiting cell exponents by including their cell and domain partitions in the cleaning list. The normalized logarithm of every surviving cell probability then changes by $o(1)$. This is the probability preservation used for the entropy limits in Section 7; no such invariance is asserted for tests that have not been homogenized.

Corollary 2.7 (Active weight after cleaning). *Let $\mathcal{A}$ be a finite collection of pairs (i, I) , where I is a dyadic temporal block and the blocks assigned to each index i are disjoint. Let $\mathcal{C}$ be a partition of $\mathcal{A}$. For a pair (i, I) , its events are (i, J) with $J \in S_i \cap I$ and have inherited weight ω_i . Fix $a \in \mathbb{R}$, $\varepsilon \geq 0$, and $0 < \vartheta_0, \sigma_N \leq 1$. Suppose each pair has between $N^{a-\varepsilon}$ and $N^{a+\varepsilon}$ events. For $C \in \mathcal{C}$ put*

$$M_C = \sum_{(i,I) \in C} \omega_i.$$

If a refinement retains at least a fraction ϑ_0 of the event mass assigned to C , then its active weight in that cell satisfies

$$\sum_{\substack{(i,I) \in C \\ S'_i \cap I \neq \emptyset}} \omega_i \geq \vartheta_0 N^{-2\varepsilon} M_C.$$

Suppose further that each cell's assigned events are partitioned into at most L_N pieces, where $L_N \geq 1$ is an integer. Discard the pieces of mass less than σ_N/L_N times their original cell's event mass. This discards at most a fraction σ_N of the total event mass. If subsequent cleaning retains at least a fraction ϑ_0 of each surviving piece's original event mass, then the active weight on every occupied piece is at least

$$\frac{\vartheta_0 \sigma_N}{L_N} N^{-2\varepsilon} M_C.$$

Here active weight counts each pair (i, I) meeting the piece once.

Proof. The original event mass assigned to C is at least $N^{a-\varepsilon} M_C$. Each pair contributes at most $N^{a+\varepsilon} \omega_i$ to its retained event mass, so division by $N^{a+\varepsilon}$ gives the first bound. The discarded pieces in a cell have total mass at most σ_N times that cell's mass. An occupied retained piece has mass at least $\vartheta_0 \sigma_N / L_N$ times the original cell's mass; the same division proves its active-weight bound. $\square$

The cell and piece partitions in this corollary may be included in the cleaning list. If $L_N = N^{o(1)}$, choose $\sigma_N \rightarrow 0$ with $\sigma_N = N^{-o(1)}$; subpower cleaning then preserves the original active-weight exponent as $\varepsilon \rightarrow 0$. For a further extensive selection of events, the cutoff can likewise be chosen subpower and smaller than its retained fraction. Under a polynomial restriction the cutoff must lie below that smaller fraction, and its polynomial loss in the active-weight bound must be retained, as in (2.11).

Convention 2.8 (Fixed tests and strict margins). Each argument is first made for a finite list of scales, widths, partitions and strictly positive tolerances. Subsequences may be selected so that its bounded logarithmic parameters converge. Only after its fixed-parameter estimates are established are the test lists increased or tolerances sent to zero, with the resolution chosen large enough for every earlier error. If a selection is defined by perturbing a parameter, its error tolerance is sent to zero faster than that fixed perturbation before the latter is allowed to shrink. Thus countable tests impose no convergence rate uniform at all differentiation points or all shrinking scale gaps.

This convention also explains discarding negligible exceptions. If a bad part of a fixed test could have event fraction $N^{-o(1)}$ along a subsequence, it would itself be an extensive configuration to which the contradiction applies. If this is impossible, its fraction is at most N^{-c} for some fixed $c > 0$ eventually. Finitely many such exceptions can be removed together; further tests are incorporated only in the subsequent slow diagonal. Spatial and angular box tests with continuous parameters can be covered by finite nets at each fixed resolution, with bounded thickening. For all limiting scale exponents it is enough to use increasingly fine meshes and monotone coverings.

2.4 Temporal cost under restriction and changes of scale

Regularization provides the profiles on which the critical inequality is imposed. We now check how their costs behave under the restrictions, limits, and changes of scale used to construct equality sequences.

Lemma 2.9 (Cost properties). *The cost Π_F is additive across logarithmic scale intervals and is zero on flat intervals. In addition to (2.6), it has the following properties.*

- (i) *If $F_j \rightarrow F$ uniformly, then $\Pi_F(t) \leq \liminf_j \Pi_{F_j}(t)$ for each fixed t .*
- (ii) *If G is obtained as in Lemma 2.6 with loss exponent at most e , then*

$$0 \leq \Pi_G(1) - \Pi_F(1) \leq pe.$$

The same estimate applies on scale subintervals with their corresponding loss accounting.

- (iii) *If a profile on a scale interval $[a, a + L]$ is normalized by $G(s) = [F(a + Ls) - F(a)]/L$, then*

$$\Pi_G(1) = \frac{1}{L} \int_a^{a+L} P_{p,q}(F'(u)) du.$$

Grafting fully occupied time below a cut grafts a flat interval and adds no cost there.

Moreover $\partial_p \Pi_F(1) = F(1)$. For $q > 0$,

$$\partial_z \Pi_F(t) = \int_0^t \mathcal{D}_q(q, F'(u)) du, \quad \mathcal{D}_q(q, e) = \frac{1}{\eta} \int_{1/2-\eta}^{1/2} \mathbf{1}_{\{q < re\}} dr. \quad (2.13)$$

Proof. For a partition $0 = t_0 < \dots < t_J = t$, Jensen's inequality gives

$$\sum_{j=1}^J (t_j - t_{j-1}) P_{p,q} \left(\frac{F(t_j) - F(t_{j-1})}{t_j - t_{j-1}} \right) \leq \Pi_F(t).$$

The supremum over refining dyadic partitions equals $\Pi_F(t)$: the interval averages of the bounded function F' converge to it in L^1 , and P is Lipschitz. Each finite sum is continuous in F for uniform convergence, proving (i). By (2.10), $G - F$ is nondecreasing and Lipschitz, so $G' \geq F'$ almost everywhere. The slope bound gives

$$0 \leq \int_0^1 [P(G') - P(F')] \leq p \int_0^1 (G' - F') \leq pe,$$

proving (ii). A change of variables proves (iii).

The p derivative is immediate. For $q > 0$ the displayed average for $\mathcal{D}_q$ exists also when $e = 0$; for $e > 0$ its possible equality set in the r integral has measure zero. It is continuous in (q, e) with $q > 0$, and bounded between zero and one. Dominated convergence gives (2.13), with the positive sign because $z = -q$. $\square$

Filling occupied blocks. Fill every occupied N^κ -block with all its fine time bins, keeping the original set of occupied blocks. In the original logarithmic units its new profile is

$$F^{\text{fill}}(s) = \begin{cases} 0, & 0 \leq s \leq \kappa, \\ F(s) - F(\kappa), & \kappa \leq s \leq 1. \end{cases}$$

Above the cut, count the original occupied cut blocks and multiply by the full N^κ bins in each; below the cut every occupied block is full. Its cost is precisely $\Pi_F(1) - \Pi_F(\kappa)$ before any further affine change of the logarithmic scale. Rebinning must use subdivisions of the same nested blocks.

2.5 Uniform bounds for ensembles

We record precisely how sequence bounds are used on families of local problems. Suppose a common upper estimate holds for every bounded-chart sequence with a specified limiting profile, and every member of an ensemble has that profile with uniform errors, the same resolution exponent bounded away from zero, and a common upper bound for its raw plank parameter. If the estimate failed uniformly by a fixed power, choosing one violating member at each resolution would give a forbidden sequence. Hence the estimate holds uniformly over the ensemble. Summing its incidence masses and separately counted support sizes gives the same upper multiplicity estimate for the aggregate ensemble. This argument neither changes the weights nor asserts bounded overlap between supports in different parents.

The same reasoning is valid for profiles approaching a compact limit when the upper bound is written with their own costs. Indeed a fixed-power failure along profiles $F_j \rightarrow F$ gives, by Lemma 2.9(i), a failure with cost at least $\Pi_F(1)$ after selecting sufficiently accurate finite realizations. This would contradict the all-sequence bound for F . These observations will be used whenever local charts or representatives are chosen after the original resolution has been fixed.

Splitting support among distinct charts need not preserve average multiplicity. In the aggregate just described, support sizes are counted separately for each chart; a lower multiplicity bound must therefore be proved for that ensemble in each application.

2.6 A differentiability point forced by failure

A failure of the desired bound at $z = 0$ will supply an interior parameter point where both h and its z derivative are positive. The two derivatives of h at that point will control the temporal deficit and the plank shapes selected in Section 5.

Lemma 2.10 (Critical parameter point). *The function $h(p, z) - z$ is separately nonincreasing in p and z . It is totally differentiable almost everywhere in the interior of its parameter domain. If I is an open interval of $p > 2$ on which $h(p, 0) > 0$, there is a point $(p, z) \in I \times (-1, 0)$ at which h is totally differentiable and*

$$h(p, z) > 0, \quad h_z(p, z) > 0.$$

At such a point $h_p \leq 0$ and $h_z \leq 1$.

Proof. Increasing p increases the cost and therefore cannot increase h . For $z_2 = z_1 + s$, $s > 0$, the dimension parameter increases by s . Starting with a valid candidate A at z_1 , use $A + s$ at z_2 . Its a exponent is unchanged and its b exponent increases by s . Writing the dimension in a subscript temporarily, we have

$$\Delta_{A+s, d+s} \geq N^{-s} \Delta_{A, d}.$$

Also q decreases, so $P_{p, q}$ increases pointwise. The new right side of (2.8) is therefore at least the old one. The candidate is still in range since $A + s \leq d + s - 1$. Thus

$$h(p, z + s) \leq h(p, z) + s, \tag{2.14}$$

including admissible endpoints. This proves the separate monotonicity of $g(p, z) = h(p, z) - z$.

Here is the passage from this monotonicity to the stated form of differentiability. On each compact rectangle in the interior, g is bounded, and its variations on coordinate lines are bounded by its range. Integration along those lines shows that both distributional first derivatives are finite measures. Thus g is locally of bounded variation. We use the approximate differentiability theorem for BV functions [1, Theorem 3.83]: away from a null set there are a value and a linear map giving a first-order approximation on sets of density one. Coordinatewise monotonicity upgrades this approximation to ordinary total differentiability, as follows.

At such a point x , fix an error tolerance and a density-one set on which the approximate linear estimate holds. For a displacement u of size r and any fixed $\varepsilon > 0$, consider small squares of side comparable to εr located coordinatewise just above and just below $x + u$. They lie within $O(r)$ of x . For sufficiently small r their area is larger than the bad part of the density-one set in that neighborhood, so each square contains a comparison point where the linear estimate holds. Monotonicity bounds $g(x + u)$ between the two comparison values. The comparison displacements differ from u by $O(\varepsilon r)$. The same sandwich first identifies the actual value $g(x)$ with its approximate value. Divide the resulting error by r , let $r \rightarrow 0$, and then let the approximation tolerance and ε tend to zero. This is total differentiability at x .

For fixed $p \in I$, the decreasing function $z \mapsto g(p, z)$ has only nonpositive singular variation. Its ordinary derivative therefore obeys

$$g(p, b) - g(p, a) \leq \int_a^b g_z(p, z) dz.$$

Adding $b - a$ gives the same inequality for the increment of h and the integral of h_z . There is no lost upward jump at $z = 0$: (2.14) gives $h(p, -t) \geq h(p, 0) - t$. At the other endpoint, $0 \leq h(p, z) \leq 1 + z$ and $h(p, -1) = 0$. Letting $a \downarrow -1$ and $b \uparrow 0$ consequently gives

$$0 < h(p, 0) \leq \int_{-1}^0 h_z(p, z) dz.$$

Fubini and almost-everywhere total differentiability yield an interior point of total differentiability with $h_z > 0$. At a differentiability point where $h = 0$, nonnegativity of h would force its gradient to vanish. Hence the selected point has $h > 0$ as well. The derivative signs follow from the two monotonicity comparisons. $\square$

If $h(p, 0) = 0$ does not occur for $p > 2$ arbitrarily close to two, its nonnegativity gives an interval immediately above two on which it is positive. Choose an open subinterval I with $p_{\min} = \inf I > 2$ and fix

$$0 < \eta \leq \min \left\{ \frac{1}{32}, \frac{p_{\min} - 2}{16} \right\}. \quad (2.15)$$

This choice is not circular: at $q = 0$ the cost and the critical exponent are independent of η . Lemma 2.10 then supplies an interior point for this fixed smoothing. For the contradiction below fix that point and put

$$h = h(p, z), \quad x = d - 1 - h, \quad y = 1 + h, \quad v = -h_p(p, z). \quad (2.16)$$

Thus $h > 0$, $h_z > 0$, $v \geq 0$, $0 \leq x \leq y$, and $x + y = d$. Strict positivity of v will follow from the positive-deficit equality and the parameter test in Section 5.

2.7 Hybrid equalities and compactness

We freeze the shape exponent in the plank parameter slightly below h and optimize only the additional power of N . The resulting equality sequences remain sharp under extensive refinements. Keeping the two exponents distinct also allows the anisotropic argument in Section 4 to force a positive temporal deficit before we pass to an exact critical equality.

Lemma 2.11 (Hybrid equality sequences). *Fix p, z, η with $h = h(p, z) > 0$, and let $0 \leq h_0 < h$. There is a sharp number H with*

$$h_0 < H \leq h$$

such that every sequence with uniform profile F satisfies

$$m \leq_{\exp} \Delta_{h_0} N^{H + \Pi_F(1)}. \quad (2.17)$$

There is a sequence with a uniform profile attaining equality in (2.17).

On any extensive refinement of such an equality sequence, followed by regularization, the profile is unchanged and

$$m' =_{\exp} m, \quad \Delta'_{h_0} =_{\exp} \Delta_{h_0}. \quad (2.18)$$

These comparisons refer to refinements of the indexed graph before any separate geometric change of coordinates.

Finally, suppose $h_{0,j} \uparrow h$ and, for each j , a hybrid equality has profile F_j . After taking a subsequence and sufficiently accurate realizations, one obtains a sequence with uniform profile F satisfying

$$m =_{\exp} \Delta_h N^{h + \Pi_F(1)}. \quad (2.19)$$

If $F_j(1) \geq \alpha_ > 0$, then $F(1) \geq \alpha_*$.*

Proof. For every uniform-profile sequence form the score

$$\limsup_{N \rightarrow \infty} \{ \log_N(m / \Delta_{h_0}) - \Pi_F(1) \},$$

and let H be the supremum of these scores. The critical bound at h and $\Delta_h \leq \Delta_{h_0}$ give $H \leq h$. If $H \leq h_0$, then the bound defining h would hold at h_0 for every sequence, contrary to $h_0 < h$. Thus $H > h_0$. The supremum definition also proves (2.17) at H itself.

Choose for each integer j a sequence with profile F_j and score at least $H - 1/j$. Profiles form a compact set for uniform convergence; pass to a subsequence with $F_j \rightarrow F$. From the j th sequence choose a realization at a resolution so large that its uniform-profile count error relative to F_j is at most $1/j$ over all occupied indices and all dyadic time scales, that all chosen resolutions increase, and that

$$\log_N(m/\Delta_{h_0}) \geq H + \Pi_{F_j}(1) - 2/j.$$

Any other finite list of prescribed tests is accommodated before this choice. The selected diagonal sequence has uniform profile F . Lower semicontinuity of the cost gives

$$\liminf \log_N(m/\Delta_{h_0}) \geq H + \Pi_F(1).$$

The universal upper bound (2.17) supplies the reverse limsup. Equality follows. This argument also explains why a downward jump of the cost at the profile limit cannot cause loss of an extremizer: it would instead improve the score, which is forbidden by the same sharp upper bound. Multiplying all weights by $\Delta_{h_0}^{-1}$ is permitted if a normalized realization is desired.

For an extensive graph refinement, its incidence mass loses only a subpower and its support is a subset of the old support. Hence $m \leq_{\text{exp}} m'$. Its plank masses cannot increase, so $\Delta'_{h_0} \leq \Delta_{h_0}$. Its profile remains F by Lemma 2.6. Applying (2.17) to the refinement sandwiches both quantities:

$$m \leq_{\text{exp}} m' \leq_{\text{exp}} N^{H+\Pi_F(1)} \Delta'_{h_0} \leq N^{H+\Pi_F(1)} \Delta_{h_0} =_{\text{exp}} m.$$

This proves (2.18).

For the final assertion, let H_j denote the corresponding sharp hybrid power. By (2.9),

$$0 \leq \log_N(\Delta_{h_{0,j}}/\Delta_h) \leq h - h_{0,j}, \quad h_{0,j} < H_j \leq h.$$

Thus a sufficiently accurate realization of the j th hybrid equality has

$$\log_N(m/\Delta_h) - h - \Pi_{F_j}(1) = o_j(1),$$

where the error can be made to tend to zero by first taking that realization far enough along its sequence. Take a uniformly convergent subsequence of F_j and a finite-test diagonal as above. Lower semicontinuity gives the lower bound in (2.19); the critical bound at h gives its upper bound. Uniform convergence also preserves the lower endpoint-deficit bound $F_j(1) \geq \alpha_*$. $\square$

Section 4 obtains that uniform positive lower bound for the hybrid deficits from the geometric inputs collected in Section 3. Until then, H and h_0 remain distinct in all local estimates. Once Lemma 4.9 is proved, Lemma 2.11 gives an exact equality with positive deficit at the fixed differentiability point.

3 Geometric inputs and their weighted forms

The local arguments use three geometric estimates. The weighted form of the Kakeya set theorem controls multiplicity for sufficiently dense shadings. Multilinear Kakeya controls incidences with three transverse directions. The planar Furstenberg theorem bounds the union of shadings after projection onto a plane. We state these inputs and derive the weighted forms needed for the indexed configurations of Section 2; the projection consequence is developed in Section 8.

Weights and shadings remain attached to individual indices, including indices with the same trace. All constants associated with a bounded coordinate chart are fixed independently of resolution. We write η_{in} for the tolerances in the input theorems below; these are unrelated to the fixed smoothing width η in the temporal cost.

For a bounded set $E \subset \mathbb{R}^j$, let $\mathcal{N}_\delta(E)$ be the number of half-open cubes of the fixed mesh $\delta\mathbb{Z}^j$ that meet E . Replacing this mesh by a translated mesh or using balls of comparable radius changes covering numbers by a dimensional constant. We say that a probability measure μ is (s, C) -Frostman down to δ if

$$\mu(B(x, r)) \leq Cr^s \quad (\delta \leq r \leq 1). \quad (3.1)$$

For a finite separated set this refers to its uniform probability measure. On bounded line charts we use the Euclidean distance between slope–intercept parameters. The constants in (3.1) may change by a fixed factor under the bounded coordinate changes used below.

3.1 The set estimate and convex clustering

A δ -tube is the δ -neighbourhood of a line segment of length one. Changing between this convention and the cylinders in the introduction costs only fixed changes of length and radius. A shading is a measurable subset $Y(T) \subset T$. For a finite set of such tubes put

$$\Delta_{\max}(\mathbb{T}) = \sup_{K \text{ convex}} \frac{\sum_{T \in \mathbb{T}: T \subset K} |T|}{|K|}, \quad \lambda(\mathbb{T}, Y) = \frac{\sum_{T \in \mathbb{T}} |Y(T)|}{\sum_{T \in \mathbb{T}} |T|}.$$

Only convex bodies of positive volume containing a tube matter in the supremum.

Theorem 3.1 (Kakeya set estimate). *For every $\varepsilon > 0$ there are $\eta_{\text{in}} > 0$ and $\delta_0 > 0$ such that, if $0 < \delta < \delta_0$,*

$$\Delta_{\max}(\mathbb{T}) \leq \delta^{-\eta_{\text{in}}}, \quad \lambda(\mathbb{T}, Y) \geq \delta^{\eta_{\text{in}}},$$

then

$$\left| \bigcup_{T \in \mathbb{T}} Y(T) \right| \geq \delta^\varepsilon \sum_{T \in \mathbb{T}} |T|. \quad (3.2)$$

This is Guth et al. [11, Theorem 1.1]. The hypothesis concerns convex clustering, rather than separation of directions. The density threshold is part of the statement and will be retained in every use.

We first extend this estimate to weighted indexed families. In particular, two indices with the same tube may have different shadings.

Lemma 3.2 (Weighted convex clustering). *Fix $R \geq 1$. For every $\varepsilon > 0$ there are $\eta_{\text{in}} > 0$ and $C < \infty$ with the following property. Let $(T_i)_i$ be finitely many δ -tubes in $B(0, R)$, where $0 < \delta < 1$, with repetitions allowed. Let $\omega_i > 0$ and let $Y_i \subset T_i$ be measurable shadings. Suppose*

$$\sum_{i: T_i \subset K} \omega_i |T_i| \leq A|K| \quad \text{for every convex body } K, \quad |Y_i| \geq \delta^{\eta_{\text{in}}} |T_i|. \quad (3.3)$$

Then

$$\sum_i \omega_i |Y_i| \leq CA\delta^{-\varepsilon} \left| \bigcup_i Y_i \right|. \quad (3.4)$$

The constants are independent of the number of indices and of their weights.

Proof. We sample the indices according to their normalized weights. A finite family of box tests will control all convex clusters in the sample, with only a logarithmic loss.

Write v_δ for the common tube volume, and normalize the weights by $v_i = \omega_i/A$. Testing the convex body T_i gives $v_i \leq 1$. Testing a fixed ball containing the family gives

$$S := \sum_i v_i \leq C_R \delta^{-2}.$$

Choose each index independently with probability v_i . The expected number of chosen indices is S .

We first reduce all convex tests to finitely many box tests. If a convex body K contains a tube from the family, put $K_R = K \cap \overline{B(0, R)}$; this intersection contains every tube counted in K . Enlarge an inscribed John ellipsoid of K_R by a dimensional factor to obtain an ellipsoid containing K_R , with volume at most a dimensional constant times $|K|$, largest axis $O_R(1)$, and every axis at least a constant times δ : the body contains a ball of radius δ . An enclosing rectangular box has the same properties. Quantize its center, axis lengths, and orthonormal frame with accuracy $c_R \delta^2$, and enlarge each side by $C_R \delta^2$. There is thus a collection $\mathcal{B}_\delta$ of at most $C_R \delta^{-C_R}$ convex boxes such that every relevant K_R is contained in some $Q \in \mathcal{B}_\delta$ satisfying

$$|Q| \leq C_R |K|, \quad |Q|/v_\delta \geq c_R.$$

The quantization error is smaller than a fixed fraction of the shortest side; this proves both containment and the volume comparison.

For $Q \in \mathcal{B}_\delta$, the number Z_Q of chosen indices whose tubes lie in Q is a sum of independent Bernoulli variables, and (3.3) gives

$$\mathbb{E} Z_Q \leq |Q|/v_\delta.$$

The elementary exponential-moment bound

$$\mathbb{P}(Z_Q \geq t) \leq \inf_{a>0} \exp((e^a - 1)\mathbb{E} Z_Q - at)$$

shows that, for a sufficiently large fixed L_R , simultaneously

$$Z_Q \leq L_R(1 + \log(1/\delta)) |Q|/v_\delta \quad (Q \in \mathcal{B}_\delta) \tag{3.5}$$

with probability at least $3/4$. Indeed each failure probability can be made smaller than δ^{C_R+2} , and a union bound applies. If $S \geq L_R(1 + \log(1/\delta))$, the Bernoulli lower-tail bound

$$\mathbb{P}(N_{\text{sample}} < S/2) \leq e^{-S/8}$$

gives at least $S/2$ chosen indices with probability at least $3/4$, after increasing L_R . This lower-tail estimate follows by applying the exponential moment method to $e^{-a N_{\text{sample}}}$. Fix a sample with both properties.

By the box comparison, its convex clustering is $O_R(1 + \log(1/\delta))$. The multiplicity of any repeated tube is bounded by the same quantity, since a box of volume $O(v_\delta)$ contains that tube. Keep one chosen index for each distinct tube. The resulting set has at least $c_R S/(1 + \log(1/\delta))$ members, and each retained shading still has the required minimum density. It is not necessary to identify different shadings on repeated tubes.

Apply Theorem 3.1 with target loss $\varepsilon/2$, and choose η_{in} no larger than its density tolerance. For all sufficiently small δ , the logarithmic clustering bound satisfies the theorem's hypothesis. Consequently

$$\left| \bigcup_i Y_i \right| \geq \frac{c_R S v_\delta \delta^{\varepsilon/2}}{1 + \log(1/\delta)}.$$

Since $\sum_i \omega_i |Y_i| \leq ASv_\delta$, this proves (3.4). If $S < L_R(1 + \log(1/\delta))$, use instead the pointwise bound $\sum_i \omega_i \mathbf{1}_{Y_i} \leq AS$ and absorb the logarithm. The bounded range of remaining δ is covered by $\sum_i \omega_i \leq C_R A \delta^{-2}$, with a larger constant. $\square$

3.2 Full-time planks and indexed shadings

The critical problem measures concentration in planks containing entire affine traces. We now convert this plank condition into the convex clustering condition of the preceding lemma, while counting only the marked times in the shading.

Let $M_i(t) = b_i + tu_i \in \mathbb{R}^2$, $0 \leq t \leq 1$, where b_i, u_i lie in a fixed bounded box. Put $\delta = D^{-1}$, where $D \geq 1$. A full-time plank of widths a, b in cell units is a set

$$\{(t, x) : 0 \leq t \leq 1, |\langle x - c - tv, e_1 \rangle| \leq a\delta, |\langle x - c - tv, e_2 \rangle| \leq b\delta\}, \quad (3.6)$$

where (e_1, e_2) is an orthonormal spatial frame and $c, v \in \mathbb{R}^2$. An index belongs to the plank when its entire trace belongs to it.

Lemma 3.3 (Weighted set estimate in a line chart). *Fix a bounded line chart and a fixed bound for the permitted enlargement of mesh cells. For every $\varepsilon > 0$ there are $\eta_{\text{in}} > 0$ and $C < \infty$ as follows. Suppose the indices have weights $\omega_i > 0$, and every full-time plank satisfies*

$$\sum_{i: M_i \subset P} \omega_i \leq Aab \quad (1 \leq a \leq b \leq D). \quad (3.7)$$

Mark a set J_i of time bins of length D^{-1} on each index, with $\#J_i \geq D^{1-\eta_{\text{in}}}$. Let E be the union of the mesh cells visited at those bin centers, or their fixed enlargements. Then

$$\sum_i \omega_i \#J_i \leq CAD^\varepsilon \#E. \quad (3.8)$$

In particular, when every index has k marked bins and $n = \sum_i \omega_i$, its raw average multiplicity satisfies $nk/\#E \leq CAD^\varepsilon$.

The plank bound also holds, up to a fixed multiplicative change in A , for fixed dilations and widths between fixed multiples of 1 and D . This follows from the endpoint partition in the proof and is not an additional hypothesis.

Proof. There are two geometric steps: a plank bound controls weighted convex clustering of thickened traces, and disjoint balls at marked bin centers convert incidence counts to shading volume.

It is harmless to allow a, b up to a fixed multiple of D in (3.7). A larger spatial coordinate can be handled by partitioning its two endpoint values, at $t = 0$ and $t = 1$, into intervals of fixed length in physical units. Linear interpolation puts every part inside a permitted full-time plank. There are only a bounded number of parts, depending on the chart. Apply the given bound to each part; the sum of their area factors is at most a fixed multiple of ab . A fixed decrease in the lower width cutoff is handled by enlargement, at a fixed cost in the area factor.

Enclose each trace in a tube T_i of radius $C_0\delta$ and a common fixed Euclidean length L . Choose L large enough that the trace, and balls of radius $C_0\delta$ about its points, lie away from the end caps. Both constants depend only on the bounded chart.

We check weighted convex clustering for these tubes. Let K contain some of them, intersect K with a fixed ambient ball containing all T_i , and take an enclosing ellipsoid $\mathcal{E}$ of volume $O(|K|)$ by dilating an inscribed John ellipsoid. Its diameter is bounded. Its horizontal slices have affine

centers and a fixed pair of principal axes; their semiaxes are a common scalar multiple of those of the maximal slice. The time diameter of $\mathcal{E}$ is at least one, because every counted tube contains its full chart trace. The ellipsoid volume is a dimensional constant times this time diameter times the maximal slice area. Thus that area is $O(|K|)$.

Circumscribe a rectangle about the maximal horizontal slice and translate it to the affine center of each slice. The resulting full-time plank contains all the counted traces and has cross-sectional area at most $C|K|$. Its spatial widths are at least a constant times δ , since a counted tube contains a spatial disk of comparable radius at every chart time. Applying (3.7) therefore gives

$$\sum_{i: T_i \subset K} \omega_i \leq CA\delta^{-2}|K|.$$

Since $|T_i| \asymp \delta^2$, the weighted convex clustering of the family is at most CA .

At each marked bin center put a ball of radius $c_0\delta$ on the trace, with $c_0 > 0$ fixed and small. Balls belonging to different bins of one index are disjoint. Their union $Y_i \subset T_i$ has volume comparable to $\#J_i\delta^3$; its density in T_i is at least $c\delta^{\eta_{\text{in}}}$. Moreover

$$\left| \bigcup_i Y_i \right| \leq C\delta^3 \#E.$$

Rescale the common length L to one and apply Lemma 3.2. Choose the present η_{in} smaller than half the density tolerance of that lemma, so fixed length and density constants are absorbed for small δ . Canceling δ^3 proves (3.8) for large D . For bounded D , a fixed collection of widest planks bounds the total weight by CAD^2 . The pointwise multiplicity bound by this total weight gives the conclusion after increasing C on that bounded range. $\square$

3.3 The planar Furstenberg estimate

The planar input will be applied to the neighbors of a reference line in a nearly coplanar configuration. Each neighbor keeps its marked times, so we need a lower bound that retains the number of shading cells as a factor. The following dual form of the quantitative planar theorem does so.

Theorem 3.4 (Quantitative planar Furstenberg estimate). *Fix $0 < s \leq 1$, $0 < t \leq 2$, and a bounded line chart. For every $\varepsilon > 0$ there are $\eta_{\text{in}} > 0$, $c > 0$, and $\delta_0 > 0$ with the following property. Let $\mathcal{L}$ be a finite δ -separated family of lines whose uniform parameter distribution is $(t, \delta^{-\eta_{\text{in}}})$ -Frostman down to δ . For every $\lambda \in \mathcal{L}$, let $Y(\lambda)$ be a δ -separated subset of a fixed bounded part of λ , with*

$$k \leq \#Y(\lambda) \leq 2k,$$

whose uniform distribution is $(s, \delta^{-\eta_{\text{in}}})$ -Frostman down to δ . Then, for $0 < \delta < \delta_0$,

$$\mathcal{N}_\delta \left(\bigcup_{\lambda \in \mathcal{L}} Y(\lambda) \right) \geq ck\delta^{-\min\{t, (s+t)/2, 1\} + \varepsilon}. \quad (3.9)$$

The same assertion holds for mesh-cell shadings within a fixed multiple of δ of the lines, with fixed changes of constants.

This follows from Ren and Wang [17, Theorem 4.1 and Definitions 1.3, 2.3, 3.1] by point–line duality. To spell out the translation, in coordinates where λ has equation $y = ax + b$, it corresponds to the parameter point (a, b) , and a shading point (x, y) corresponds to the dual line $b = y - ax$.

Incidence is preserved. Along a bounded-slope original line, the shading coordinate x is bi-Lipschitz to arclength, so its Frostman bound is exactly the required bound for the incident dual directions. Discretizing introduces only fixed changes of cell size and bounded overlaps. In fact, a parameter cell contains only a bounded number of the separated original lines, and a dual parameter cell contains only a bounded number of separated shading points from any one original line. Retaining representatives preserves relative Frostman counts up to fixed factors. The nice configuration of Ren–Wang therefore has $M \asymp k$ selected incident tubes at each base cell; the number of distinct dual parameter cells is comparable to the covering number in (3.9). Choose the displayed η_{in} smaller than half the source theorem’s tolerance to absorb the fixed factors. A finite division into coordinate charts handles all bounded planar patches. Their theorem permits the constant $\delta^{-\eta_{\text{in}}}$; this quantitative tolerance is why subpower losses are allowed here.

The cited theorem is stated at dyadic scales. A comparable dyadic scale changes mesh sizes and covering numbers only by fixed factors, which are absorbed in the displayed constants.

The equivalent shaded-tube formulation in Wang and Wu [19, Theorem 0.6, Equation (0.8)] gives the same exponent. Indeed its density parameter is $\lambda \asymp k\delta$; division of the area bound by δ^2 gives the factor k in (3.9).

Corollary 3.5 (Weighted planar families). *Fix $0 < s \leq 1$, $0 < t \leq 2$, and a bounded line chart. For every $\varepsilon > 0$ there are $\eta_{\text{in}} > 0$, $c > 0$, and $\delta_0 > 0$ as follows. Let $0 < \delta < \delta_0$ and let finitely many, possibly repeated, line indices i carry positive weights ω_i . Suppose their normalized line-parameter measure is $(t, \delta^{-\eta_{\text{in}}})$ -Frostman down to δ . For each index i , let Y_i be a δ -separated shading of its line in a fixed bounded chart, with $k \leq \#Y_i \leq 2k$ and with its uniform distribution $(s, \delta^{-\eta_{\text{in}}})$ -Frostman down to δ . Then*

$$\mathcal{N}_\delta \left(\bigcup_i Y_i \right) \geq ck \delta^{-\min\{t, (s+t)/2, 1\} + \varepsilon}.$$

The constants are independent of the number of indices, their repetitions, and their weights.

Proof. It suffices to select separated line parameters whose uniform distribution inherits the weighted Frostman bound, and to keep one original shading at each selected parameter.

Push the normalized weights to the $O(\delta^{-2})$ parameter cells. Discard cells of mass below δ^4 ; for small δ the lost mass is less than $1/2$. There are $O(1 + \log(1/\delta))$ remaining dyadic mass levels, so one level has mass at least $c/(1 + \log(1/\delta))$. On this level the uniform distribution of occupied cells has Frostman constant at most $C\delta^{-\eta_{\text{in}}}(1 + \log(1/\delta))$: compare the comparable masses of the cells in an enlarged ball with the original probability. Color the parameter mesh by J congruence classes, where J is a fixed constant, and retain a class containing at least $1/J$ of the retained cells. Choose one actual line and its original shading from each retained cell. The resulting line parameters are separated, the shadings retain their Frostman bounds, and their union is contained in the original union. Apply Theorem 3.4, having chosen the current η_{in} smaller than half its allowed tolerance. $\square$

Remark 3.6. The same conclusion in normalized-log estimates holds when the shading cardinalities are comparable only up to subpowers of δ^{-1} . Thin every shading to the common minimum cardinality. Its relative Frostman constant increases by at most the ratio of the old and new cardinalities. That ratio, and the changes just used to regularize line weights, are subpowers and are absorbed by the fixed positive tolerance in Theorem 3.4. The case $t < s$ is included: the minimum in (3.9) is then t .

3.4 Multilinear Kakeya with quantitative transversality

The remaining input limits overlap among three families with transverse directions. We retain its dependence on the determinant of those directions, since the local arguments allow that determinant

to decrease with the resolution.

Theorem 3.7 (Endpoint multilinear Kakeya). *For $j = 1, 2, 3$, let $T_{j,a}$ be finitely many infinite cylinders of radius one in $\mathbb{R}^3$, with unit axial directions $v_{j,a}$. Suppose*

$$|\det(v_{1,a_1}, v_{2,a_2}, v_{3,a_3})| \geq \theta > 0 \quad \text{for every } (a_1, a_2, a_3).$$

For nonnegative weights $\omega_{j,a}$ put $\mu_j = \sum_a \omega_{j,a} \mathbf{1}_{T_{j,a}}$ and $M_j = \sum_a \omega_{j,a}$. Then

$$\int_{\mathbb{R}^3} (\mu_1 \mu_2 \mu_3)^{1/2} \leq C \theta^{-1/2} (M_1 M_2 M_3)^{1/2}. \quad (3.10)$$

In particular this holds for finite length- D tubes with no additional dependence on D .

Proof. For unit weights this is Guth [10, Theorem 1.3, p. 264], the endpoint form of multilinear Kakeya introduced in Bennett et al. [2, Theorem 1.15]. Integer weights are represented by repetition. There is no distinctness requirement; alternatively one can first separate repeated cylinders by distinct parallel translations tending to zero and use Fatou's lemma. Off their boundaries, the indicators then converge to the repeated indicators, and directions are unchanged.

For rational weights clear denominators separately in the three families; the factors cancel on the two sides of (3.10). Approximate real weights monotonically from below by rational weights. Monotone convergence proves the displayed estimate. Truncating the cylinders can only decrease the integrand. $\square$

Corollary 3.8 (Cube counts and small caps). *For unit mesh cubes Q , let $m_j(Q) = \sum_{a: T_{j,a} \cap Q \neq \emptyset} \omega_{j,a}$. Under the hypotheses of Theorem 3.7,*

$$\sum_Q (m_1(Q) m_2(Q) m_3(Q))^{1/2} \leq C \theta^{-1/2} (M_1 M_2 M_3)^{1/2}. \quad (3.11)$$

If the directions lie in caps of radius at most $c\theta$ about centers whose determinant is at least θ , the same conclusion holds with another absolute constant, for sufficiently small c .

Proof. Increase each cylinder's radius to $1 + \sqrt{3}$. Every cube meeting the original cylinder lies entirely in the enlarged cylinder, so the enlarged multiplicity dominates $m_j(Q)$ everywhere on Q . Integrate over the disjoint cubes and use Theorem 3.7, followed by the fixed radius rescaling. Finally the determinant of three unit vectors changes by at most a constant times the sum of their perturbations. Caps of radius $c\theta$ therefore give cross-family determinant at least $\theta/2$ when c is small enough. $\square$

For later use, a cap partition of mesh r has $O(r^{-2})$ members. Summing (3.11) over triples of caps costs at most $O(r^{-6})$. Thus choosing r to be a fixed positive power of θ introduces only a polynomial loss in θ^{-1} ; in particular the choice $r = \theta^2$ is permissible as $\theta \rightarrow 0$.

4 Local bounds, planar filling, and positive temporal deficit

We work at the fixed critical parameter point from Lemma 2.10. The arguments in this section first apply to a hybrid equality from Lemma 2.11, with

$$0 < h_0 < H \leq h, \quad m =_{\text{exp}} \Delta_{h_0} N^{H + \Pi_F(1)}.$$

They also apply to an exact equality, when $H = h_0 = h > 0$. Our goal is to obtain an exact equality with positive total deficit and an initial interval of zero deficit. Zero deficit means full temporal branching in exponent; the actual filling of time blocks will be a separate operation.

The argument has two geometric stages. Multilinear Makeya and the planar Furstenberg estimate control the support cost of filling short time blocks, forcing an initial flat interval in every equality profile. The weighted set estimate then captures plate packets that attain the local mass bound on those flat scales. Anisotropic rescaling of aligned plates rules out arbitrarily small total deficit for hybrid equalities near h . Its strict gain comes from $H - h_0 > 0$; when $H = h_0$, the same rescaling will produce new exact equalities on shorter time intervals in Section 5.

We abbreviate Π_F to Π and set

$$p_* = p - \frac{1}{2} > \frac{3}{2}, \quad x_H = d - 1 - H, \quad y_H = 1 + H.$$

In particular $x_H + y_H = d$, $x_H \geq 0$, and

$$\Pi(u) \geq p_* F(u), \quad \Pi(1) \geq p\alpha - q. \quad (4.1)$$

The first inequality follows from $\mathcal{D}(q, e) \leq e/2$, and the second from $\mathcal{D}(q, e) \leq q$.

All clustering and multiplicity estimates use the inherited weights ω_i . The probability measures in the planar Frostman argument will have their normalizing denominators displayed. An extensive regular refinement preserves F , m , and Δ_{h_0} in exponent, by Lemma 2.11; its sharp-bound comparison also applies when $H = h_0 = h$. We use this stability after each of the finitely many restrictions at a fixed stage of the slow diagonal.

In local geometric arguments, lengths such as a, b, K are in original fine-cell units. Thus a temporal block of K bins has actual duration K/N , and a transverse width a means actual width a/N . Containment of an index in a plank over a block always means containment of its entire affine segment over that block.

4.1 The local mass bound and the isotropic scale cut

An index is active on a temporal block if that block contains one of its retained marked bins. A packet on the block is a set of active indices assigned to a plank containing their entire affine segments over that block. Its mass is the sum of their inherited weights. Disjoint packet assignments mean disjoint assigned index sets on each block; their containing planks may overlap.

Proposition 4.1 (Local bound). *After an extensive regular restriction, the mass of active original indices contained in any plank of widths $1 \leq a \leq b \leq K$ over a dyadic temporal block of $K = N^\kappa$ bins satisfies*

$$n_{I,W} \leq_{\exp} T(\kappa, a, b), \quad T(\kappa, a, b) = mN^{-\Pi(\kappa)} K^{-H} a^{d-1-H} b^{1+H}. \quad (4.2)$$

The bounds hold simultaneously at every fixed limiting scale and shape, under the finite-test and slow-diagonal convention of Section 2, and are inherited by further restrictions.

Suppose conversely that, on an extensive graph, disjoint indexed-block assignments to planks of common widths a, b have masses at least $T(\kappa, a, b)$ in exponent. Set

$$\mathbf{a} = \log_N a, \quad \mathbf{u} = \log_N(K/b), \quad \chi = \mathbf{a} + \mathbf{u}. \quad (4.3)$$

If $\chi < 1$, the isotropic scale cut described below gives an ensemble at resolution $N' = Nb/(Ka) = N^{1-\chi}$ with matching sharp upper and lower bounds. At $H = h_0 = h$ this is an ensemble of exact

critical equalities, from which a realizing representative sequence can be selected. Its temporal profile is

$$G(v) = \begin{cases} 0, & 0 \leq v \leq (\kappa - \chi)/(1 - \chi), \\ \frac{F(\chi + (1 - \chi)v) - F(\kappa)}{1 - \chi}, & (\kappa - \chi)/(1 - \chi) \leq v \leq 1. \end{cases} \quad (4.4)$$

Proof. The rescaling and its cell count. We first describe the operation that determines the threshold T . Suppose active indices on selected K -blocks have been assigned to packets of widths a, b . Suppose also that the indices of each packet lie in one cell of an isotropic grid in matrix space, of side b/K ; call this cell its full-time parent. Subtract the parent center matrix and divide space by b/K , leaving time unchanged. Use resolution

$$N' = \frac{Nb}{Ka}. \quad (4.5)$$

The new matrices lie in a bounded patch. A new spatial fine cell corresponds to a old spatial fine cells, and an original K -block contains b/a new time bins.

Fill each selected block with those new bins, keeping all selected blocks of one original index together in its full-time parent. A packet now has transverse widths $1, b/a$. It occupies at most $C(b/a)^2$ space-time cells: at most Cb/a spatial cells per bin, over at most Cb/a bins. If its inherited index mass is M , its incidence mass is at least cMb/a . Thus a collection of disjoint indexed-block assignments, each of mass at least M , has aggregate raw multiplicity at least a constant times Ma/b . We count the supports in different parents separately and bound each union by the sum of its packet supports. Overlaps within a parent only improve this lower bound.

This calculation explains the factor a/b in the comparison below. To use the sharp upper bound we must also preserve the original time tree above scale K . We now construct the extensive parent ensemble for which that profile comparison is valid.

Capturing and routing an excessive family. Fix a proposed excess N^ε and a finite scale and shape test. On each temporal block, greedily assign all remaining active indices in a plank with mass greater than $N^\varepsilon T(\kappa, a, b)$, and remove their events on that block. Continue until no such plank remains. The plank tests may be meshed and thickened by a fixed factor, so the procedure is finite. Its assignments are disjoint in original indices on each block. Suppose the removed events form an extensive graph. Pigeonholing dyadic widths costs only a subpower, so retain a common pair a, b .

Two matrices whose segments stay in the same local plank differ in spatial slope by $O(b/K)$: evaluate the difference at the endpoints of the block and divide its $O(b/N)$ transverse diameter by the duration K/N . Their intercepts likewise differ by $O(b/K)$, since all original times belong to a bounded interval. Each packet therefore meets only boundedly many full-time isotropic matrix-grid parents of side b/K . Partition its indices among those parents. For a fixed width type each original matrix has one parent, so an index is not duplicated once for each selected block.

Before this subdivision, every assigned index-block contributes $KN^{-F(\kappa)+o(1)}$ events. Include the assignment cells and their parent subdivisions among the regularization tests. The inherited-weight cleaning of Lemma 2.6 and Corollary 2.7 preserves the threshold $N^{\varepsilon-o(1)}T$ on surviving packets: delete active packets retaining less than an inverse subpower fraction of their original incidence, and retain nondeficient parent subdivisions. Upper branching is inherited on each original index and starting block. Total incidence retention and bounded routing then preserve the whole profile F on the parent ensemble.

Apply the rescaling and filling just described. Its cell count gives aggregate raw multiplicity at least

$$N^{\varepsilon-o(1)}T(\kappa, a, b)\frac{a}{b}. \quad (4.6)$$

The sharp upper bound. A new full-time plank of widths A, B pulls back to one of widths aA, aB in original fine units. These are legitimate original tests: $1 \leq aA \leq aB \leq aN' = Nb/K \leq N$. Since the shape exponents for Δ_{h_0} sum to d , each parent's raw parameter obeys

$$\Delta_{h_0, \text{new}} \leq_{\text{exp}} a^d \Delta_{h_0}. \quad (4.7)$$

The old occupied time tree above scale K is unchanged; below it we have grafted full time intervals of b/a bins. The profile is exactly (4.4) in normalized logarithmic coordinates. In particular its temporal cost, expressed in original $\log N$ units, is $\Pi(1) - \Pi(\kappa)$.

The sharp hybrid upper bound in each parent is consequently at most

$$\begin{aligned} a^d \Delta_{h_0}(N')^H N^{\Pi(1) - \Pi(\kappa)} &=_{\text{exp}} m N^{-\Pi(\kappa)} K^{-H} a^{d-H} b^H \\ &= T(\kappa, a, b) \frac{a}{b}. \end{aligned} \quad (4.8)$$

The same upper bound holds uniformly for the ensemble. Otherwise a sequence of failing parent representatives would contradict the definition of the sharp exponent after regularizing to their common limiting profile. This contradicts (4.6).

If $N' = N^{o(1)}$, then $\chi \rightarrow 1$. Because $\chi \leq \kappa \leq 1$, one has $\kappa \rightarrow 1$ and $\Pi(1) - \Pi(\kappa) = o(1)$. The full-box mass bound in a parent, at most $a^d \Delta_{h_0}(N')^d$, supplies the same exponent upper bound as (4.8). Thus this endpoint also cannot carry the proposed fixed excess.

It follows that removing excessive assignments costs no extensive piece. Perform these removals successively for a finite mesh of scales and shapes, regularizing after each removal. The observation at the start of the section preserves equality for the next test. Nearby exponents are controlled by enlarging widths and extending to nearby nested temporal blocks; endpoint extrapolation costs only the prescribed mesh error. Let the mesh and error tolerances tend to zero in the stated order. This proves (4.2).

For the converse assertion, start with the extensive saturated assignments in the statement and perform the same routing, rescaling, and filling, with packet mass at least T in exponent and no fixed excess. The lower bound is now Ta/b and the upper bound is exactly the same expression. When $\chi < 1$, the derived resolution grows with a positive exponent and the profile in (4.4) is well-defined. Equality of the ensemble bounds provides either the ensemble formulation or realizing representatives. At $H = h_0 = h$ these are equalities for the original critical problem. $\square$

4.2 Multilinear slack and a common plane on each short block

We first isolate two geometric consequences of the local bound. They will be used both in the flat-delay argument and in the later alignment argument.

Lemma 4.2 (Slack and cap mass). *Every equality under consideration satisfies*

$$N^{H+(p-2)\alpha} \leq_{\text{exp}} \frac{k^2 m}{n}. \quad (4.9)$$

At a fine incidence, the raw mass of active lines whose directions lie in any cap of radius C/L , $1 \leq L \leq N$, is at most

$$N^{o(1)} m L^{-H} N^{-\Pi(\log_N L)}. \quad (4.10)$$

The constant C is fixed; the exponent error is uniform under the finite-test convention.

Proof. The full bounded patch is covered by boundedly many planks with both widths comparable to N , so $n \leq_{\text{exp}} \Delta_{h_0} N^d$. Since $k =_{\text{exp}} N^{1-\alpha}$ and $m =_{\text{exp}} \Delta_{h_0} N^{H+\Pi(1)}$, the ratio in (4.9) is at least $N^{q+H+\Pi(1)-2\alpha}$ in exponent. Inequality (4.1) gives the claim.

Lines meeting a fixed fine cell with slope diameter $O(1/L)$ stay within a bounded-width plank about one of them throughout the dyadic L -block containing that cell. They are active on that block. Apply (4.2) with $a, b = O(1)$, using bounded coverings to handle fixed enlargements. Bounded slope charts identify spatial slope diameter and spherical direction diameter up to constants. This proves (4.10). $\square$

The concentration of directions near a plane is the planiness principle familiar from Katz et al. [14]; see also the related multilinear consequence in Guth [10, Corollary 1.4]. For filling time blocks, we need the direction plane to stay fixed along each reference index throughout the block. We obtain this control by applying the multilinear estimate on balls of the block scale; the positive slack makes the incidence in balls with three rich transverse directions negligible.

Lemma 4.3 (Common short-block plane). *There is $\kappa_{\text{ml}} > 0$, depending only on the positive slack in (4.9) and the multilinear input, with the following property. For each fixed $0 < \kappa < \kappa_{\text{ml}}$, put $K = N^\kappa$. On an extensive regular central incidence graph, each active reference index-block can be assigned one direction plane such that:*

1. *the same plane is used at all of its retained times;*
2. *at each such time there is original neighboring incidence mass at least $mN^{-o(1)}$, in directions within $O(K^{-1})$ of that plane;*
3. *after discarding a negligible part of this neighboring mass, the neighbor direction is separated from the reference direction by at least $N^{-o(1)}$.*

All reference and neighbor segments over the block lie in a common plate of dimensions $C \times CK \times CK$ in fine units. The neighboring indices retain their original marked-time shadings and their inherited raw cap bounds.

Proof. Choose a lattice cover by balls of radius a sufficiently large fixed multiple of K in fine space-time units. Its overlap is bounded. The radius is large enough that one of the balls assigned to a reference index-block contains its whole segment and the whole K -block segment of every bounded-slope line meeting it at a fine cell. Only boundedly many choices are needed for any index-block.

By (4.2), the active mass near such a ball is at most mK^{C_0} , for an absolute constant C_0 : cover its time extent by boundedly many K -blocks and its transverse extent by boundedly many planks with $a, b \asymp K$. Multiplying by the at most CK events of each active line gives an incidence upper bound mK^{C_0+1} for the ball.

Divide directions into caps of radius K^{-20} and call a cap rich at the ball if its active mass there is at least mK^{-100} . There are $O(K^{40})$ caps. A ball is bad if three rich caps have direction determinant at least K^{-10} . Apply Theorem 3.7 to the unshaded tubes at spatial resolution K/N , summing over the possible cap triples. The transversality loss, the cap count, and the thresholds are fixed powers of K . Therefore, for a fixed constant C_1 and any fixed $\varepsilon > 0$,

$$\#\{\text{bad balls}\} \lesssim_\varepsilon N^\varepsilon K^{C_1} (n/m)^{3/2}. \quad (4.11)$$

Indeed, the multilinear summand at a tested ball is at least $(mK^{-100})^{3/2}$, and the product of the three full tube masses is at most $n^{3/2}$. Bounded ball-to-cube comparisons do not change the estimate.

The fraction of all weighted incidence in bad balls is consequently at most

$$N^\varepsilon K^{C_2} \left(\frac{n}{k^2 m} \right)^{1/2}.$$

Choose first κ_{ml} small enough that the K^{C_2} factor uses less than one quarter of the positive slack exponent, and then choose ε smaller still. By (4.9), bad balls carry a power-small incidence fraction. The same estimate applies to the bounded enlargements and choices of balls just used.

At a remaining ball, any three rich directions have determinant $O(K^{-10})$. Choose a maximally separated pair. If its separation is at most K^{-1} , all rich directions are within $O(K^{-1})$ of any plane containing one of them. Otherwise the norm of the pair's cross product is bounded below by c/K , and dividing the determinant bound by this norm places every rich direction within $O(K^{-9})$ of their spanning plane. In either case there is a single plane with the claimed $O(K^{-1})$ tolerance. Its normal has spatial component bounded below: it is approximately perpendicular to a direction of the form $(1, u)$ with bounded u .

The total mass in light caps at the ball is at most CmK^{-60} . Cells whose original multiplicity is less than $mN^{-\varepsilon_1}$ carry at most $N^{-\varepsilon_1}$ of all incidence, because their number is at most $|E|$. Let ε_1 decrease slowly to zero. Remove central incidences requiring bad balls and central incidences in the light caps of their assigned ball. At each relevant fine cell the light-cap bound is negligible compared with $mN^{-o(1)}$; the bounded number of possible adjacent balls does not alter that conclusion. Choose the ball once for each retained reference index-block, and keep this choice fixed at all its times. Regularize this central graph, preserving F .

For neighbors at a retained time use the original rich incidences at that fine cell. Their total mass is at least $mN^{-o(1)}$. The cap bound (4.10) with a radius $N^{-\gamma}$ shows that neighbors within that angle of the reference have mass at most $mN^{-H\gamma+o(1)}$. First take any fixed small $\gamma > 0$, then diagonalize $\gamma \downarrow 0$ after the previous errors are smaller. This leaves separation at least $N^{-o(1)}$ and does not use an estimate at an uncontrolled shrinking scale.

Finally, all these directions are $O(K^{-1})$ from the one chosen plane, and each neighbor meets the reference within bounded fine distance. Over a block of length K , their displacement normal to the affine plane through the reference remains bounded. Their tangential and temporal displacements are $O(K)$. A plate of the stated dimensions contains them all, including the filled reference segment. Neighbor shadings have not been restricted in this step, so their original profile and cap upper bounds remain available. $\square$

4.3 Planar filling and the flat delay

The common short-block plane supplies a plate containing each full reference segment. We will use its neighboring shadings to show that the original union has substantial density in this plate, and then fill the reference segment. Two estimates control the support added by this operation. The planar Furstenberg theorem supplies the plate density; the following maximal estimate converts that density into a bound for the union of all filled segments.

Lemma 4.4 (Plate Nikodym bound). *For $K \geq 2$, let $\mathcal{N}_K$ be the maximal averaging operator over rectangular plates of dimensions $1 \times K \times K$, with arbitrary orientations and translations, whose averaging plate contains the query point. Fixed changes in these dimensions are allowed. Then*

$$\|\mathcal{N}_K f\|_{L^2(\mathbb{R}^3)}^2 \lesssim (1 + \log K) \|f\|_{L^2(\mathbb{R}^3)}^2. \quad (4.12)$$

Consequently, if every point of a measurable set E^+ belongs to one such plate in which E has density at least $\rho > 0$, then

$$|E^+| \lesssim \rho^{-2} (1 + \log K) |E|. \quad (4.13)$$

The same statement holds for fine-cell counts after bounded thickening.

Proof. Take any measurable linearized choice P_x of an averaging plate containing x , and put

$$Af(x) = \frac{1}{|P_x|} \int_{P_x} f(z) dz.$$

The symmetric kernel of AA^* is

$$\mathcal{K}(x, y) = \frac{|P_x \cap P_y|}{|P_x||P_y|}.$$

Let $\vartheta \in [0, \pi/2]$ be the angle between the unoriented central-plane normals of the two plates. Intersecting two unit slabs gives transverse cross-sectional area $O(1/\sin \vartheta)$, while the available extent along their intersection line is $O(K)$. The trivial overlap bound is $O(K^2)$. Thus

$$|P_x \cap P_y| \lesssim \min(K^2, K/\vartheta) \lesssim \frac{K^2}{1 + K\vartheta}. \quad (4.14)$$

Fix P_x and let $\mathcal{P}_x$ be its central plane. If the plates intersect, choose a point in their intersection. Since $y \in P_y$, the normal displacement from that point to y , measured relative to $\mathcal{P}_x$, is at most $C(1 + K\vartheta)$. It follows that

$$\text{dist}(y, \mathcal{P}_x) \leq C(1 + K\vartheta).$$

Also $|x - y| \leq CK$, since both plates contain their respective query points and have a common point. Because $|P_x|, |P_y| \asymp K^2$, (4.14) gives

$$0 \leq \mathcal{K}(x, y) \lesssim \frac{\mathbf{1}_{\{|x-y| \leq CK\}}}{K^2(1 + \text{dist}(y, \mathcal{P}_x))}. \quad (4.15)$$

In coordinates tangent and normal to $\mathcal{P}_x$, the tangential area in this ball is $O(K^2)$. Integrating the normal coordinate therefore proves

$$\sup_x \int \mathcal{K}(x, y) dy \lesssim 1 + \log K.$$

Symmetry gives the same column bound. Schur's test yields $\|AA^*\|_{2 \rightarrow 2} \lesssim 1 + \log K$, proving the claimed squared norm bound uniformly in the linearization.

For a finite family of plate parameters, choose a maximizing plate measurably at each point. The uniform bound passes to the supremum over that family. A countable dense family suffices, by continuity of plate integrals under translation and rotation for locally integrable functions, and monotone convergence then gives (4.12). Apply it to $f = \mathbf{1}_E$ and use $\mathcal{N}_K \mathbf{1}_E \geq \rho$ on E^+ to obtain (4.13). Replacing counted cells by their unit cubes and allowing fixed plate enlargements proves the discrete version. $\square$

To obtain the plate density, sample a marked reference time and then a separated neighboring line at that time. If the time distribution has Frostman exponent s and the conditional direction distribution has exponent a_{ang} , their joint line-parameter law has exponent $t = s + a_{\text{ang}}$. The planar theorem contributes $\mu = \min\{1, (s + t)/2, t\}$ to the union size. The next lemma compares the resulting support cost of filling with the temporal cost $\Pi(\kappa)$ that filling removes.

Lemma 4.5 (Planar-prefix extension). *Let $0 < \kappa < \kappa_{\text{ml}}$ be fixed as in Lemma 4.3, and put $K = N^\kappa$. Suppose there are constants $s \in (0, 1]$ and $\gamma \geq 0$ such that*

$$F(\kappa) - F(u) \leq (1 - s)(\kappa - u), \quad F(u) \geq \gamma u \quad (0 \leq u \leq \kappa). \quad (4.16)$$

Set

$$a_{\text{ang}} = H + p_*\gamma, \quad t = s + a_{\text{ang}}, \quad \mu = \min\{1, (s+t)/2, t\}. \quad (4.17)$$

Then $a_{\text{ang}} \leq 1$, so $0 < t \leq 2$. Every equality configuration under consideration must satisfy

$$\Pi(\kappa) \leq F(\kappa) + 2\kappa(1 - \mu). \quad (4.18)$$

In particular, a strict reverse inequality yields a contradiction. There is no requirement that γ or $F(\kappa)/\kappa$ have a positive lower bound. The small-scale range is uniform over equalities with H bounded below by a fixed positive constant.

Proof. The planar neighbor measure. Use the central graph and the fixed block planes of Lemma 4.3. Choose a retained reference index-block. Normalize its block length to one and project orthogonally onto the corresponding affine plane. The projection changes its relevant unit directions by an additive $O(K^{-1})$. The reference-neighbor separation is $N^{-o(1)}$, which is much larger than K^{-1} for this fixed $\kappa > 0$; that separation is therefore preserved up to fixed factors. Also, projected direction diameter $O(r)$ implies original direction diameter $O(r + K^{-1}) = O(r)$ whenever $r \geq K^{-1}$.

We use physical time, rather than distance along the reference, as the graph coordinate on this plane. After a translation let its unit normal be $\nu = (\nu_0, \nu_{\text{sp}})$, where $|\nu_{\text{sp}}| \geq c > 0$. Put $n_{\text{sp}} = \nu_{\text{sp}}/|\nu_{\text{sp}}|$ and choose a unit spatial vector e_{sp} perpendicular to it. The coordinates $(\sigma, y) = (t, e_{\text{sp}} \cdot x)$ on the plane have inverse

$$(t, x) = \left(\sigma, y e_{\text{sp}} - \frac{\nu_0}{|\nu_{\text{sp}}|} \sigma n_{\text{sp}} \right),$$

so they are uniformly bi-Lipschitz. For a relevant original direction $(1, u)$, the projected time component is $1 - \nu_0 \nu \cdot (1, u) = 1 + O(K^{-1})$, while its projected y component is $e_{\text{sp}} \cdot u$. All projected lines are consequently bounded-slope graphs in these coordinates. This also preserves lengths, time resolution, and shading counts up to fixed factors.

First, the marked-time probability distribution on the reference block is s -Frostman at resolution K^{-1} . To check this, an interval of relative length $r \in [K^{-1}, 1]$ is covered by boundedly many dyadic blocks of $L \asymp rK$ bins. Put $u = \log_N L$. The relative number of reference events it contains is at most

$$N^{o(1)} \frac{L N^{-F(u)}}{K N^{-F(\kappa)}} \leq N^{o(1)} r^s, \quad (4.19)$$

by (4.16). The same estimate holds for the original marked times of every neighboring line on this block. The errors are subpowers of K because $\kappa > 0$ is fixed.

Construct a probability measure on planar neighbor lines as follows. Choose a marked reference time with the preceding distribution, then choose one of its coplanar separated neighbors with probability proportional to its original raw weight at that fine cell. The normalizing neighboring mass is at least $m N^{-o(1)}$. For any cap of radius $r \in [K^{-1}, 1]$, use (4.10) with $L \asymp 1/r$ and (4.1) to obtain the conditional probability bound

$$\mathbb{P}(\text{neighbor direction in that cap} \mid \text{time}) \leq N^{o(1)} r^H N^{-p_* F(\log_N(1/r))} \leq N^{o(1)} r^{a_{\text{ang}}}. \quad (4.20)$$

The coplanar directions fit in $O(K)$ caps of radius K^{-1} . Their normalized mass is one, so summing (4.20) at this radius gives $1 \leq N^{o(1)} K^{1-a_{\text{ang}}}$. Hence $a_{\text{ang}} \leq 1$ in the limit.

We claim that the resulting planar line-parameter probability measure is t -Frostman. A line-parameter ball of radius r restricts the projected slope to an interval of size $O(r)$, and therefore the original three-dimensional direction to a cap of size $O(r)$: the lost normal component is only $O(K^{-1}) \leq O(r)$. It also restricts the encounter time with the reference to an interval of length

$N^{o(1)}r$. For the latter assertion, subtract the projected reference's affine graph by a bounded shear in the physical-time coordinates just defined, and write a projected neighbor as $y = c + v\sigma$. The neighbor-reference angle is at least $N^{-o(1)}$, so $|v| \geq N^{-o(1)}$. The encounter condition is $|c + v\sigma| \lesssim K^{-1}$. Varying (c, v) in an r -ball changes $-c/v$ by at most $N^{o(1)}r$, provided the ball is smaller than a fixed fraction of $|v|$. For larger balls the same assertion follows from the trivial unit bound on the time interval. The fine encounter error contributes at most $N^{o(1)}K^{-1} \leq N^{o(1)}r$.

Use (4.19) for that encounter interval and (4.20) at each admissible time. Their product is at most $N^{o(1)}r^{s+a_{\text{ang}}}$, as required. This argument is conditional probability followed by integration; it does not assume that the encounter time and direction are independent.

Plate density and the filled support. The line-parameter measure is now t -Frostman. We apply the planar theorem to the neighbors' shadings, and use the resulting plate density to fill the reference blocks. Each neighbor index carries its original shading on the entire current block. It consists of $KN^{-F(\kappa)+o(1)}$ cells, with the s -Frostman bound (4.19). Repetitions of a neighbor through different sampled times merely add its line weight; they do not shorten this shading. The bounded projection preserves its cell count and Frostman estimates. Thus the weighted and repeated-line formulation in Corollary 3.5 applies to this probability law and these shadings. It gives at least

$$N^{-F(\kappa)+o(1)}K^{1+\mu} \quad (4.21)$$

planar cells in their union. More explicitly, one first fixes any small theorem loss, includes all required cell and conditional mass tests at that tolerance, and then lets the loss decrease. The strict inequality tested below is fixed before these choices. This is sufficient for the subpower notation in (4.21).

All the original neighboring segments lie in the plate from Lemma 4.3. Its thickness is bounded, so a projected cell receives only boundedly many relevant normal layers. In particular (4.21) lower-bounds the number of original union cells in this plate. The plate has volume comparable to K^2 . It therefore has original-union density at least

$$\rho \geq N^{-F(\kappa)+o(1)}K^{\mu-1}. \quad (4.22)$$

It contains the full reference segment, including times that were not originally marked. The same construction applies to every retained reference index-block.

Fill all such reference blocks with every fine time bin, keeping each original index and its inherited weight. Write E^+ for the new support. Every point of E^+ lies in a plate with density (4.22) in a fixed enlargement of the original support. Lemma 4.4 gives

$$|E^+| \leq N^{2F(\kappa)+2\kappa(1-\mu)+o(1)}|E|. \quad (4.23)$$

The squared density loss in this formula is retained as a power of N .

The number of incidences increases by $N^{F(\kappa)+o(1)}$. Indeed each retained occupied block had $KN^{-F(\kappa)+o(1)}$ times and now has K . The filled profile is zero below κ and equals $F(u) - F(\kappa)$ above it. Its cost is $\Pi(1) - \Pi(\kappa)$, while its global raw clustering parameter is at most Δ_{h_0} . The support estimate gives

$$m^+ \geq mN^{-F(\kappa)-2\kappa(1-\mu)-o(1)}.$$

The sharp hybrid upper bound gives $m^+ \leq mN^{-\Pi(\kappa)+o(1)}$. These are incompatible with a strict reverse inequality to (4.18). This proves the lemma. $\square$

Lemma 4.6 (Positive flat delay). *Every hybrid or exact equality with $H > 0$ has a profile that is identically zero on an interval $[0, \tau]$ with $\tau > 0$.*

Proof. Suppose that there is no such interval, and put $e = \liminf_{u \downarrow 0} F(u)/u$. Then $F(u) > 0$ for every $u > 0$. For any sufficiently small $\varepsilon > 0$, choose a small $r_0 > 0$ so that $F(u) \geq (e - \varepsilon)u$ for $0 \leq u \leq r_0$. There are arbitrarily small $r \leq r_0$ where $F(r) < (e + \varepsilon/2)r$. The continuous function $F(u) - (e + \varepsilon)u$ has a negative minimum on $[0, r]$. Choose a minimizing point $\kappa > 0$. It is a record minimum on $[0, \kappa]$, and hence

$$F(\kappa) - F(u) \leq (e + \varepsilon)(\kappa - u), \quad F(u) \geq \max(e - \varepsilon, 0)u \quad (0 \leq u \leq \kappa). \quad (4.24)$$

We may take κ as small as needed for Lemma 4.5.

At a typical fine incidence the neighboring mass is at least $mN^{-o(1)}$ and lies within $O(K^{-1})$ of one plane. Cover these directions by $O(K)$ caps of radius K^{-1} and use (4.10) together with (4.24) and (4.1). This gives $1 \leq N^{o(1)}K^{1-H-p_*\max(e-\varepsilon, 0)}$, hence $H + p_*\max(e - \varepsilon, 0) \leq 1$, without first requiring a positive time exponent. Letting $\varepsilon \downarrow 0$ yields $H + p_*e \leq 1$. In particular $e < 1$, and we may arrange $s = 1 - e - \varepsilon > 0$. Use (4.24) in Lemma 4.5, with

$$\gamma = \max(e - \varepsilon, 0), \quad t = 1 - e - \varepsilon + H + p_*\gamma.$$

If $e = 0$, choose ε small compared with H . Then $s = 1 - \varepsilon$, $t = 1 + H - \varepsilon$, and $\mu = 1$. Since $F(\kappa) > 0$,

$$\Pi(\kappa) - F(\kappa) \geq (p_* - 1)F(\kappa) > 0,$$

contradicting (4.18). This argument does not need a lower bound for $F(\kappa)/\kappa$.

Suppose $e > 0$. At $\varepsilon = 0$ one has

$$s = 1 - e, \quad t = 1 + H + (p_* - 1)e > 1,$$

and therefore

$$2(1 - \mu) = \max\{0, (2 - p_*)e - H\}.$$

The gap between $(p_* - 1)e$ and the right-hand side is strictly positive. If the maximum is zero this is immediate; otherwise the gap equals $(2p_* - 3)e + H > 0$. By continuity, for sufficiently small fixed ε ,

$$(p_* - 1)(e - \varepsilon) > 2(1 - \mu).$$

But (4.24) and (4.1) now give

$$\Pi(\kappa) - F(\kappa) \geq (p_* - 1)F(\kappa) \geq (p_* - 1)(e - \varepsilon)\kappa > 2\kappa(1 - \mu),$$

again contradicting (4.18). Both cases are impossible, proving the flat delay. $\square$

We have proved that an equality has an initial interval with zero temporal deficit. On any fixed scale in that interval, its shadings have subpower density loss, so the weighted set estimate applies. Together with the local bound, it forces the following plate shape.

Lemma 4.7 (Saturated plates on a flat scale). *At every fixed positive scale exponent κ with $F|_{[0, \kappa]} = 0$, an extensive regular restriction admits disjoint indexed-block assignments to plates with widths $1, K$, up to subpowers, where $K = N^\kappa$. Each active plate packet has raw index mass at least mK in exponent.*

Proof. Split into temporal K -blocks and bounded-overlap position charts of diameter CK fine cells containing complete block segments. Each original index-block uses boundedly many charts. The normalized resolution is K , and the local shading density is $N^{-o(1)}$ because $F(\kappa) = 0$. Count

charts separately. Their total incidence is extensive and their aggregate support is at most a fixed multiple of the old support, so every extensive remainder has aggregate raw multiplicity at least m in exponent.

Fix a small $\varepsilon > 0$. Greedily remove and assign all remaining local lines in a plank whenever its density $n_W/(ab)$ is at least $mN^{-\varepsilon}$. If an extensive remainder survives, its clustering parameter is at most $mN^{-\varepsilon}$ in every local chart. The weighted set estimate, Lemma 3.3, applies with local resolution K and arbitrarily small fixed theorem loss: choose that loss below $\varepsilon/(2\kappa)$, then take N large enough for the subpower local shading density to meet its hypothesis. It bounds the remainder's multiplicity by at most $mN^{-\varepsilon/2}$ in exponent. Uniformity over charts follows by representative selection. This contradicts its extensive incidence and inherited support. Thus the captured graph is extensive, with disjoint index ownership on each block.

Combine its lower bound $mN^{-\varepsilon}ab$ with (4.2), where $\Pi(\kappa) = 0$. It gives

$$N^{-\varepsilon} \leq N^{o(1)} K^{-H} a^{-q-H} b^H,$$

or, in logarithmic coordinates,

$$(q + H) \log_N a + H \log_N(K/b) \leq \varepsilon + o(1).$$

Both terms are nonnegative and $H > 0$. Let the capture tolerance decrease to zero after each finite set of tests. The captured planks have $a = N^{o(1)}$, $b = KN^{o(1)}$, and their mass is at least $mKN^{-o(1)}$. Cleaning the assignment and chart partitions preserves these lower bounds on occupied packets, as in Proposition 4.1. $\square$

4.4 Anisotropic normalization with inherited weights

The next transformation uses alignment of the saturated plates. It preserves their thin spatial coordinate while compressing time and the long spatial coordinate by the angular scale θ . At fixed packet mass, the incidence count involves $K\theta$ time bins and the support bound involves $(K\theta)^2$ cells. Their ratio gives a raw multiplicity gain of θ^{-1} . The lemma also records the pulled-back plank bound and the remaining temporal profile.

Lemma 4.8 (Anisotropic normalization). *Suppose that on an extensive regular graph, over larger blocks of $K_1 = N^{\kappa_1}$ bins, there are disjoint indexed-block assignments to saturated plates on a flat scale $K = N^{\kappa} \leq K_1$. Their widths are $1, K$ up to subpowers and their raw active index masses are at least mK in exponent. Suppose the thin spatial axes of the assigned plates along each original index and larger block are within θ of one fixed axis for that index and block, up to subpower factors, where*

$$K^{-1} \leq \theta \leq 1 \quad \text{in exponent}, \quad \lambda = \log_N(1/\theta) \leq \kappa.$$

There is an ensemble of bounded-chart configurations at resolution

$$N' = K_1\theta \tag{4.25}$$

whose aggregate raw multiplicity is at least m/θ in exponent. Each derived index retains its original weight, and each original index and larger block contributes to only subpower many charts.

Suppose also that all original active lines on a larger block obey the local clustering bound $n_W \leq_{\text{exp}} Ca^{x'}b^{y'}$ for $1 \leq a \leq b \leq K_1$, where $0 \leq x' \leq y'$. Then every derived chart has full-time clustering parameter at most

$$C\theta^{-y'} \tag{4.26}$$

for those same shape exponents. If $\kappa_1 - \lambda > 0$, its temporal profile, uniformly over the ensemble, is

$$G(v) = \frac{F(\lambda + (\kappa_1 - \lambda)v)}{\kappa_1 - \lambda}, \quad 0 \leq v \leq 1. \quad (4.27)$$

Its cost in original $\log N$ units is $\Pi(\kappa_1)$; only the initial flat interval $[0, \lambda]$ has been deleted.

Proof. The coordinate map and multiplicity gain. Replacing θ by $\min(\theta, 1)$ changes it only by a subpower factor, so we may take $0 < \theta \leq 1$ in the geometric construction. First consider a chart centered on one affine line, with normal-tangent coordinates z_n, z_t and time $r \in [0, K_1]$, all in old fine-cell units. The chart bounds we will need are $|z_n| \lesssim \theta K_1$, $|z_t| \lesssim K_1$, together with the corresponding bounds on displacement over the larger block. For such a chart use normalized coordinates

$$s = \frac{r}{K_1}, \quad X = \frac{z_n}{\theta K_1}, \quad Y = \frac{z_t}{K_1}, \quad (4.28)$$

at resolution $N' = K_1 \theta$. Equivalently, in the new fine-cell units the coordinate map is

$$(r, z_n, z_t) \mapsto (\theta r, z_n, \theta z_t). \quad (4.29)$$

The new slopes and intercepts are bounded. An old normal fine cell remains a fine cell; old time and tangent cells are grouped in intervals of length $1/\theta$.

Suppose a fine plate's axis differs from the chart normal by $|\delta| \leq \theta/10$, as the angular partition below will ensure. Its thin condition is transformed from $|\cos \delta z_n + \sin \delta z_t - c(r)| \lesssim N^{o(1)}$ to

$$|\cos \delta z'_n + (\sin \delta / \theta) z'_t - c'(r')| \lesssim N^{o(1)}.$$

The coefficients have bounded or subpower size and the normal coefficient is bounded below. Hence it has subpower fine thickness and tangential extent at most $N^{o(1)} K \theta$ in the new chart. Fill its selected K -intervals. Each now spans $K \theta$ new bins, up to rounding, and occupies at most $N^{o(1)} (K \theta)^2$ cells. If its retained index mass is at least $m K N^{-o(1)}$, its incidence mass is at least $m K N^{-o(1)} \cdot K \theta$. Thus disjoint indexed-block assignments with these retained masses give aggregate raw multiplicity at least m/θ in exponent. Packet support overlaps can only improve this lower bound.

Charts and preservation of the time tree. We now construct charts with the stated bounds, keeping each original index and its selected time blocks together and retaining the packet masses required by the cell count. Use angular bins of width $\theta/10$ for the unoriented thin spatial axis. Since the axes used along any one original index and larger block lie within subpower times θ of its fixed axis, that index-block meets only subpower many bins. In each angular bin choose a normal-tangent frame. Subdivide matrix space, expressed over the larger interval, into parent cells of widths $\theta K_1, K_1$ in normal and tangent position and also in the normal and tangent displacement over that interval. A fixed matrix has only boundedly many adjacent grid parents in each bin.

An entire fine plate packet meets only subpower many such parents. Indeed its thin axis belongs to the angular bin and differs from the bin normal by at most $\theta/10$. Its cross-sectional spread measured in that normal is at most $N^{o(1)}(1 + \theta K)$, and its tangential spread is at most $N^{o(1)} K$. Divide the endpoint differences by the fine-block duration, then extrapolate over the larger interval. The resulting spreads are at most

$$N^{o(1)}(K_1/K + \theta K_1) \leq N^{o(1)} \theta K_1 \quad \text{in the normal coordinate,}$$

and $N^{o(1)} K_1$ in the tangent coordinate. This uses $\theta \geq K^{-1}$ in exponent. The same bounds hold for intercepts after subtracting a parent center. Thus the asserted parent routing follows from endpoint extrapolation, rather than from the number of fine subblocks used by the index.

Within a chosen larger block, bin, and matrix parent, keep all selected fine intervals of one original index together as a single derived index. Do not create a separate derived index for each fine interval. This preserves inherited weights and ensures that a full-plank test in a chart counts a subset of original active lines on the larger block. Include packet subdivisions and parent assignments in the finite cleaning. Since every original index-block has subpower routing, Corollary 2.7 shows that surviving packets retain active mass $mKN^{-o(1)}$.

The routing also preserves the time tree. On every fixed logarithmic subrange of the larger block, each derived time tree is contained in its original tree. Its branching exponents therefore cannot increase. The chart ensemble retains an extensive part of the weighted incidence, while its total available original index-block mass has increased by only a subpower. After uniform regularization there can be no loss in the total branching exponent. The nonnegative losses on each fixed subrange must consequently all be zero. This is the ensemble profile-preservation argument of Lemma 2.6, with its routing hypothesis now verified. Thus the inherited prefix is $F|_{[0, \kappa_1]}$. The charts and retained packets now meet the hypotheses of the first calculation. Applying (4.29) and filling the selected blocks gives the asserted lower multiplicity m/θ .

The plank parameter and temporal cost. For the clustering calculation, pull back a new full-time plank of spatial widths a, b by the inverse of $(z_n, z_t) \mapsto (z_n, \theta z_t)$. Its cross-section is a parallelogram contained in a rotated rectangle of side lengths $A \leq B$ with

$$AB \lesssim ab/\theta, \quad B \lesssim b/\theta.$$

For example, singular-value axes of the linear image of its inscribed ellipse give this rectangle with fixed comparison constants. Its smallest width is bounded below by a constant times a , and $B \lesssim b/\theta \leq K_1$; thus these are permitted old tests after bounded coverings or clipping. Since $0 \leq x' \leq y'$, the pulled-back mass is at most

$$\begin{aligned} CA^{x'} B^{y'} &= C(AB)^{x'} B^{y'-x'} \\ &\lesssim C(ab/\theta)^{x'} (b/\theta)^{y'-x'} = C\theta^{-y'} a^{x'} b^{y'}. \end{aligned}$$

Each chart index is an original index counted only once there, so no extra weight factor enters this inequality. This proves (4.26).

Finally, the fine scale $1/\theta = N^\lambda$ lies in the flat prefix, since $\lambda \leq \kappa$. Filling the selected K -blocks changes neither the limiting profile nor its cost. Rebinning at $1/\theta$ deletes the initial flat prefix of length λ and translates the remaining logarithmic scale axis. After dividing by the new total logarithmic length $\kappa_1 - \lambda$, this is precisely (4.27). Its cost in original units is $\Pi(\kappa_1) - \Pi(\lambda) = \Pi(\kappa_1)$. When $\kappa_1 - \lambda = 0$ in the limit, the packet and clustering bounds still have their asserted exponents; an upper estimate can use total mass instead of a positive-resolution profile. $\square$

4.5 An equality with positive total deficit

The two rescalings retain complementary parts of the time profile. The isotropic cut keeps its branching above K and fills the scales below that block. The anisotropic normalization keeps the prefix through K_1 and removes only its initial flat range $[0, \lambda]$. We will use both on exact equalities in the next section. First we use the anisotropic operation in the hybrid case, where the strict inequality $H > h_0$ gives a contradiction if the plates can be aligned over the full time interval.

Lemma 4.9 (Positive-deficit equality). *For $h_0 \uparrow h > 0$, the total deficits $\alpha = F(1)$ of hybrid equalities are bounded below by a positive constant, uniformly once h_0 is sufficiently close to h . Consequently there is an exact equality at $H = h_0 = h$ whose limiting profile satisfies $F(1) > 0$.*

Proof. We show that a sufficiently small fixed upper bound on α is incompatible with any such hybrid equality. All fixed tolerances in the next paragraphs will be chosen independently of $H - h_0$. Restrict to $h_0 \geq h/2$, so that $H \geq h/2$.

We will capture an extensive incidence graph in nearly maximally elongated global slabs, then align the saturated local plates with the fixed slab axis assigned to each index. Anisotropic normalization will make the sharp hybrid upper bound smaller than the retained raw multiplicity.

Fine-scale planarity and global slab capture. We first establish a fine-scale version of the multilinear discard. Choose a small constant $c_* > 0$, depending only on the positive lower bound $h/2$ and the multilinear input. At a fine cell use direction caps of radius N^{-2c_*} , declaring a cap rich if its active mass is at least mN^{-10c_*} . Mark a cell bad if three rich caps have determinant at least N^{-c_*} . The same multilinear count as in (4.11), now at fine resolution, gives

$$\#\{\text{bad fine cells}\} \lesssim_\varepsilon N^{\varepsilon+C_3c_*} (n/m)^{3/2}$$

for a fixed C_3 . Local bound (4.2) at $K = 1$ bounds active mass at a fine cell by $mN^{o(1)}$. By (4.9), the fraction of incidence in bad cells is at most

$$N^{\varepsilon+C_3c_*+o(1)} \left(\frac{n}{k^2m} \right)^{1/2} \leq N^{\varepsilon+C_3c_*-h/4+o(1)}.$$

Choose c_* and then ε so this is a negative fixed power. At every remaining cell all triples of directions in rich caps have determinant $O(N^{-c_*})$. The total mass in light caps is at most CmN^{-6c_*} , since there are $O(N^{4c_*})$ caps. These choices are independent of any later flat scale.

Choose a fixed $e_1 > 0$ sufficiently small compared with c_*h and h , and require α to be smaller still, as well as small enough for Lemma 3.3 with target loss $e_1/2$. Greedily assign and remove all original full-time indices in any global plank S of widths a, b with density

$$n_S/(ab) \geq mN^{-e_1}. \quad (4.30)$$

If an extensive remainder survives, its profile remains F , so its shading density is $N^{-\alpha+o(1)}$. All its plank densities are at most mN^{-e_1} . The weighted set estimate then gives multiplicity at most $mN^{-e_1/2}$, contrary to extensivity and the inherited support bound. Thus the assigned graph is extensive. Each original index has exactly one global assigned slab, and all of its retained times keep that label.

Compare (4.30) with the original Δ_{h_0} bound and the hybrid equality. Put $A = \log_N a$, $B = \log_N b$ just for this calculation. Then

$$(q + h_0)A + h_0(1 - B) + (H - h_0) + \Pi(1) \leq e_1 + o(1). \quad (4.31)$$

In particular

$$a \leq N^{O(e_1/h_0)}, \quad b \geq N^{1-O(e_1/h_0)}.$$

If (4.31) has no feasible shape, the capture already gives a contradiction; otherwise these width bounds hold on every retained slab.

The assigned slab S has at least $n_S k$ weighted incidences and support in $O(abN)$ cells. Its average assigned multiplicity is therefore at least $mN^{-e_1-\alpha-o(1)}$. In each slab delete cells with assigned multiplicity smaller by an inverse subpower factor. Their incidence is bounded by that factor times the slab incidence, by the support count and (4.30). Summing over slabs and cleaning the corresponding partitions gives the following lower bound at every retained query incidence:

$$\mu_S \geq mN^{-e_1-\alpha-o(1)}. \quad (4.32)$$

This is a lower bound for the incidence from its own assigned slab. No independence from any later assignment is asserted.

Alignment with saturated local plates. The global slab assigned to an index has a fixed axis. We next compare it with the axes of the local saturated plates used by that index. By Lemma 4.6, this particular hybrid equality has a positive flat delay. Now choose a fixed

$$0 < \xi \ll c_*$$

smaller than that delay, and set $K = N^\xi$. Lemma 4.7, applied within the extensive S -assigned graph, supplies saturated local plates W of widths $1, K$ up to subpowers and mass at least mK in exponent. Each such packet has mK^2 incidence in exponent and at most $K^2 N^{o(1)}$ support cells. The analogous low-cell deletion therefore gives, at its retained query incidences,

$$\mu_W \geq mN^{-o(1)}. \quad (4.33)$$

Retain (4.32) as another partition test. Both lower bounds thus hold on the same extensive central graph. Fine multilinear bad cells can be discarded from it by their power-small original incidence bound, followed by the same subpower deficiency cleaning.

At such a query cell, remove directions in the global light caps when choosing comparison directions. Their mass CmN^{-6c_*} is negligible compared with both (4.32) and (4.33), by the choices of e_1 and α . Among the remaining directions in S there are two separated by at least $N^{-c_*/10}$. Otherwise one cap of this radius would contain all their mass, whereas (4.10) with $L = N^{c_*/10}$ bounds that cap mass by $mN^{-Hc_*/10+o(1)}$. Choose $e_1 + \alpha < Hc_*/20$ to contradict (4.32). Likewise there are two remaining directions in W separated by at least $K^{-1/2}$: the cap bound with $L = K^{1/2}$ is $mK^{-H/2}N^{o(1)}$, which is smaller than (4.33) for this fixed $\xi > 0$.

We justify how these two pairs compare the actual plate axes. A plank with spatial thin axis n_0 , thickness a_0 , and duration K_0 confines $n_0 \cdot u$ to an interval of length $O(a_0/K_0)$. Its direction plane has normal proportional to $(-v_0, n_0)$, where v_0 is the central normal slope. Since the original slopes are bounded, this normal has spatial component bounded below after normalization. Thus all directions in S are within $N^{-1+O(e_1/h_0)}$ of its actual direction plane; all directions in W are within $K^{-1+o(1)}$ of its direction plane.

Let v_1, v_2 be the separated rich pair from S . Every rich direction is within

$$O\left(\frac{N^{-c_*}}{|v_1 \wedge v_2|}\right) = O(N^{-9c_*/10})$$

of their span, by the absence of a bad triple. The span of v_1, v_2 is within $N^{-1+O(e_1/h_0)+c_*/10}$ of the actual S plane. If w_1, w_2 is the separated rich pair from W , their span is within $N^{-9c_*/10+\xi/2+o(1)}$ of the span of v_1, v_2 , and within $K^{-1/2+o(1)}$ of the actual W plane. Consequently the two actual direction planes differ by at most

$$N^{-1+O(e_1/h_0)+c_*/10} + N^{-9c_*/10+\xi/2+o(1)} + N^{-\xi/2+o(1)}. \quad (4.34)$$

With the fixed choices above and $\xi \ll c_*$, this is smaller than $\theta = N^{-\xi/4}$. Taking the normalized spatial components of the plane normals is a uniformly Lipschitz map here. Hence the thin spatial axes of W and of the index's assigned global slab S differ, up to sign, by at most $C\theta$.

The hybrid contradiction. For each original index its S axis is fixed over all time. Thus the hypotheses of Lemma 4.8 hold globally, with $K_1 = N$ and the finer flat scale $K = N^\xi$. All previous discards were extensive or removed a power-small incidence fraction. The fixed losses e_1 and α were used only to obtain the query lower bounds and identify the axes; they are not losses in the retained

incidence or packet mass. After subpower cleaning and routing, the lower raw multiplicity supplied by that lemma is therefore m/θ , without an extra factor $N^{-e_1-\alpha}$.

We now use the strict gap between the hybrid exponents. Take $(x', y') = (d - 1 - h_0, 1 + h_0)$ in the anisotropic clustering bound; these satisfy $0 \leq x' \leq y'$. The new resolution is $N\theta$, and the temporal cost in original units is still $\Pi(1)$. The sharp hybrid upper estimate becomes

$$\begin{aligned} m_{\text{new}} &\leq_{\text{exp}} \Delta_{h_0} \theta^{-(1+h_0)} (N\theta)^H N^{\Pi(1)} \\ &=_{\text{exp}} m\theta^{H-1-h_0}. \end{aligned} \tag{4.35}$$

Its ratio to the lower bound m/θ is θ^{H-h_0} , a fixed negative-power saving in N because $H - h_0 > 0$ and $\xi > 0$ are fixed for this hybrid equality. This is a contradiction. The possibly small value of $\xi(H - h_0)$ causes no problem: that hybrid and its flat scale are fixed before its realizing resolution tends to infinity.

Passage to an exact equality. The choices of c_*, e_1 and the upper bound on α depended only on h, p, q and the external theorem tolerances, not on $h - h_0$. This proves a uniform positive lower bound for α as $h_0 \uparrow h$. Finally, for every configuration,

$$N^{-(h-h_0)} \Delta_{h_0} \leq \Delta_h \leq \Delta_{h_0},$$

because $1 \leq b/a \leq N$ in each plank test. Thus the hybrid equality exponents tend to the exact critical one. Apply the profile compactness and equality diagonal from Lemma 2.11. A downward jump of the convex cost would improve the critical score and contradict the sharp upper bound, so it cannot spoil equality. Uniform convergence of the profiles preserves their positive endpoint lower bound. The resulting exact equality has $F(1) > 0$, as claimed. $\square$

5 Construction of a stationary equality

We work at the differentiability point supplied by Lemma 2.10, and use the exact, positive-deficit equality from Lemma 4.9. Thus the upper exponent and the exponent in the plank parameter are both h . Throughout this section,

$$x = d - 1 - h \geq 0, \quad y = 1 + h, \quad x + y = d, \quad y - x = q + 2h > 0.$$

The word equality refers to a realizing sequence in the normalized logarithmic sense of Section 2. All local masses retain the original index weights. In particular, taking an extensive regular refinement does not renormalize those weights.

The two rescalings from Section 4 retain complementary parts of a temporal profile. An isotropic cut keeps the scales above a chosen block, with full time grafted below them; an anisotropic cut keeps the scales within that block and removes an initial flat interval. At exact criticality, saturated packets make both rescalings attain the sharp inequality. We call the resulting models outer and inner equalities.

Parameter derivatives first select the packet shapes, and a packing argument makes their normals coherent enough for the inner rescaling. We then compare the delays and deficits of the two resulting profiles. This forces a profile that is flat and then linear, and whose normalized outer models reproduce the same profile. That self-reproduction, together with a constant rate of change of packet widths, is the stationarity constructed here.

5.1 Derivative capture and outer equality models

At logarithmic scale κ , put $K = N^\kappa$ and write the local bound from Proposition 4.1 as

$$T(\kappa, a, b) = mN^{-\Pi(\kappa)}K^{-h}a^xb^y, \quad 1 \leq a \leq b \leq K. \quad (5.1)$$

A saturated assignment at this scale is a disjoint assignment of active indices to planks on each K -block whose occupied packets have mass $=_{\text{exp}} T(\kappa, a, b)$, with one common width type. Disjointness concerns the assigned indices on a fixed temporal block; geometric planks need not be disjoint.

Here and below a prefix means the logarithmic scale interval $[0, \kappa]$, realized on every occupied physical K -block; it does not mean only the physical interval starting at time zero. The next lemma uses changes of the critical parameters to select saturated shapes. The p derivative measures temporal deficit, while the z derivative also detects the thin spatial width.

Lemma 5.1 (Derivative capture). *Every equality profile has deficit $\alpha \leq v = -h_p$. In particular, $v > 0$. Fix an equality, a prefix $\kappa > 0$, and a perturbation direction $(\dot{p}, \dot{z})$. On an extensive refinement there are saturated assignments with limiting shape parameters*

$$\mathbf{a} = \log_N a, \quad \mathbf{u} = \log_N(K/b), \quad \chi = \mathbf{a} + \mathbf{u},$$

satisfying

$$(F(\kappa) - v\chi)\dot{p} + \left(h_z\chi + \int_0^\kappa \mathcal{D}_q(q, F'(s)) ds - \mathbf{a}\right)\dot{z} \geq 0. \quad (5.2)$$

Consequently there are selections with $\chi \leq F(\kappa)/v$ and selections with $\chi \geq F(\kappa)/v$. These assertions remain valid after any previously imposed extensive regular refinement.

Proof. Decrease p by a positive amount ε . Differentiability gives $h(p - \varepsilon, z) = h + v\varepsilon + o(\varepsilon)$, with $v \geq 0$. Increasing the clustering exponent cannot increase Δ_h : for each plank, its dependence on that exponent is through $(a/b)^h$. The cost changes by $-\alpha\varepsilon$. Applying the perturbed upper bound to an equality therefore gives $\alpha \leq v$. Since a positive-deficit equality exists, $v > 0$.

For the local assertion fix the perturbation size first. Denote the nearby parameters by primes, and let Π' denote the cost with the changed integrand and the same profile. On each prefix block greedily assign and remove the remaining active indices in planks whose mass is at least

$$mN^{-\text{tol} - \Pi'(\kappa)}K^{-h'}a^{d'-1-h'}b^{1+h'}. \quad (5.3)$$

Here $\text{tol} > 0$ is fixed during this operation. If the captured events were not extensive, the remainder would be extensive. Regularize that remainder to its inherited profile and apply the perturbed critical bound separately on the prefix blocks and bounded spatial charts. Its plank parameter is bounded by the threshold used in the greedy removal, so this bound gives aggregate raw multiplicity at most $mN^{-\text{tol}+o(1)}$. The extensive remainder has aggregate raw multiplicity at least $mN^{-o(1)}$, a contradiction. Spatial charts may be assigned by position at the block center; bounded slopes give bounded overlap in their support counts.

The captured graph is therefore extensive. Pigeonhole a common dyadic width type and use Lemma 2.6 to preserve the packet lower counts. Comparing (5.3) with (5.1) yields

$$\Pi'(\kappa) - \Pi(\kappa) + (h' - h)\chi - (d' - d)\mathbf{a} \geq -\text{tol}. \quad (5.4)$$

The first term records the change in temporal cost. The remaining terms record the resolution lost in the cut and the change of its thin-width exponent. The shapes lie in the compact triangle $\mathbf{a}, \mathbf{u} \geq 0, \mathbf{a} + \mathbf{u} \leq \kappa$. Take the perturbation to zero along the specified direction, with tol of smaller

order, and pass to a convergent sequence of shapes. First choosing the resolution sufficiently large at each fixed perturbation makes the preceding exponent errors smaller than tol . The captured masses are squeezed to saturation in the limit.

Since $z = -q$ and $d = 2 + z$,

$$\partial_p \Pi(\kappa) = F(\kappa), \quad \partial_z \Pi(\kappa) = \int_0^\kappa \mathcal{D}_q(q, F'(s)) ds, \quad d_z = 1.$$

The averaging in $\mathcal{D}$ makes its q derivative available at the interior parameter in use. Differentiating (5.4) proves (5.2). Taking $(\dot{p}, \dot{z}) = (1, 0)$ and $(-1, 0)$ gives the two final selections. The same argument applies to a preceding extensive regular refinement because its profile, equality exponents, and inherited local upper bounds are unchanged. $\square$

We record explicitly the equality obtained from a saturation. This also fixes the scale bookkeeping for the iteration. If $\chi < 1$, the isotropic parent construction in Proposition 4.1 has resolution

$$N_{\text{out}} = \frac{Nb}{Ka} = N^{1-\chi}.$$

Its physical cell width in the old cell units is $a = N^a$. The raw multiplicity supplied by a saturated packet is

$$T(\kappa, a, b) \frac{a}{b} = m N^{-\Pi(\kappa) - h\chi + da} \leq_{\text{exp}} m_{\text{out}}. \quad (5.5)$$

Pulling back an arbitrary full-time plank multiplies both widths in cell units by a , so its parameter is at most $\Delta_h a^d$. Filling the selected K -blocks drops exactly the cost below κ . Thus the critical upper estimate is

$$\Delta_h a^d N_{\text{out}}^h N^{\Pi(1) - \Pi(\kappa)} = m N^{-\Pi(\kappa) - h\chi + da}.$$

Consequently the outer model is an equality. In normalized log coordinates its profile is

$$F_{\text{out}}(r) = \begin{cases} 0, & 0 \leq r \leq (\kappa - \chi)/(1 - \chi), \\ \frac{F(\chi + (1 - \chi)r) - F(\kappa)}{1 - \chi}, & (\kappa - \chi)/(1 - \chi) \leq r \leq 1. \end{cases} \quad (5.6)$$

In particular its deficit is $(\alpha - F(\kappa))/(1 - \chi)$. The full-time isotropic parents have one owner per index. Later local plank bounds pull back with the same multiplication of widths, and their active indices belong to the corresponding physical blocks.

5.2 Finite chains and orientation control

Call an equality nontrivial if its total deficit $F(1)$ is positive. By Lemma 4.6, such an equality has an initial flat interval whose endpoint τ satisfies $0 < \tau < 1$.

Lemma 5.2 (Compatible finite cuts). *Fix a nontrivial equality and a finite increasing mesh $\tau = \kappa_0 < \kappa_1 < \dots < \kappa_J < 1$. There is an extensive refinement with simultaneous saturated assignments at these physical prefixes. Their cumulative shape parameters have nonnegative increments in $\mathbf{a}$ and $\mathbf{u}$, and can be selected so that*

$$\Delta \chi_i \leq \frac{F(\kappa_i) - F(\kappa_{i-1})}{v}, \quad \chi_i \leq F(\kappa_i)/v. \quad (5.7)$$

At each prefix the assignments have disjoint index ownership on each physical block. Every retained event carries all its assignments. Their used packet masses have the saturation exponents even after pulling the finite chain back to the original event graph. Any fixed finite list of additional partition tests can be imposed at the same time.

Proof. At the initial flat prefix, Lemma 5.1 gives $\chi_0 = 0$. Thus $a = 1$, $b = N^\tau$, and the packets are plates of mass $=_{\text{exp}} mN^\tau$. Cut to the outer model, keeping its full ensemble of matrix parents with inherited weights, rather than selecting a single parent. Apply derivative capture at the next physical prefix in this ensemble, and continue.

The parameter calculation for one step explains the choice of cuts. Suppose the preceding cumulative parameters are $(\mathbf{a}_{\text{old}}, \mathbf{u}_{\text{old}})$, with $\chi_{\text{old}} = \mathbf{a}_{\text{old}} + \mathbf{u}_{\text{old}}$, and put $L = 1 - \chi_{\text{old}}$. If the next cut in the normalized outer model has shape $(\mathbf{a}', \mathbf{u}')$, its pullback has

$$(\mathbf{a}, \mathbf{u}) = (\mathbf{a}_{\text{old}}, \mathbf{u}_{\text{old}}) + L(\mathbf{a}', \mathbf{u}').$$

Indeed spatial cell widths multiply. Physical time is unchanged, but in the new cell units

$$K_{\text{new}} = KN^{-\chi_{\text{old}}}, \quad b_{\text{new}} = bN^{-\mathbf{a}_{\text{old}}}, \quad \frac{K_{\text{new}}}{b_{\text{new}}} = \frac{K}{b}N^{-\mathbf{u}_{\text{old}}}.$$

This also gives the asserted increment in $\mathbf{u}$; both increments are nonnegative. If the preceding physical cut was at κ_{old} , the outer deficit up to the next physical scale κ is $[F(\kappa) - F(\kappa_{\text{old}})]/L$, by (5.6). The first selection in Lemma 5.1 therefore gives

$$\chi - \chi_{\text{old}} \leq [F(\kappa) - F(\kappa_{\text{old}})]/v.$$

Iteration gives (5.7). Since F is 1-Lipschitz and $v > 0$ is fixed, adjacent changes in both shape parameters are bounded by a constant times the mesh size.

We give the retention details for one step and then for its pullback. Each current parent has bounded normalized matrix coordinates, the common profile, and the inherited upper bounds. Summing support counts with the different parents counted separately preserves the aggregate lower multiplicity in (5.5). The greedy argument of Lemma 5.1 applies in each parent and physical block with this common raw threshold. If it could not capture an extensive piece of the ensemble, the uniform perturbed upper bound on the remainder, summed over the parents, would contradict that aggregate lower estimate. Retain a common width type among the captured packets, and reunite the selected blocks of each index in its current full-time parent. There is one such parent per index, so this reunion does not duplicate its weight. Regularization preserves the profile and the packet masses, after which the next isotropic parent transformation is defined.

All the derived blocks used at the next stage are coarser physical blocks than those already filled. A filled time is present only because its index occupied the earlier, finer block. Hence activity in a derived block implies original activity in the same physical block. The pulled back upper bounds therefore count subsets of the original active indices, as required by Proposition 4.1.

For the pullback, consider an earlier registered pair consisting of an index and its occupied cut block. Before filling, this pair supplied a common number in exponent of selected events; after filling and rebinning, it supplies another common number. The index weight is unchanged. An extensive selection of the later events therefore hits an extensive weight of these pairs. Keep all the preceding selected events of each hit pair. A later assignment is constant on that earlier pair: its temporal block is a coarser member of the same nested dyadic grid, and its ownership decision concerns the index on that block. Thus this pullback preserves all later assignment memberships. Repeating the operation restores an extensive graph of original events carrying all assignments in the finite chain.

One must also preserve the masses of the used parts of earlier packets. At any fixed earlier level, its assignments are disjoint on each physical block, and every assigned index had the same event count in exponent there. A packet's event mass is consequently a common factor times its index mass. Include the packet partitions at every preceding level in the deficiency deletion of Lemma 2.6. For a finite list, choose the deletion cutoff smaller than the total retained proportion, while still

subpower. The charge from each partition is at most its cutoff times the original total event mass. Iterating deficient cell deletion over the finite list leaves an extensive graph and retains each packet's lower mass exponent. Any other fixed finite list of tests is included in this same deletion procedure. When only an upper packing count is needed, one may also keep the whole earlier assigned index set, which remains a valid active set for the original local bound.

Selections obtained by a derivative limit are made before the next cut: for each fixed perturbation and tolerance perform the finite selection, then take a subsequence attaining its limiting shape. Only finitely many such choices occur in the present lemma. At any stage at which a pure outer equality is needed, rather than its full ensemble, one may choose a representative parent subsequence: the inherited upper bounds are uniform at a fixed positive remaining length, and at least one parent attains the aggregate lower estimate in exponent. The construction itself continues with the ensemble. Finally, each $\kappa_i < 1$ and $\chi_i \leq \kappa_i$ leave positive outer length, so no zero-length iteration is used. $\square$

In what follows, a limit over finer meshes always has the following order. Fix finitely many scales, tests, and comparison tolerances; carry out Lemma 5.2; then take the resolution sufficiently large to absorb their exponent errors. Increase the finite list and refine the mesh only along a subsequent slow diagonal. Thus no estimate requires a preassigned resolution error to be smaller than every shrinking scale gap at once.

At a cut of scale s , the packet's aspect exponent is $s - \chi$. The selected bound $\chi \leq F(s)/v$ makes it at least $s - F(s)/v$. The next packing argument shows that the corresponding angular tolerance controls how much the packet normal can change at that step. Along a chain the largest tolerance, hence the smallest aspect exponent, controls the total change. This is why a running minimum enters the inner rescaling.

Lemma 5.3 (Packing and inner equality). *For a nontrivial equality with delay τ , fix $\tau < \kappa < 1$ and set*

$$D_*(\kappa) = \min_{\tau \leq s \leq \kappa} (s - F(s)/v). \quad (5.8)$$

Then $0 \leq D_(\kappa) \leq \tau$. On an extensive refinement, the normals of the assigned initial plates used by one index in a κ -block lie within $N^{-D_*(\kappa)+o(1)}$ of one normal for that index and block. If $\kappa - D_*(\kappa) > 0$, there is an inner equality of effective length exponent $\kappa - D_*(\kappa)$, initial delay*

$$\frac{\tau - D_*(\kappa)}{\kappa - D_*(\kappa)}, \quad \text{and deficit} \quad \frac{F(\kappa)}{\kappa - D_*(\kappa)}. \quad (5.9)$$

The corresponding conclusions hold at endpoint prefixes by limits from within the required positive lengths.

Proof. Packing adjacent packets. Use the finite chains of Lemma 5.2. Write δ for their mesh size. Consider one smaller-block packet of widths (a, b) and the larger-block packets meeting it along assigned indices at dyadic angle $\theta \gg a/b$. Adjacent widths differ by $N^{O(\delta)}$. Each meeting is witnessed by an affine index belonging to both packets, so their centers are tied to that same segment throughout the smaller block. At a fixed time, a strip of length comparable to b tilted by θ has normal spread $O(a + \theta b) = O(\theta b)$ relative to the smaller packet. Extrapolating the shared segment to the larger block costs another factor $N^{O(\delta)}$. Thus all these larger packets lie in a plank of widths $N^{O(\delta)}(\theta b, b)$, aligned with the smaller packet. Their assigned index sets on the larger block are disjoint. Dividing the local upper mass of this enclosing plank by their saturated mass bounds their number by

$$N^{O(\delta)+o(1)} \theta^x (b/a)^x.$$

On the smaller block, the two thin conditions bound transverse motion along their intersection by $O(a/\theta)$: solving the two strip inequalities divides by the sine of their angle. Their intersection is therefore contained in a plank of widths $N^{O(\delta)}(a, a/\theta)$. Figure 1 shows the two width calculations. Apply the local upper bound again and divide by the smaller packet's saturated mass. The fraction of its assigned indices accounted for at this angle is at most

$$N^{O(\delta)+o(1)} [\theta^x (b/a)^x] \frac{a^x (a/\theta)^y}{a^x b^y} = N^{O(\delta)+o(1)} \left(\frac{a}{\theta b} \right)^{y-x}. \quad (5.10)$$

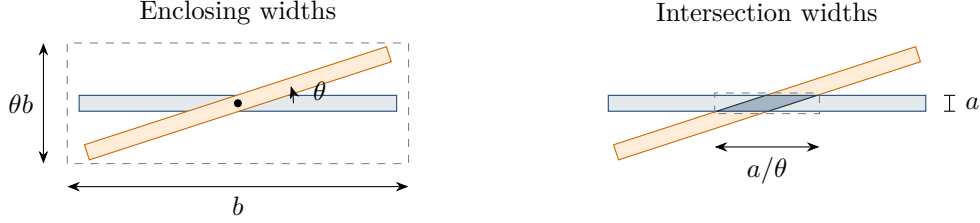


Figure 1: The transverse geometry in the adjacent-packet estimate, schematically at comparable scales. A common assigned affine index ties the packet centers together; the dot in the left panel represents its trace. For $\theta \gg a/b$, the enclosing normal width is of order θb , while the intersection of the two thin strips has length of order a/θ . Endpoint extrapolation and the adjacent scale change contribute the $N^{O(\delta)}$ factors in the proof. All dimensions in the diagram are understood up to constant factors.

The factors omitted in the displayed quotient are changes of time length, profile cost, and width type across one mesh step; their logarithms are $O(\delta)$. If an extrapolated width passes the full spatial size, cover it by the corresponding $N^{O(\delta)}$ bounded planks. The same bound holds for event mass because the active indices in a packet have uniform event counts on its block.

Since $y - x > 0$, angles larger by a fixed positive exponent than a/b can be discarded with a power-small loss, after making δ sufficiently small. Include the earlier packet partitions in the subsequent cleaning. At cut i the aspect exponent is $\kappa_i - \chi_i$, which is at least $\kappa_i - F(\kappa_i)/v$. Summing the angular errors along the finite chain, rather than adding their exponents, bounds the total angle by $N^{-D_*(\kappa)+O(\delta)+o(1)}$. The final packet has one owner per index on its physical block, so all retained initial plates of that index on the block compare to the same final normal. Sending the mesh to zero in the specified order proves the orientation assertion.

The inner equality. The reverse selection in Lemma 5.1 has $\chi \geq F(s)/v$, while every admissible shape has $\chi \leq s$. Therefore $s - F(s)/v \geq 0$. Its value at $s = \tau$ is τ , proving the bounds on D_* . Apply Lemma 4.8 over the N^κ -block with angular scale $\theta = N^{-D_*}$. Its fine assigned plates are the flat-prefix plates of length N^τ , and $D_* \leq \tau$ ensures $\theta \geq N^{-\tau}$. The effective resolution is $N^{\kappa-D_*}$ and the raw lower multiplicity is $mN^{D_*-o(1)}$. By the local upper bound its full-time plank parameter is at most

$$mN^{-h\kappa-\Pi(\kappa)+yD_*}.$$

Only an initial flat log interval of length D_* is dropped. Multiplying this parameter by the critical factor $N^{h(\kappa-D_*)+\Pi(\kappa)}$ gives $mN^{(y-h)D_*} = mN^{D_*}$, exactly the lower exponent. The normalized profile thus gives (5.9). As for outer parents, one can extract a representative equality from the aggregate chart ensemble whenever this statement is used as a fact about equality profiles. $\square$

5.3 A positive minimum delay and the canonical profile

Proposition 5.4 (Canonical equality profile). *There is an exact equality whose profile is*

$$F(\kappa) = \beta(\kappa - \tau)_+, \quad 0 < \tau < 1, \quad \ell = 1 - \tau, \quad \beta = v/\ell \leq 1. \quad (5.11)$$

Its deficit is v , and τ is the least initial delay among all equalities with deficit v .

Proof. Attaining the maximal deficit. First, equalities with deficit v exist. Start with any positive-deficit equality and put $g(s) = s - F(s)/v$. If the running minimum of g from τ remained equal to τ throughout its nontrivial part, Lemma 5.3, applied at a prefix with positive deficit, would produce a nontrivial equality with zero initial delay. This contradicts Lemma 4.6. There is therefore a new minimum $D < \tau$, attained at a prefix κ with $g(\kappa) = D$. At this prefix the inner deficit is

$$\frac{F(\kappa)}{\kappa - D} = v.$$

Let $\mathcal{E}_v$ denote the nonempty class of equality profiles with deficit v , and set

$$\tau_* = \inf_{F \in \mathcal{E}_v} \sup\{r : F = 0 \text{ on } [0, r]\}.$$

A positive lower bound for the delay. We claim that $\tau_* > 0$. Suppose otherwise, and choose $F_j \in \mathcal{E}_v$ with initial delays $\tau_j \rightarrow 0$. Equality compactness, as in Lemma 2.11, gives a subsequence converging uniformly to an equality F_∞ with deficit $v > 0$. The cost is lower semicontinuous; a strict downward jump would give a violation of the critical inequality, so this limiting configuration is still an equality. By Lemma 4.6, F_∞ has a flat interval of positive length.

The small-prefix multilinear argument underlying Lemma 4.5 has slack depending only on the fixed critical parameters. In particular, its allowed prefix length can be chosen independently of the equality profile: the estimate used there is

$$\frac{k^2 m}{n} \geq N^{h+q+\Pi(1)-2\alpha-o(1)} \geq N^{h+(p-2)\alpha-o(1)} \geq N^{h-o(1)}. \quad (5.12)$$

Indeed $n \leq_{\exp} \Delta_h N^d$, $m =_{\exp} \Delta_h N^{h+\Pi(1)}$, and $\Pi(1) \geq p\alpha - q$. Choose fixed $0 < a_0 < b_0$ inside both that allowed prefix range and the flat interval of F_∞ .

For a sufficiently late j ,

$$\tau_j < a_0, \quad F_j(b_0) < \frac{h}{4}(b_0 - a_0).$$

Choose a minimizer κ_j of $F_j(u) - (h/4)u$ on $[0, b_0]$. At b_0 this function is strictly less than $-ha_0/4$, whereas for $u \leq a_0$ it is at least $-ha_0/4$. Hence $\kappa_j > a_0 > \tau_j$, so $\delta_j = F_j(\kappa_j) > 0$. The minimizing property gives

$$F_j(\kappa_j) - F_j(u) \leq \frac{h}{4}(\kappa_j - u), \quad 0 \leq u \leq \kappa_j. \quad (5.13)$$

Fix this particular j , its prefix κ_j , and its positive δ_j before passing to the realizing resolution sequence.

Apply Lemma 4.5 to this fixed equality and prefix, with $s = 1 - h/4$ and $\gamma = 0$. Its uniform small-prefix range includes κ_j , by the choice of b_0 . The record inequality (5.13) is precisely its temporal hypothesis; the other hypothesis is simply $F_j \geq 0$. Its angular exponent is h , so

$$t = 1 + 3h/4, \quad \mu = \min\{1, (s + t)/2, t\} = 1.$$

The lemma gives $\Pi(\kappa_j) \leq F_j(\kappa_j) = \delta_j$. But $\Pi(\kappa_j) \geq (p - 1/2)\delta_j > \delta_j$, since $\delta_j > 0$ and $p > 2$. This is a contradiction. The positive gap is fixed after choosing j and before taking its realizing resolution to infinity; no uniform lower bound for the numbers δ_j is needed. We have proved $\tau_* > 0$.

Rigidity from the inner and outer delays. We now determine a profile attaining this infimum. For any $F \in \mathcal{E}_v$ with initial delay τ , $g(1) = 0$. Given $D' \in (0, \tau)$, let s be the first passage below D' after τ :

$$s = \inf\{r > \tau : g(r) < D'\}.$$

Continuity gives $\tau < s < 1$, $g(s) = D'$, and $\min_{\tau \leq r \leq s} g(r) = D'$. There are values below D' arbitrarily soon to the right of s . The inner equality from Lemma 5.3 has deficit v , so its delay is at least τ_* :

$$\tau - D' \geq \tau_*(s - D'). \quad (5.14)$$

Any saturated outer cut at s has $\chi \leq s < 1$ and deficit $(v - F(s))/(1 - \chi)$. By Lemma 5.1 this is at most v , whence $\chi \leq F(s)/v$. The reverse selection of that lemma therefore attains $\chi = F(s)/v = s - D'$, and its outer deficit is exactly v . Moreover F increases arbitrarily soon to the right of s : $g(r) < g(s)$ implies $F(r) - F(s) > v(r - s)$. Formula (5.6) consequently gives the exact outer delay $D'/(1 - s + D')$, not merely a lower bound for it. Its membership in $\mathcal{E}_v$ yields

$$D' \geq \tau_*(1 - s + D'). \quad (5.15)$$

The two inequalities compare complementary rescalings of the same profile, both with deficit v . For a sequence approaching the least delay, they will force the same first-passage location.

Choose $F_j \in \mathcal{E}_v$ with $\tau_j \rightarrow \tau_*$. The Lipschitz bound gives $v \leq 1 - \tau_j$, so $\tau_* \leq 1 - v < 1$. For each fixed $D' \in (0, \tau_*)$, the two passage inequalities give

$$1 + D' - D'/\tau_* \leq s_j \leq D' + (\tau_j - D')/\tau_*.$$

Thus

$$s_j \longrightarrow 1 - \frac{1 - \tau_*}{\tau_*} D', \quad F_j(s_j) = v(s_j - D').$$

Pass to a compact equality limit F . The rational values of D' in $(0, \tau_*)$ suffice; continuity then determines the profile at every point above τ_* . Below τ_* all these profiles vanish. We obtain

$$F(\kappa) = \frac{v}{1 - \tau_*} (\kappa - \tau_*)_+.$$

Its delay is exactly τ_* and its slope is at most one by the Lipschitz bound. This proves (5.11). $\square$

Figure 2 shows the canonical profile and the aspect exponent that controls its angular scale.

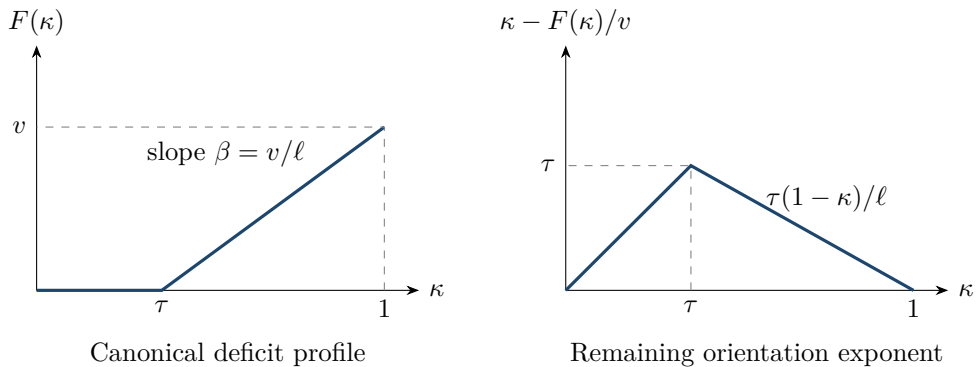


Figure 2: The canonical profile in Proposition 5.4. After the flat interval, its slope is constant. For $\kappa \geq \tau$, the associated quantity $\kappa - F(\kappa)/v$ decreases linearly from τ to zero; at the stationary prefix $\kappa = 1 - \ell c$ it equals τc . This identity gives the angular scale $\Theta_c = N^{-\tau c}$. The vertical scales in the two panels are independent.

5.4 Tracking a stationary shape

Fix the canonical equality from Proposition 5.4. For the remainder of the proof, ℓ always denotes $1 - \tau$; the logarithmic loss of an outer cut remains χ .

Lemma 5.5 (Canonical shape tracking). *For the canonical profile there are compatible finite chains, and slow mesh limits of them, with cumulative shape parameters*

$$\kappa = \tau + \ell b', \quad \chi = b', \quad \mathbf{a} = w b', \quad w = h_z + \ell \mathcal{D}_q(q, \beta) \in (0, 1], \quad 0 \leq b' < 1. \quad (5.16)$$

The equalities in the parameters allow errors tending to zero with the mesh. Packet mass, ownership, and orientation conclusions of Lemmas 5.2 and 5.3 remain valid.

Proof. An outer cut of a deficit- v equality again has deficit at most v . Applied to the canonical profile, this forces $\chi \leq b'$. The reverse derivative selection attains $\chi = b'$. Formula (5.6) then shows that the outer profile is the same canonical profile: its flat delay is

$$\frac{\tau + \ell b' - b'}{1 - b'} = \tau,$$

and its nonflat slope is still β . The same statement applies incrementally in every current outer model, in the original logarithmic units.

For one mesh increment $\Delta b'$, the two coefficients in (5.2), written in those units, are

$$v(\Delta b' - \Delta \chi), \quad h_z \Delta \chi + \ell \mathcal{D}_q(q, \beta) \Delta b' - \Delta \mathbf{a}.$$

Every available saturation has $\Delta \chi \leq \Delta b'$ by the outer deficit bound. Take either sign of $\dot{z}$ and take $\dot{p} < 0$ with an arbitrarily large ratio $|\dot{p}|/|\dot{z}|$. First fix that ratio and perform derivative capture. Along a subsequence as the ratio increases, the nonnegative quantity $\Delta b' - \Delta \chi$ must tend to zero; otherwise its negative $\dot{p}$ contribution cannot be offset by the bounded second coefficient. The two signs of $\dot{z}$ therefore provide choices with $\Delta \chi = \Delta b'$ and, respectively,

$$\Delta \mathbf{a} \leq w \Delta b', \quad \text{or} \quad \Delta \mathbf{a} \geq w \Delta b'.$$

Since $0 \leq \Delta \mathbf{a} \leq \Delta \chi$, this implies $0 \leq w \leq 1$. Its defining expression and $h_z > 0$ give $w > 0$.

It is unnecessary to realize an exact intermediate shape on a single step. If the running value of $\mathbf{a}$ is above $w b'$, choose the next increment from the lower side; if below, choose from the upper side. The error can then overshoot zero by at most the mesh size, because both $\Delta \mathbf{a}$ and $w \Delta b'$ lie in $[0, \Delta b']$. This tracks the desired total shape within one mesh increment. The current outer profile stays canonical after each choice, so both sides are available at every step. All perturbation limits are resolved on a fixed finite chain before taking its mesh diagonal, and Lemma 5.2 preserves the required earlier assignments. $\square$

We now express the tracked packets in physical scales. Reverse the cut parameter by $c = 1 - b'$ and put

$$\rho = 1 - w, \quad B_* = h + \ell P_{p,q}(\beta), \quad R_c = N^{-\ell c}, \quad \Theta_c = N^{-\tau c}.$$

The corresponding logarithmic scale and cumulative shape are

$$\kappa = 1 - \ell c, \quad \chi = 1 - c, \quad \mathbf{a} = w(1 - c), \quad \mathbf{u} = \rho(1 - c).$$

For $0 \leq c \leq 1$, use nested dyadic time partitions whose blocks I_c have physical length R_c , up to dyadic rounding. For $0 < c \leq 1$, write B_c for its assigned packet. Its thin and thick spatial widths in original cell units are

$$a = N^{w(1-c)}, \quad b = N^{w+(\tau-w)c}. \quad (5.17)$$

At $c = 1$ these are the initial flat-scale plates. As c decreases, the time blocks grow and the packets record the successive outer cuts.

For the probability statements, replace the finer events of each occupied index- I_1 pair by one atom. More precisely, let $\mathcal{J}_i$ be the occupied I_1 blocks on index i and set

$$\Omega_* = \{(i, J) : J \in \mathcal{J}_i\}, \quad Z_* = \sum_i \omega_i \# \mathcal{J}_i, \quad \mathbb{P}\{(i, J)\} = \frac{\omega_i}{Z_*}.$$

The atom (i, J) is placed at the center t_J of J and carries the spatial normal of the initial plate assigned to that index on J . We restrict these normals to a fixed direction chart. After a fixed orthonormal spatial change of coordinates, write their covectors as $(1, \vartheta)$ with ϑ bounded. In the displayed law, $\mathcal{J}_i$ and Z_* refer to the retained pairs. Denote the atom's time and normal coordinates by $e = (t, \vartheta)$, while retaining i as a separate random variable. Nested dyadic quantization of t and ϑ at widths R_c and Θ_c gives labels e_c . The original uncollapsed equality supplies m , Δ_h , and all active-index plank bounds; this collapsed law supplies event probabilities.

In an event frame define the four matrix forms

$$X = (1, \vartheta) \cdot M(t), \quad Y = M(t)_2, \quad U = (1, \vartheta) \cdot u, \quad V = u_2. \quad (5.18)$$

Thus X, U are normal position and velocity, and Y, V are the fixed complementary coordinates. This chart is generally oblique. Dividing (5.17) by N gives the spatial widths; dividing again by the time length R_c gives the velocity widths. We write these four scales as

$$\begin{aligned} L_X(c) &= N^{-\rho-wc}, & L_Y(c) &= L_X(c)\Theta_c^{-1}, \\ L_U(c) &= L_X(c)R_c^{-1}, & L_V(c) &= L_X(c)(R_c\Theta_c)^{-1}. \end{aligned} \quad (5.19)$$

The proposition verifies that these widths remain valid in every used event frame, and records the masses and branching of the same packets.

Proposition 5.6 (Stationary data). *There is a realizing sequence of canonical equalities and compatible packet assignments for which the preceding collapsed law, event labels, and matrix forms have the following properties. All local upper bounds and original index weights are retained.*

- (i) *Along an index, occupied time blocks have regular branching exponent $s_0 = 1 - \beta \geq 0$. More precisely, for fixed $c < b \leq 1$, an occupied I_c contains*

$$N^{\ell(1-\beta)(b-c)+o(1)} \quad (5.20)$$

occupied I_b blocks. The event normals on that index in I_c lie in a range of size $\Theta_c N^{o(1)}$.

- (ii) *For $0 < c \leq 1$ there are assigned packets B_c on the occupied I_c blocks, with raw active index mass at least*

$$n_c = \Delta_h N^{(B_*+\tau)c+dw(1-c)} \quad (5.21)$$

in exponent. On each fixed time block the assigned index sets are disjoint, and each retained index has one packet owner there. The used normals of a packet differ from its spatial normal by at most $\Theta_c N^{o(1)}$.

- (iii) *In every used event frame, the widths of B_c in the four forms (5.18) are bounded by $L_X(c), L_Y(c), L_U(c), L_V(c)$ up to subpower expansion. Equivalently, their negative log exponents are*

$$\rho + wc, \quad \rho + (w - \tau)c, \quad \rho + (w - \ell)c, \quad \rho + (w - 1)c, \quad (5.22)$$

- (iv) *The data and packet lower bounds coexist on any prescribed finite list of scales and tests, and hence on countable lists by slow diagonals. They persist, in exponent, on typical nondeficient parts after extensive regular refinements. All packet memberships are memberships of original active indices on their physical intervals. In a chart registered at I_c , the packet at a finer I_b is determined by the index, chart, and I_b ; a matrix cell in that chart is determined by the index and chart.*

Proof. Choose the finite chains from Lemma 5.5 and use the scales defined above. On the canonical profile the running aspect exponent is decreasing above τ , and at $\kappa = 1 - \ell c$ its value is

$$\kappa - F(\kappa)/v = \tau c.$$

The proof of Lemma 5.3, which applies unchanged to the tracked finite chains, shows that every initial plate used by this packet has its normal within $\Theta_c N^{o(1)}$ of the packet normal. Since a packet is assigned only once to each index on its block, this also gives the range assertion for all event normals on that index and block. The canonical profile gives the number of occupied subblocks: subtracting its deficits between the prefixes $1 - \ell c$ and $1 - \ell b$ yields (5.20).

We now verify the width bounds in the event frames. The widths in (5.19) are first computed in the packet's own normal frame. Changing to one of its used event normals adds at most $\Theta_c N^{o(1)}$ times the transverse coordinate; the ratios of transverse to normal widths are exactly Θ_c^{-1} . Changing the time center within I_c adds at most R_c times the velocity coordinate. Both changes preserve the four widths up to subpower factors.

For this profile,

$$\Pi(\kappa) = \ell(1 - c)P_{p,q}(\beta), \quad m =_{\text{exp}} \Delta_h N^{B*}.$$

Substituting (5.17) in (5.1), the exponent of the packet mass relative to Δ_h is

$$\begin{aligned} B_* - \ell(1 - c)P_{p,q}(\beta) - h(1 - \ell c) + xw(1 - c) + y(w + (\tau - w)c) \\ = dw(1 - c) + c(\ell P_{p,q}(\beta) + h\ell + y\tau) \\ = dw(1 - c) + (B_* + \tau)c, \end{aligned}$$

using $\ell + \tau = 1$ and $y = 1 + h$. This proves (5.21).

We next verify the passage to the collapsed law. Before collapse, the number of finer events in every occupied index- I_1 pair has a common exponent. Thus an extensive selection of atoms of weight ω_i pulls back to an extensive selection of the uncollapsed graph. Restricting the normal to one of finitely many direction charts also retains an extensive piece, and normalized covectors and unit normals are comparable on that chart. The orientation and time ranges proved above imply that one index and I_c use only subpower many e_c bins. Whenever the chart restriction or these labels split a packet, retain nondeficient occupied pieces. The packet lower bounds therefore survive for the law specified before the proposition.

For completeness, scale interpolation does not require new unrelated assignments. At a finite mesh, use the packets already constructed. A nearby intermediate interval is assigned the packet of its nearest coarser mesh interval, restricted to the indices active in the smaller interval. Extrapolation, width changes, and subdivision have $N^{O(\text{mesh})}$ factors. Regular time counts and deficiency deletion preserve the packet lower masses on typical occupied pieces. Along the slow diagonal these factors are subpower. Alternatively, include any specified intermediate scales in the next finite mesh. Thus a new finite or countable list of scale tests can be accommodated by further refinement, with the same limiting data.

Finally include, in the finite cleaning list, the assignment partitions from every level in use. Lemma 5.2 then gives all the simultaneous memberships, the original active-index interpretation,

and stability under extensive regular refinement in (iv). On a fixed physical I_b , ownership was a decision for an index on that block; after registration in a chart it is therefore constant on the tuple specified in (iv). A cell of that chart's fixed matrix partition is likewise a function only of the index and chart. The construction needs outer steps only for $b' < 1$, or $c > 0$; it makes no cut at zero remaining length. The endpoint $c = 1$ is the initial assignment and has already been included. $\square$

We have proved $0 < \tau < 1$, $0 < \ell < 1$, $v > 0$, and $\beta = v/\ell > v$. The value $w = 1$ is allowed, so that ρ may vanish. At this stage $\beta = 1$ has not been excluded; the angular estimate in the next section will show that $s_0 = 1 - \beta$ is positive.

6 Two estimates for the stationary configuration

We use the underlying equality and the collapsed event law of Proposition 5.6. They have different purposes: plank bounds always count inherited weights of original active indices, whereas probabilities below are probabilities of indexed events. In particular, restricting an event label does not restrict the reference index mass in a plank test.

We prove two estimates. Lemma 6.2 bounds the concentration of normal labels along an index and gives $\beta < 1$. Lemma 6.3 bounds the additional normal-position entropy inside a matrix cell after conditioning on a finer event label. The first estimate follows by turning angular concentration into a configuration with excessive temporal deficit. For the second, we compare the mass of a queried packet with the original active-index mass in its matrix cell, and then convert that comparison to entropy.

Throughout this section put

$$P = P_{p,q}, \quad x = d - 1 - h, \quad y = 1 + h, \quad x + y = d, \quad y - h = 1.$$

The parameters $0 < \tau, \ell < 1$, $\tau + \ell = 1$, $v > 0$, $\beta = v/\ell \leq 1$, and $0 < w \leq 1$ are fixed. Thus $\beta > v$, including before we exclude $\beta = 1$.

Recall the four matrix widths $L_X(c), L_Y(c), L_U(c), L_V(c)$ from (5.19). For a coarse event label $e_c = (t_c, \vartheta_c)$, evaluate the four forms at its chosen center. For a real offset a , let $Q(c, a)$ be their joint matrix-cell label at those widths multiplied by N^{-a} . The grids are dyadic, with the usual bounded enlargements at rounded scales. Conditional on e_c , this is a partition of the matrix indices; repetitions at the same matrix retain their individual weights. All limits first fix the scale parameters and any strict exponent margin, and then let $N \rightarrow \infty$. Finite lists of such tests precede the countable diagonals.

6.1 Mass accounting for the selections

Both proofs select events and then fill the occupied time blocks they meet. We must retain the packet mass and control the change in the time profile under the same selection. The following form of finite cleaning does this even when the selection has a fixed power loss. Its essential distinction is between original packet masses, against which that loss is measured, and the restricted graph on which new probability labels are computed.

Lemma 6.1 (Cleaning registered events). *Let a finite weighted event graph have total weight W , and let a restriction retain weight at least $N^{-E+o(1)}W$. Fix finitely many partitions of the graph. The cells may include packet assignments, index-time blocks, and joint labels. After a further restriction of subpower cost, every surviving cell of each partition retains at least $N^{-E-o(1)}$ of its*

original weight. The same assertion holds after splitting each original cell among subpower many registered charts, with that number absorbed in the error.

Suppose, in addition, that the original time trees have a common profile F on a log interval $[0, \Lambda]$ in $\log N$ units, and each original index and starting interval is routed through only subpower many charts. If the restricted trees are made uniform, with profile G , then

$$G(t) - G(s) \geq F(t) - F(s), \quad 0 \leq G(\Lambda) - F(\Lambda) \leq E, \quad 0 \leq s < t \leq \Lambda. \quad (6.1)$$

In particular an extensive restriction preserves the profile, and

$$0 \leq \int_0^\Lambda P(G') - \int_0^\Lambda P(F') \leq pE. \quad (6.2)$$

These assertions concern inherited index weights.

Proof. We distinguish two reference graphs in the cleaning. For the new homogenization tests use the finite label construction of Lemma 2.6: these are its within-index integer counts and conditional-probability tests with polynomially many children. Perform its negligible conditional-probability tail deletions before computing the labels. Arbitrary-cardinality original packet partitions are retained only as cleaning tests; no absolute weight label is imposed on their cells. Let W be the original event weight and let R be the restricted, routed graph after those tail deletions and before the finite labels are selected. Write $M_0 = |R|$; thus $M_0 \geq N^{-E-o(1)}W$. Compute the prescribed rounded labels on R , including their joint and domain tests for conditional counts, and retain one common finite list of labels. All graph masses denoted by $|\cdot|$ here and in the registration rule below are weighted event masses. Denote this label class by A , and write $M = |A|$. The number of label lists is subpower, so $M \geq N^{-o(1)}M_0 \geq N^{-E-o(1)}W$. Counts within index fibers are event counts, with the inherited index weight cancelling in their ratios. No bound on the number of distinct weighted indices is required.

There are two collections of reference masses. Original packet, index-time, and chart tests use their masses in the original graph; the sum of these references is at most $L_{\text{old}}W$. New regularization tests use their masses in R , the graph on which their labels were computed; their reference sum is at most $L_{\text{new}}M_0$. Here $L_{\text{old}}, L_{\text{new}} = N^{o(1)}$ include the finite number of partitions and any subpower chart copies. In particular, one does not substitute the possibly much smaller post-label cell mass for an R -reference without recomputing its label.

Choose $\epsilon_N \downarrow 0$ slowly enough to dominate the label loss $\log_N(M_0/M)$, the error in $M \geq N^{-E-o(1)}W$, and the $\log_N L_{\text{old}}$ and $\log_N L_{\text{new}}$, with margins tending to infinity after multiplication by $\log N$. Starting from A , delete all events in a deficient original-reference cell when its current mass is less than $N^{-E-\epsilon_N}$ times that reference. For a new regularization cell or domain, use instead $N^{-\epsilon_N}$ times its R -reference. Continue either type of deletion until no deficient cell remains. A cell is charged only once, because it is empty after deletion. Even with cascades, the two total charges are at most

$$L_{\text{old}}N^{-E-\epsilon_N}W = o(M), \quad L_{\text{new}}N^{-\epsilon_N}M_0 = o(M).$$

The final graph therefore has mass $(1 - o(1))M$. Every surviving old cell retains at least $N^{-E-o(1)}$ of its original mass. Every surviving new cell retains $N^{-o(1)}$ of its R -mass and is a subset of that cell, so the rounded exponents selected from R remain valid, including conditional exponents obtained by ratios of joint and domain masses. This proves the mass assertion: the new labels incur only a subpower loss in addition to the original loss E . The finite-test diagonal allows further prescribed tests.

For the profile assertion fix a finite mesh of time scales and uniformize the branching counts on its gaps. A subtree cannot have more children on any gap than the original tree. Since the

branching exponent on $[s, t]$ is $(t - s) - (F(t) - F(s))$, this proves the first inequality in (6.1). The total available index weight after routing is at most $N^{o(1)}$ times its original value. Comparing this with the retained weighted event count shows that the number of events per surviving uniform index is at least $N^{-E-o(1)}$ times the original number. This proves the endpoint inequality. Refine the fixed mesh only after these comparisons hold. The Lipschitz profile limits give (6.1) on every interval, hence $G' \geq F'$ almost everywhere and $\int(G' - F') \leq E$. The function P is increasing and has Lipschitz constant at most p , which proves (6.2). $\square$

Here is the registration rule used below. Before any restriction, an occupied time block J on an assigned index has a common number $m_J = N^{\kappa+o(1)}$ of finer events, uniformly in the blocks under consideration. Given a selected set of finer events, first register every occupied token (i, J) that it hits, giving that token weight ω_i , and complete its block. Any preassigned chart or other label on which the predicate depends remains part of its token; completion never changes that label. Subpower many such tags enlarge the original reference totals by only a subpower factor. If T_0 is the original token weight and T the weight of the registered tokens, then

$$|\text{selected finer events}| \leq N^{\kappa+o(1)}T, \quad |\text{original finer events}| = N^{\kappa+o(1)}T_0.$$

Consequently a restriction of finer-event weight $N^{-E-o(1)}$ registers at least $N^{-E-o(1)}T_0$ token weight. Completing these tokens precedes chart splitting and incurs no further loss. Route whole tokens through their geometric charts; each original index and starting block has only subpower many such charts. If it is necessary to prune chart portions, compare them with their masses in this restricted, completed token graph, rather than with the original unrestricted block's finer-event mass. Portions below $N^{-\epsilon_N}$ of their restricted unsplit reference have total weight $o(T)$ after choosing ϵ_N slowly. A predicate constant on (i, J) is unchanged by completion or later filling. The original packet references are still the unrestricted assigned token masses, and keep the $N^{-E-o(1)}$ threshold in the preceding two-budget cleaning. Thus this registration retains both the original packet lower bounds and the single event-retention loss E .

6.2 Angular nonconcentration

Lemma 6.2 (Angular Frostman bound). *There is $\gamma > 0$, depending only on the fixed stationary parameters, with the following property. For every fixed $0 < c < b \leq 1$, outside a set of index- I_c fibers of event probability tending to zero, every angular interval J of length Θ_b satisfies*

$$\frac{\#\{e \text{ on } i : t(e) \in I_c, \vartheta(e) \in J\}}{\#\{e \text{ on } i : t(e) \in I_c\}} \leq N^{-\gamma\tau(b-c)+o(1)}. \quad (6.3)$$

It is enough to test a fixed grid with bounded enlargements. The bound also holds conditional on typical nondeficient e_c bins in the fiber, and after extensive regular refinements. In particular

$$0 < \beta < 1, \quad s_0 = 1 - \beta > 0. \quad (6.4)$$

Proof. Angular concentration would produce a configuration whose normalized temporal deficit is at least β . Since $\beta > v = -h_p$, decreasing p makes the critical upper bound for that configuration strictly smaller than its packet lower bound. We quantify how close the two bounds remain after selecting a narrow angular bin: a selection loss E decreases the packet mass by at most E in exponent and increases its temporal cost by at most pE .

Write $g = b - c > 0$ and

$$S = n_b N^{-\tau b}, \quad K' = N^{b-\ell c}, \quad \theta = \Theta_b = N^{-\tau b}.$$

First pass to the isotropic outer parents associated to B_b , as in Lemma 5.2. Their effective resolution count is N^b . The assigned B_b plates have time and long spatial length $N^{\tau b}$ in these cell units, unit thin width, and mass at least $SN^{\tau b - o(1)}$. On the longer interval I_c , which has K' bins in this model, the local bound of Proposition 4.1 pulls back to

$$n(\text{plank of widths } a_1, b_1) \leq N^{o(1)} SN^{-\ell g P(\beta)} (K')^{-h} a_1^x b_1^y. \quad (6.5)$$

For example, the isotropic physical cell width is $N^{w(1-b)}$ in original cell units. Substitution in the original local bound gives the coefficient

$$m N^{-\ell(1-c)P(\beta)} N^{-h(1-\ell c)} N^{dw(1-b)} = SN^{-\ell g P(\beta)} (K')^{-h}$$

in exponent, using $m =_{\text{exp}} \Delta_h N^{B^*}$ and (5.21). Thus (6.5) counts original active indices; it is not a bound only for indices occupying a selected angular bin.

Suppose that fibers violating (6.3) account for an extensive event piece. On each such index and I_c choose one violating bin and keep its events. This costs at most

$$E = \gamma \tau g$$

in log probability, apart from a vanishing error. First register and complete every occupied I_b token hit by this selection, before any chart-portion cutoff. The preceding registration rule gives token weight at least $N^{-E - o(1)}$ of the original token graph. The normal of its assigned B_b packet is within $\Theta_b N^{o(1)}$ of the selected bin, since the packet contains the event that registered the token and its normal label is constant on the token. Completing the block does not change its packet owner. Route these whole tokens through the outer parents and bin choices. Each packet has only subpower many relevant choices, and each index on I_c has only subpower many charts. Any deficient chart portions are pruned relative to this restricted token graph. Now apply the two-budget cleaning of Lemma 6.1: original packet and time-tree references use the loss E , while the new homogenization references, computed after registration and routing, use only a subpower loss. The final token graph still has weight $N^{-E - o(1)}$ of the original one, and each surviving packet has raw assigned mass at least

$$SN^{\tau b - E - o(1)}. \quad (6.6)$$

The uniform token trees are subtrees of the original occupied I_b trees below I_c , with subpower many routed copies. Thus their endpoint deficit loss is at most E and their cost increase is at most pE . Packet loss E and cost increase pE are the two consequences of this same retained graph; the selection is not performed a second time.

Apply the geometric construction in the proof of Lemma 4.8 on I_c , with angular width θ . Here we use only its bounded-chart routing, rectangle pullback, and cell-count calculation. That calculation is linear in the packet mass, which here is the reduced value $SN^{\tau b - E - o(1)}$ from (6.6). We do not use that lemma's profile-preservation conclusion for an extensive graph: the endpoint loss E and cost increase pE were proved by Lemma 6.1 and are used explicitly in (6.9) below. The resulting normalization has resolution

$$N_{\text{new}} = K' \theta = N^{\ell g}. \quad (6.7)$$

The selected plates become bounded-cell packets: their old time and long spatial length multiplied by θ is one, and their orientation error is $N^{o(1)} \theta$. Filling their occupied I_b blocks before this normalization therefore gives aggregate raw multiplicity at least the quantity in (6.6). The full-plank parameter of the new models is at most

$$\Delta_{\text{new},h} \leq N^{o(1)} SN^{-\ell g P(\beta)} (K')^{-h} \theta^{-y}. \quad (6.8)$$

Here the factor θ^{-y} follows by pulling back a rectangle: its area expands by θ^{-1} and its largest width by at most θ^{-1} , and $a^x b^y = (ab)^x b^{y-x}$ with $0 \leq x \leq y$.

We next quantify the time cost. Prior to selection the occupied I_b blocks below I_c have constant deficit slope β over a log interval of length ℓg . Filling and then normalizing removes only the flat range below them. The restricted uniform time tree cannot gain branching on any fixed gap. Its normalized deficit α_{new} is therefore at least β . By (6.2), its cost, expressed in $\log N$ units, is at most

$$\ell g P(\beta) + pE + o(1). \quad (6.9)$$

The comparison uses the whole routed ensemble: the number of chart copies per original index and I_c is subpower, so the total branching loss is at most E , as in the proof of Lemma 6.1.

Without the controlled losses, the sharp upper bound from Definition 2.3 and (6.8) matches $SN^{\tau b}$ exactly. Indeed its exponent relative to S is

$$-\ell g P(\beta) - h(b - \ell c) + y\tau b + h\ell g + \ell g P(\beta) = \tau b.$$

Thus (6.6) and (6.9) place the new models within $(p+1)E + o(1)$ of their sharp upper bound in $\log N$ units.

Choose a small fixed $t_* > 0$ with $p - t_* > 2$ and

$$0 \leq h(p - t_*, z) - h(p, z) \leq \frac{v + \beta}{2} t_*. \quad (6.10)$$

This is possible by differentiability and $\beta > v$. Increasing h at fixed d cannot increase the plank parameter, because its value on a plank is multiplied by $(a_1/b_1)^{h'-h} \leq 1$. Also $P_{p-t_*,q}(e) = P_{p,q}(e) - t_*e$. Applying the critical bound at $(p - t_*, z)$ to these same models therefore saves at least

$$N_{\text{new}}^{(\beta-v)t_*/2}$$

relative to the old sharp upper estimate. Choose once and for all

$$0 < \gamma < \min \left\{ \frac{1}{2}, \frac{\ell(\beta - v)t_*}{8(p+1)\tau} \right\}. \quad (6.11)$$

The saving then strictly exceeds the error $(p+1)E$. This contradicts the aggregate lower multiplicity. Uniformity over bounded parent charts follows from the uniform all-configuration critical bound; otherwise a parent subsequence would violate that bound.

It follows that the violating fibers are not an extensive piece. In particular their probability tends to zero. This proves (6.3) for each fixed gap. The choice (6.11) is independent of $b - c$: both the selection loss and the strict saving were proportional to that gap before normalization. Fix finitely many gaps first and absorb their errors; the diagonal convention then gives all required scale tests.

Within an index and I_c the normals occupy only subpower many e_c bins. Delete bins retaining less than a subpower fraction of the fiber, by the same deficiency argument. Dividing by their remaining mass changes (6.3) by only a subpower factor. An extensive subsequent regular refinement has the same property by Lemma 6.1.

Finally suppose $\beta = 1$. For any fixed $0 < c < 1$, an occupied index- I_c fiber has only $N^{o(1)}$ occupied I_1 blocks by (5.20), and there is one collapsed event per such block. One angular bin of width Θ_1 therefore has at least $N^{-o(1)}$ of its event mass. This contradicts (6.3) at $b = 1$, since $\gamma\tau(1 - c) > 0$. Hence $\beta < 1$; positivity follows from $v > 0$. $\square$

6.3 Conditional concentration in the normal position

We next compare a queried packet with the original active-index mass in its matrix cell. This reference mass includes indices whose shadings do not carry the query's finer event label. A lower bound for such a reference mass can nevertheless bound the query's entropy: entropy is at most cross-entropy against any positive reference probability. The proof first establishes the geometric mass bound and then applies this inequality to the normal-position observation.

For a finite random variable write $H_N(Z | L) = H(Z | L) / \log N$, using natural logarithms. If $L = e_{c+\sigma}$, let X_L be the normal-position form evaluated at the center of that finer label. Let

$$[X_L]_{c,a+\Delta}$$

denote its grid bin at width $L_X(c)N^{-a-\Delta}$. Its grid may be translated by a quantity determined by $Q(c, a)$ and L ; changing between such grids changes the conditional entropy by $O(1)$.

Lemma 6.3 (Conditional X estimate). *Fix $0 < c < 1$, $0 < a < w(1 - c)$, and $\sigma > 0$ with $c + \sigma \leq 1$. For all sufficiently small fixed $\Delta > 0$,*

$$\limsup_{N \rightarrow \infty} H_N([X_{e_{c+\sigma}}]_{c,a+\Delta} | Q(c, a), e_{c+\sigma}) \leq x\Delta. \quad (6.12)$$

For example it suffices to take

$$0 < \Delta < \frac{1}{4} \min \left\{ a, w(1 - c) - a, \frac{\tau a}{w}, \min(\tau, \ell)\sigma \right\}. \quad (6.13)$$

The assertion holds for the event law on any extensive regular refinement, with the original active-index masses still available as reference masses. The limit in N precedes limits in Δ or σ .

Proof. The geometric argument uses coarse matrix cells within registered charts. We will replace each cell by its center, weighted by the full original active-index mass in that cell, and regard the selected original indices as its children. After selecting a common ratio of child mass to representative mass, their normalized event laws are comparable. The critical inequality bounds the representatives' multiplicity, whereas packet saturation gives a lower bound for the children's mass in a typical spatial cell. These bounds force a typical packet to contain a sufficiently large fraction of its queried cell's reference mass. We first construct the charts in which this comparison is made.

Put

$$g = b - c = \frac{a + \Delta}{w}, \quad D_0 = N^g, \quad D = N^\Delta, \quad Z' = N^{\tau g}, \quad S' = \frac{n_b}{Z'}. \quad (6.14)$$

The conditions (6.13) imply $c < b < 1$ and $\Delta < \tau g$, so a positive flat range remains in the time filling used below.

Chart geometry. Fix e_c and its interval I_c . In its frozen frame normalize time by R_c , spatial normal and transverse position by $R_c\Theta_c, R_c$, and velocity by the corresponding spatial units divided by R_c . In these coordinates a $Q(c, a)$ cell has isotropic matrix diameter, up to dimension constants,

$$N^{-\rho(1-c)-a}.$$

Take isotropic matrix parents of diameter $N^{-\rho(1-b)}$, subtract their centers, and divide the spatial and velocity coordinates by that diameter. The resulting charts are bounded and have resolution D_0^{-1} . In each such parent, the diameter of a cell of $Q(c, a)$ is

$$N^{-\rho(b-c)-a} = N^{-g+\Delta} = D/D_0. \quad (6.15)$$

We used $wg = a + \Delta$ in the equality. One may choose compatible nested grids; if a prescribed cell meets a parent boundary, splitting it among boundedly many parents adds only $O(1)$ entropy at the end. Hereafter C denotes the resulting cell within its registered chart, including this subdivision when needed.

A B_b packet fits in subpower many of these charts. Its used normals lie in $\Theta_b N^{o(1)}$, hence in subpower many e_c bins, and its four width bounds extrapolated from I_b to I_c have matrix diameter at most $N^{-\rho(1-b)+o(1)}$ in the I_c normalization. Use the packet's own normal and its own time center for the following four widths after parent normalization. Their negative log exponents are

$$(g, \ell g, \tau g, 0). \quad (6.16)$$

In the normalized I_c chart the change of normal is $(\vartheta_P - \vartheta_c)/\Theta_c = O(1)$, and the change of time center is $(t_P - t_c)/R_c = O(1)$. These are bounded shears. They need not preserve the four coordinate widths in the frozen e_c frame; they preserve the following geometric bounds, up to fixed factors. At resolution D_0 the packet is a plate of thin spatial width one cell, long spatial width Z' , and time length Z' bins, with raw assigned mass at least $S'Z'N^{-o(1)}$. Each index and I_c is in only subpower many registered charts. After chart splitting, Lemma 6.1 preserves these packet masses on typical occupied pieces.

The reference-mass claim. For each cell C in a registered chart define

$$n_C = \sum_{\substack{i \text{ active on the original } I_c \\ M_i \in C}} \omega_i. \quad (6.17)$$

For a packet $P = B_b$ on an I_b block, write $n(C \cap P)$ for the inherited mass of its assigned active indices that lie in C . We will prove that, outside a set of query events of probability tending to zero,

$$n(C \cap P) \geq n_C D^{-x-o(1)}. \quad (6.18)$$

The claim also holds if the query events were first restricted to an extensive regular piece, with the same full original active mass in (6.17). This reference mass belongs to the chart cell C just defined.

The critical comparison uses the following inherited bound. Let Δ_0 denote the full-time plank parameter in one of these bounded normalized charts, with exponents x, y . Its physical normal cell width in original cell units is $N^{w(1-b)}$; the transverse cell width is $\Theta_c^{-1} N^{w(1-b)}$. Pulling back an arbitrarily oriented rectangle multiplies its area by $N^{2w(1-b)} \Theta_c^{-1}$ and its largest width by at most $N^{w(1-b)} \Theta_c^{-1}$. Since $0 \leq x \leq y$, the original local estimate gives

$$\begin{aligned} \Delta_0 &\leq N^{o(1)} m N^{-\ell(1-c)P(\beta)} N^{-h(1-\ell c)} N^{dw(1-b)} \Theta_c^{-y} \\ &=_{\text{exp}} S' D_0^{-B_*}. \end{aligned} \quad (6.19)$$

For the last equality, its exponent relative to Δ_h is $cB_* + \tau c + dw(1-b)$; the same exponent results from $S' D_0^{-B_*}$ using (5.21). All these bounds count original indices active on I_c , whether or not those indices carry the particular normal label used to name the chart. An upper bound for a subcollection is therefore inherited without changing the reference weights.

Registering the failed queries. Suppose the reference-mass claim fails. After a subsequence, for some fixed $\epsilon > 0$ the events satisfying

$$n(C \cap P) < n_C D^{-x} N^{-\epsilon} \quad (6.20)$$

form an extensive piece. Register tuples $(i, \mathfrak{c}, J)$, where $\mathfrak{c}$ specifies the chart, including e_c, I_c and the parent, and $J = I_b$ is occupied. Give each tuple weight ω_i . The cell C is fixed by $(i, \mathfrak{c})$, and its packet owner is fixed by $(i, \mathfrak{c}, J)$, by Proposition 5.6(iv). Consequently (6.20) is constant on the tuple. This constancy is the reason that replacing its finer events by a token and later filling J does not change the failed test.

Before selection each occupied J has a common number of original events in exponent. Register every tuple $(i, \mathfrak{c}, J)$ hit by a bad query and complete its J block in that fixed chart, before chart-portion pruning. The chart component is retained throughout: the failed predicate depends on it. Because the predicate is constant on this tuple, completion preserves it. The bad tuples consequently have extensive inherited weight in the original token graph. Any chart cutoff is taken relative to the restricted completed token graph; there are only subpower many chart portions per index and I_c . Clean the packet partitions, using the original token masses for packet references and the graph where labels were computed for new regularization references. The registered assigned mass of each surviving packet in a chart is then at least $S'Z'N^{-o(1)}$. The assigned index sets are disjoint for a fixed chart and J .

Frequency selection. We next make the selected child mass comparable to the reference mass by a common factor. This will allow a multiplicity estimate sampled under the representative law to be used for the child events. On the retained tuple graph define

$$w(C, J) = \sum_{\substack{(i, \mathfrak{c}, J) \text{ retained} \\ M_i \in C}} \omega_i.$$

This is at most n_C . Select one common exponent of $w(C, J)/n_C$ across the ensemble. Here is the finite justification even when the original weights are arbitrarily unequal. There are polynomially many charts, cells, and time blocks, and each $n_C \leq n$. If their total number is at most N^A , ratios below N^{-A-3} contribute at most $N^{-3}n$ to the tuple weight and may be discarded. The extensive original token graph has weight at least $nN^{-o(1)}$: every original index had a common total event count, and the restriction cannot create more events per index. Partition the remaining logarithmic ratios in $[0, A+3]$ on a mesh tending slowly to zero. There are subpower many levels, so one retains an extensive level. After further cleaning the (C, J) cells, using their masses at this level as references, there is an exponent $r \geq 0$ such that

$$\frac{w(C, J)}{n_C} = N^{-r+o(1)} \quad (6.21)$$

uniformly on all surviving pairs. It is the selected $w(C, J)$, not n_C , that is used as the reference for this deficiency deletion. Therefore the sum of reference masses in this cleaning partition is the tuple weight, not a possibly much larger sum of n_C 's. Include the packet cells and all prescribed time-tree tests in the same deletion process. The packet lower masses, the ratio (6.21), and the inherited uniform child branching then hold simultaneously.

Representatives and their event law. Fill every registered tuple over its whole interval J and rebin time at the coarser resolution

$$(D_0/D)^{-1}.$$

Each J contains Z'/D such bins, up to the harmless dyadic rounding, and this number has positive exponent. The operation drops only part of the full-branching range grafted below J . For each occupied C put one representative matrix at its center, of weight n_C . Occupy that representative at

every coarse time bin belonging to a registered pair (C, J) . Charts are counted separately. A child tuple at a coarse bin is assigned to the spatial cell hit by its representative at that bin center; this is the hashing used in the argument.

On each representative event the total weight of its children is $w(C, J)$. By (6.21), the ratio of that weight to the representative weight is $N^{-r+o(1)}$, uniformly. The same common factor compares the total masses. Thus the two normalized event laws are mutually dominated up to subpower factors. In particular, an extensive set of representative events pulls back to an extensive set of interpolated child events. This statement does not identify the individual index weights n_C with the child weights.

The representative multiplicity bound. We apply the critical inequality to this representative configuration. Its plank parameter is inherited from the reference indices; its temporal cost will be bounded using the selected children. The full-plank parameter for this representative model is at most

$$\Delta_{\text{rep},h} \leq N^{o(1)} S' D_0^{-B_*} D^d. \quad (6.22)$$

Indeed, by (6.15), every original line counted in n_C stays within a bounded coarse cell of the representative throughout the whole normalized time interval. A coarse plank containing the representative therefore contains all these lines after bounded enlargement. In the old D_0 cell units its widths are at most constant multiples of Da_1, Db_1 . The cells C partition the original active indices within a chart, so summing their weights does not duplicate any index in that chart. Apply (6.19) and use $x + y = d$ to obtain (6.22).

We claim that the sum of the occupied representative weights in a typical hashed space–time cell is at most

$$S' D^{d-h} N^{o(1)}. \quad (6.23)$$

If cells with a strict excess carried extensive representative incidence weight, restrict to those cells. Their aggregate support is at most their weighted incidence count divided by the proposed excess threshold. Make the representative time profiles uniform over the ensemble, retaining an extensive piece.

It remains to bound the cost of this profile. Up to the creation of representatives, all selections (the bad predicate, chart routing, frequency-level selection, and their regularizations) are extensive restrictions of inherited-weight child tokens. The reference mass n_C used to define the frequency ratio never becomes a child index weight. Filling each registered J adds the same Z'/D coarse events per token. It follows from Lemma 6.1, on every tested gap, that the interpolated children have profile

$$F_{\text{ch}}(\kappa) = \beta(\kappa - (\tau g - \Delta))_+, \quad 0 \leq \kappa \leq g - \Delta. \quad (6.24)$$

The bottom interval is filled and has zero cost, and the remaining interval has length ℓg .

Keep the extensively restricted and regularized representative graph fixed from now on. Pull its event restriction back to its children. Equation (6.21) compares normalized representative and child event weights by subpower factors on every event, so this pullback is an extensive restriction of the child graph. Uniformize these children, again extensively, and retain their profile by the same lemma. This last pruning removes only child events; it does not alter the already uniform representative graph. Every surviving child block is therefore still occupied in that fixed representative graph. Descendant inclusion can be compared within that very block on each tested gap, regardless of whether different children of the same representative occupied disjoint branches before pruning.

Each surviving child index stays in one cell C in its chart. Its surviving times form a subtree of the surviving times of that representative. On any fixed log gap, choose an occupied block of this

child. Uniformity says that the representative's number of occupied descendants of that block has its common representative branching exponent; it is at least the child's number. Therefore, if G_{rep} is the representative deficit profile in these same log units,

$$G_{\text{rep}}(t) - G_{\text{rep}}(s) \leq F_{\text{ch}}(t) - F_{\text{ch}}(s) \quad (0 \leq s < t \leq g - \Delta).$$

First make this comparison on the finite tested gaps and then use the Lipschitz profile limits. It implies $G'_{\text{rep}} \leq F'_{\text{ch}}$ almost everywhere. Since P is increasing, the representative cost is at most

$$\int_0^{g-\Delta} P(G'_{\text{rep}}) \leq \ell g P(\beta). \quad (6.25)$$

This comparison uses the children on every gap, not just their total number of events.

Apply the sharp critical bound at resolution D_0/D using (6.22) and (6.25). The resulting upper bound for raw average multiplicity is

$$\begin{aligned} N^{o(1)} [S' D_0^{-B_*} D^d] (D_0/D)^h N^{\ell g P(\beta)} \\ = S' D^{d-h} N^{o(1)}, \end{aligned}$$

because $B_* = h + \ell P(\beta)$. This contradicts the strict excess on the restricted support. Hence (6.23) holds typically. At each fixed excess margin its exceptional fraction is power-small, by Convention 2.8. The subpower comparison of the two event laws therefore makes it negligible for the child events as well.

The packet comparison. Fix a surviving packet and one of its filled coarse time bins. Its mass is at least $S' Z' N^{-o(1)}$. Its child positions lie in a rectangle of thin width one and long width Z' in D_0 cell units. Replacing a child by its representative moves it by $O(D)$ such units, by (6.15). Hence at this time the packet hashes into at most

$$N^{o(1)} O(Z'/D)$$

spatial cells. In its typical hashed cell its assigned child mass is therefore at least $S' D N^{-o(1)}$. To make this last typicality explicit, cells of packet mass below $S' D N^{-\epsilon/4}$ have total weight at most $N^{o(1)} (Z'/D) S' D N^{-\epsilon/4}$ at this time, which is a vanishing fraction of its packet mass. Sum this estimate over packets and filled bins; assignments are disjoint on each block.

Every child remaining in the test satisfies (6.20). Partition the mass of one packet in a hashed cell according to C . Its contribution from C is at most the original $n(C \cap P)$, hence is at most $n_C D^{-x} N^{-\epsilon}$. All eligible C representatives are occupied in that hashed cell. Outside the representative exceptional set their total n_C is at most $S' D^{d-h} N^{\epsilon/4}$. The packet mass in that cell is therefore at most

$$S' D^{d-h-x} N^{-3\epsilon/4} = S' D N^{-3\epsilon/4},$$

using $d - h - x = 1$. This contradicts the preceding lower bound on a set of child events of probability tending to one. It proves (6.18). Choosing the strict margin through a slow diagonal gives its $o(1)$ form and a vanishing exceptional event fraction. All selections used only extensive restrictions and the original upper bounds, so the same proof applies after any extensive regular refinement of the query law.

From reference mass to normal position. The reference-mass claim is now proved. We use full-time packet containment to place those reference indices near the query in the finer label's normal-position bins, and then apply cross-entropy. Return to original coordinates. Let M be the queried matrix and $\widetilde{M}$ any reference index in $C \cap P$ counted in (6.18). Write $C_j = L_j(c)N^{-a}$ for the four cell widths. Full-time containment in the packet over I_b gives, at the actual query time t and in the packet normal ϑ_P ,

$$|(1, \vartheta_P) \cdot (M - \widetilde{M})(t)| \leq N^{-\rho - wb + o(1)} = C_X N^{-\Delta + o(1)}.$$

This uses containment of each assigned line throughout I_b , not an assumption that each of its shadings meets the query time.

Inside C , replacing its frozen frame by an actual $e \in e_c$ changes difference bounds only by dimension constants: the identity

$$X_{t_c + \delta t, \vartheta_c + \delta \vartheta} = X_{t_c, \vartheta_c} + \delta t U_{t_c, \vartheta_c} + \delta \vartheta Y_{t_c, \vartheta_c} + \delta t \delta \vartheta V \quad (6.26)$$

has $|\delta t| \lesssim R_c$, $|\delta \vartheta| \lesssim \Theta_c$, and $C_U = C_X/R_c$, $C_Y = C_X/\Theta_c$, $C_V = C_X/(R_c \Theta_c)$. Changing ϑ_P to the query normal adds at most $\Theta_b N^{o(1)} C_Y$, whose ratio to C_X is $N^{-\tau g + o(1)}$. Changing the query to the center of $L = e_{c+\sigma}$ adds, relative to C_X , at most

$$N^{-\tau \sigma + o(1)} + N^{-\ell \sigma + o(1)} + N^{-\sigma + o(1)}. \quad (6.27)$$

The three terms are the normal, time, and mixed terms of (6.26). Our choices make $\tau g, \tau \sigma, \ell \sigma$ strictly greater than Δ . Consequently all the reference indices counted in (6.18) lie within a subpower number of additional-precision X_L bins of the queried bin.

Reference cross-entropy. For each fixed base (C, L) let $p_{C,L}$ be the histogram obtained by evaluating X_L on *all* indices in (6.17), with their weights divided by n_C . Let $q_{C,L}$ be the actual conditional distribution of the query bin under the event law. These distributions need not agree. The preceding geometry and (6.18) say that, outside a vanishing event fraction, the $p_{C,L}$ mass in a subpower enlargement of the queried bin is at least $D^{-x - o(1)}$.

Choose an integer $r_N = N^{o(1)}$ large enough for that enlargement on the good events. Replace $p_{C,L}$ by its convolution with the uniform distribution on $\{-r_N, \dots, r_N\}$. At a good query bin the new probability is at least $D^{-x} N^{-o(1)}$. The image of a full C under X_L has length $O(C_X)$ by (6.26); hence there are only $N^{\Delta + o(1)}$ possible bins, including the enlargement. Take the equal-weight mixture of the convolved histogram and the uniform law on this finite support, and call it $\tilde{p}_{C,L}$. It is a positive reference probability on every possible query bin. On good events

$$-\log \tilde{p}_{C,L}(X_L) \leq (x\Delta + o(1)) \log N,$$

and on every event it is $O(\log N)$. The vanishing exceptional fraction thus has vanishing normalized contribution.

Nonnegativity of relative entropy, on each conditional base, gives

$$\begin{aligned} H([X_L]_{c,a+\Delta} \mid C, L) &\leq \mathbb{E}[-\log \tilde{p}_{C,L}([X_L]_{c,a+\Delta})] \\ &\leq (x\Delta + o(1)) \log N. \end{aligned}$$

The auxiliary conditioning can now be removed with a conditional count, rather than the global number of charts. Given $Q(c, a)$ and L , the coarser label e_c and its interval I_c are determined. In this fixed normalization, the $Q(c, a)$ cell has diameter $N^{-g+\Delta}$ times its isotropic parent diameter, so

it meets only boundedly many parent cells. The possible bounded grid and boundary choices have the same property; any further subpower routing multiplicity contributes only $o(\log N)$. Hence their conditional entropy given $Q(c, a)$, L is $o(\log N)$, even if the global collection of parents is polynomial. No I_b or packet-owner label is conditioned upon in the reference cross-entropy: those labels were only used to prove the reference-mass inequality, and the histogram itself uses only C, L . The chain rule therefore gives (6.12) with exactly its stated conditioning. No independence of the matrix and the finer event label has been used. $\square$

Both lemmas are formulated with fixed positive gaps. Their constants and strict margins can first be imposed on any finite collection of scale and offset tests. The cleaning procedure and the stationary construction then permit countable diagonals, with errors tending to zero before a tested gap is shrunk. In particular Lemma 6.3 supplies an upper bound for a conditional entropy increment; it does not identify that increment with a Frostman exponent without the additional joint-cell regularizations used below.

7 Entropy increments in moving frames

The stationary packet masses determine how many event–matrix cells are occupied at each scale. We express the change in this count through the information gained or lost in the four matrix coordinates. Their widths change at different rates, so each coordinate must be measured conditional on the coordinates already observed. The resulting entropy demand will be compared with the planar projection constraints in Section 8.

The use of entropy increments to record projection growth across scales is related to the local entropy averages method of Hochman and Shmerkin [12]. Here we use finite conditional entropies in the stationary moving frames.

7.1 The limiting entropy function

Use the weighted event law of Proposition 5.6. Each retained collapsed pair (i, I_1) has weight ω_i , and probabilities are obtained by dividing by the total weight of these pairs. Thus the index marginal is proportional to ω_i times the number of its retained pairs; conditional on the index, those pairs are equally likely. The matrix belonging to the sampled index is denoted by M . All finite entropies in this section are Shannon entropies, with natural logarithms. We write

$$H_N(Z | W) = \frac{H(Z | W)}{\log N}, \quad I_N(Z; Z' | W) = \frac{I(Z; Z' | W)}{\log N}.$$

After taking the subsequences below, H_∞ and I_∞ denote their limits. Changing any observation by a partition of bounded mutual overlap changes its ordinary entropy by $O(1)$, and hence changes its limiting normalized entropy by zero.

Scalar observation at threshold r means quantization in a nested dyadic grid at length N^{-r} , with the integer dyadic depth rounded. Recall that X, U are the normal-position and normal-velocity forms, while Y, V are the second position and velocity components in the fixed spatial chart. In particular, V is an absolute velocity observation, unchanged by the event frame. The packet widths (5.19) give the thresholds

$$(k_X, k_Y, k_U, k_V) = (0, \tau, \ell, 1), \quad r_i(c, a) = \rho + (w - k_i)c + a. \quad (7.1)$$

Increasing the offset a refines all four coordinates equally. Increasing c changes the threshold for coordinate i at rate $w - k_i$. Let $Q(c, a)$ be their joint observation at these thresholds, in the frame

of a fixed representative of the event label e_c . Representatives can be bin centers. On every compact interval of $c \in (0, 1)$, all four thresholds lie strictly between 0 and 1 for sufficiently small positive offsets a . Negative offsets are also allowed when $w < 1$. More explicitly, the domain of interior full-cell tests is

$$\mathcal{U} = \{(c, a) : 0 < c < 1, -\rho(1 - c) < a < w(1 - c)\}. \quad (7.2)$$

When $w = 1$, offset zero is the left boundary of this domain.

Apply Lemma 2.6 also to joint observations from a countable dense family of c 's, offsets, extra label depths, and full or partial coordinate observations. Include unequal-depth observations and the preceding full cells. Their occupied joint cells have common mass exponents, and their conditional refinement counts have common exponents. Consequently a limiting conditional entropy is also the limiting logarithm of the corresponding typical conditional count. This is why later entropy identities yield covering and ball-mass estimates. Mean entropy alone would not give those conclusions. At every finite stage there are only finitely many tests. A new finite list can be added before the next sufficiently large value of N is chosen. The regularization and deficiency deletions preserve all earlier limits and the stationary packet data. We use this slow diagonal throughout the section.

Here are the comparison estimates that extend the limiting full-cell tests to continuous parameters. Changing the frame by $(\Delta t, \Delta \vartheta)$ changes the forms, including their values on matrix differences, according to

$$\begin{aligned} V' &= V, & U' &= U + \Delta \vartheta V, & Y' &= Y + \Delta t V, \\ X' &= X + \Delta t U + \Delta \vartheta Y + \Delta t \Delta \vartheta V. \end{aligned} \quad (7.3)$$

If both frame labels refine the same coarse label e_c , then $|\Delta t| \lesssim N^{-\ell c}$ and $|\Delta \vartheta| \lesssim N^{-\tau c}$. These factors are exactly the ratios of the widths in (7.1). Conditional on a common finer label, each full cell in either frame therefore meets boundedly many full cells in the other frame. Changing thresholds or label precisions by at most ϵ costs $O(\epsilon) + o(1)$ in normalized entropy: the extra label and grid refinements have at most $N^{O(\epsilon)}$ possibilities.

A partial coordinate test requires a slightly stronger comparison. Start inside a full cell and give both frames extra label precision $\sigma > 0$. For a refinement gap of size b with $b \ll \min\{\ell, \tau\}\sigma$, errors from the unobserved coordinates in (7.3) are smaller than the requested widths. Known coordinates cause only known translations, which do not affect intersection counts. Partial rectangular tests then have the same stable comparison under nearby parameters. All continuous partial tests below are used in this regime. The limit in N is taken first, with σ fixed, then the derivative gap tends to zero, and only afterward does σ tend to zero.

The stable full-cell comparisons define a locally Lipschitz function

$$\mathcal{H}(c, a) = \lim_{N \rightarrow \infty} H_N(e_c, Q(c, a)). \quad (7.4)$$

At offset zero, each saturated packet occupies only subpower many joint event–matrix cells. The local upper mass bound gives the reverse count, so the stationary mass formula determines their entropy:

$$\mathcal{H}(c, 0) = \lim_{N \rightarrow \infty} \log_N(n/n_c) + s_0 \ell c, \quad \partial_c \mathcal{H}(c, 0) = dw - B_* - \tau + s_0 \ell. \quad (7.5)$$

To see both directions of the first equality, a full matrix cell has active index mass at most $n_c N^{o(1)}$ by the local bound, and a time bin at scale c has relative mass at most $N^{-s_0 \ell c + o(1)}$ on each index. Thus each occupied joint cell has event probability at most $(n_c/n) N^{-s_0 \ell c + o(1)}$. Conversely, the disjoint assigned packets at each time block have mass at least n_c in exponent. Summing their assigned masses, and then summing the regular time counts, gives at most $(n/n_c) N^{s_0 \ell c + o(1)}$ occupied joint

cells. Each packet splits among only subpower many corresponding grid cells and normal labels. The two estimates prove the equality. Differentiating (5.21) proves its derivative formula.

The same local bound applied to cells whose four widths are multiplied by N^{-a} gives

$$\mathcal{H}(c, a) \geq \mathcal{H}(c, 0) + da \quad (7.6)$$

whenever the tested widths remain in range. This uses their total homogeneity $x + y = d$. It applies on both sides of zero if $w < 1$, and on the positive side if $w = 1$. Write $g = \partial_a \mathcal{H}$, defined almost everywhere.

7.2 A differentiation lemma for entropy gaps

We next express the change in $\mathcal{H}$ as c increases. A scale increment b adds at least $s_0 \ell b$ of time information and changes the four matrix thresholds by $(w - k_i)b$. We will construct conditional rates j_G for groups G of equal-speed coordinates and prove

$$\partial_c \mathcal{H} \geq s_0 \ell + \sum_G (w - k_G) j_G.$$

Here j_G measures the additional information in G after the groups with smaller k_i have been observed. Equal speeds must be kept together because their refinements occur at the same depths.

Two issues must be resolved before this calculation is valid. The coordinate thresholds move by different amounts, so we need a first-order formula for rectangles with unequal depths, uniformly over all intervals in a shrinking neighborhood. Also, a change of frame mixes coordinates; we first condition on a finer event label to control this mixing and then remove that extra information. The next lemma supplies the uniform differentiation needed for the rectangular observations.

Lemma 7.1 (Uniform differentiation of a superadditive gap kernel). *Let $K(s, t)$, $s < t$ in an interval, be a nonnegative kernel satisfying*

$$K(s, u) \geq K(s, t) + K(t, u), \quad 0 \leq K(s, t) \leq C(t - s).$$

There is a measurable ϕ , $0 \leq \phi \leq C$, such that for almost every a , and for every fixed $C_0 < \infty$,

$$\sup_{[s, t] \subset [a - C_0 b, a + C_0 b]} |K(s, t) - \phi(a)(t - s)| = o(b) \quad (b \downarrow 0). \quad (7.7)$$

Proof. For a finite partition $\mathcal{P}$ of $[s, t]$, sum the values of K on its intervals, and set

$$m(s, t) = \inf_{\mathcal{P}} \sum_{[u, v] \in \mathcal{P}} K(u, v).$$

Concatenation of partitions gives $m(s, u) \leq m(s, t) + m(t, u)$. Inserting t into any partition of $[s, u]$ can only decrease its sum, by superadditivity of K . Taking infima gives the reverse inequality. Thus m is additive and $0 \leq m(s, t) \leq C(t - s)$. It is the interval function of an absolutely continuous measure, so $m(s, t) = \int_s^t \phi(r) dr$ for some $0 \leq \phi \leq C$.

The excess $E = K - m$ is nonnegative and superadditive. Its infimum of partition sums is zero on every interval. Fix a compact interval J and $\epsilon > 0$. Consider points in its interior that admit arbitrarily small centered intervals I with $E(I) > \epsilon|I|$. Choose a finite partition of J with sum of excesses less than an arbitrary $\eta > 0$. Except at its finitely many endpoints, the small test intervals can be confined to individual partition cells. The Vitali covering theorem supplies a

disjoint countable selection of these intervals covering the exceptional set up to a null set. For every finite selection, superadditivity and nonnegativity give

$$\epsilon \sum_I |I| < \sum_I E(I) \leq \sum_{J' \in \mathcal{P}} E(J') < \eta.$$

Pass to the countable union. Its length, and hence the measure of the exceptional set, is at most η/ϵ . Since η was arbitrary, this set is null. Exhaust the domain by compact intervals and take countably many positive rational ϵ . It follows that the excess on centered intervals has upper density zero almost everywhere.

At a point with this property that is also a Lebesgue point of ϕ , the excess on any subinterval of $[a - C_0 b, a + C_0 b]$ is bounded by the excess on the entire surrounding interval. The additive part has error at most

$$\int_{a-C_0 b}^{a+C_0 b} |\phi(r) - \phi(a)| dr = o(b).$$

This proves (7.7), first for positive integral C_0 and therefore for every fixed C_0 . $\square$

Fix c and an extra label depth $\sigma > 0$ with $c + \sigma < 1$. Work in the $e_{c+\sigma}$ frame, and condition on that label. For an axis subset S , let S_t denote its observation at offset t and let Q_s denote the full observation at offset s , both with the threshold base c . Define

$$K_S^\sigma(s, t) = H_\infty(S_t \mid Q_s, e_{c+\sigma}), \quad s < t. \quad (7.8)$$

Use only a short offset interval compared with σ , as above. The kernel is bounded between zero and $|S|(t - s)$, and is superadditive. Indeed, split the entropy at S_t in the entropy of S_u given Q_s ; then condition the second increment further on the remaining coordinates of Q_t . Lemma 7.1 gives densities $\phi_S^\sigma(c, a)$ with the uniform expansion (7.7).

This yields a rectangular layering rule. Start with a full observation at a_0 . To reveal a rectangular refinement, move through the finitely many depths at which its set of required axes changes. On a layer $[d_1, d_2]$, call the required axes S . The existing information contains the full starting cell and S_{d_1} , and is contained in the full observation at d_1 . Therefore the entropy contributed on that layer is between

$$K_S^\sigma(d_1, d_2) \quad \text{and} \quad K_S^\sigma(a_0, d_2) - K_S^\sigma(a_0, d_1). \quad (7.9)$$

If all depths are within $O(b)$ of a differentiation point a , both bounds equal $\phi_S^\sigma(c, a)(d_2 - d_1) + o(b)$. One may sum the finitely many layers, or subtract two such rectangular entropy formulas. The latter operation permits nuisance coordinates to be conditioned upon at a different depth. The error is uniform over the endpoints of these rectangles in a fixed $O(b)$ window.

7.3 Removing the extra event precision

The total information in an extra label of depth σ is $O(\sigma)$. This does not control an entropy increment divided by its length b when b tends to zero first. We instead bound the extra label's information on the same shrinking interval. Its density will tend to zero as σ decreases.

Lemma 7.2 (Local removal of label conditioning). *Let $\sigma \downarrow 0$ through a fixed countable sequence of extra label depths. For almost every (c, a) , all the rates $\phi_S^\sigma(c, a)$ converge to rates $\phi_S(c, a)$. The rectangular layering rule holds with these limiting rates, with the order of limits stated above. Moreover*

$$\phi_\emptyset = 0, \quad \phi_{\{X, Y, U, V\}} = g. \quad (7.10)$$

For two nested extra labels, changes in conditional increments are controlled by the incremental mutual information of a full observation over a containing gap. Its density tends to zero almost everywhere as the total extra label depth tends to zero.

Proof. Use first the fixed e_c frame and define

$$F_\sigma(a) = I_\infty(Q(c, a); e_{c+\sigma} \mid e_c).$$

Nested grids and the chain rule show that F_σ is nondecreasing, and that its increment on $[a_1, a_2]$ is exactly

$$F_\sigma(a_2) - F_\sigma(a_1) = I_\infty(Q(c, a_2); e_{c+\sigma} \mid Q(c, a_1), e_c). \quad (7.11)$$

It is bounded by $4(a_2 - a_1)$, and its total variation on any fixed interior interval is at most $H_\infty(e_{c+\sigma} \mid e_c) = O(\sigma)$. If $\sigma_1 \leq \sigma_2$, the increment measure for σ_1 is at most that for σ_2 , again by the chain rule and label nesting. Their bounded densities therefore have decreasing versions along $\sigma \downarrow 0$. Their integrals tend to zero, so their density limit is zero almost everywhere. This monotonicity is the reason that the conclusion is almost everywhere, rather than just in L^1 .

We spell out how to use this measure for partial observations, since mutual information need not decrease under arbitrary conditioning. Write $L_0 = e_c$ and take nested extra labels $L_1 = e_{c+\sigma_1}$ and $L_2 = e_{c+\sigma_2}$, where $0 < \sigma_1 \leq \sigma_2$. First use the L_1 frame for every partial observation. Let A consist of the full starting cell and any preceding nuisance observations, and let Z be the partial increment being revealed. Choose containing full L_0 -frame observations $R = Q(c, a - Cb)$ and $T = Q(c, a + Cb)$. Given L_1 , the observation A determines R , and (T, L_1) determines (A, Z) , up to bounded overlap. Adjoin the bounded-overlap observations at ordinary entropy cost $O(1)$. The chain rule gives

$$\begin{aligned} I(Z; L_2 \mid A, L_1) &\leq I(T; L_2 \mid R, L_1) + O(1) \\ &\leq I(T; L_2 \mid R, L_0) + O(1). \end{aligned} \quad (7.12)$$

For the second inequality use $L_1 \subset L_2$: its difference is the nonnegative term $I(T; L_1 \mid R, L_0)$. After dividing by $\log N$ and taking the limit, the last line is $F_{\sigma_2}(a + Cb) - F_{\sigma_2}(a - Cb)$. The first line bounds exactly the change in $H(Z \mid A, L_1)$ on adding L_2 . Thus nuisance data are permitted because they are determined by the containing full ending grid and the older label; no arbitrary-conditioning assertion is used. A difference of rectangular entropies is treated by applying this bound to its finitely many terms.

Only after conditioning on L_2 do we change the partial observations to the L_2 frame. Full starting cells in the two frames have bounded mutual overlap. For partial refinements retain this full base cell and take $b \ll \min\{\ell, \tau\}\sigma_1$. The unresolved base coordinates in (7.3) acquire the extra factors $N^{-\ell\sigma_1}$ or $N^{-\tau\sigma_1}$, which put all their errors inside the ending widths. Known coordinate centers cause only known translations. The joint partial tests therefore have bounded mutual overlap given L_2 and the base cell. This comparison costs $O(1)$ ordinary entropy, which vanishes in the N limit before any division by b .

Apply Lemma 7.1 and differentiate at a common Lebesgue point of the information densities. The difference of two σ -rates is bounded by a constant times the information density for their total label refinement. That density tends to zero almost everywhere. The rates are thus Cauchy there, proving convergence and the limiting layering rule. For the full subset, conditional increments differ from increments of $\mathcal{H}(c, a)$ by exactly the information in (7.11). This proves (7.10). Taking a countable intersection for subsets, extra depths, and stable tests, and then using Fubini in c , gives the assertions almost everywhere in (c, a) . The stable comparison estimates give measurable versions of all the functions used here. $\square$

Order the coordinates by increasing speed k_i , keeping tied speeds together. Let $G_1, \dots, G_m$ be these groups and $S_j = G_1 \cup \dots \cup G_j$, with $S_0 = \emptyset$. Define

$$j_{G_j} = \phi_{S_j} - \phi_{S_{j-1}}, \quad \mathcal{M} = \sum_{j=1}^m k_{G_j} j_{G_j}. \quad (7.13)$$

The conditioning in these rates can be read directly from the chain rule. In the $e_{c+\sigma}$ frame, for $r > 0$ sufficiently small,

$$\begin{aligned} K_{S_j}^\sigma(a, a+r) - K_{S_{j-1}}^\sigma(a, a+r) \\ = H_\infty((G_j)_{a+r} \mid Q_a, (S_{j-1})_{a+r}, e_{c+\sigma}). \end{aligned} \quad (7.14)$$

Dividing by r , then taking $r \downarrow 0$ and finally $\sigma \downarrow 0$, gives j_{G_j} . Thus this rate measures new information in G_j after the full old cell and the refinements of all earlier groups are known. In particular,

$$0 \leq j_{G_j} \leq |G_j|, \quad \sum_j j_{G_j} = g. \quad (7.15)$$

In every case write $a_1 = j_X$ and $v_1 = j_V$. For distinct speeds put $y_1 = j_Y$ and $u_1 = j_U$. If $\tau = \ell = 1/2$, keep the middle pair together and denote its rate by $m_1 = j_{\{Y,U\}}$. In every case the last group is the single axis V .

The geometric estimate from Section 6 now has a direct interpretation in terms of the first rate.

Corollary 7.3 (Normal-position rate). *For almost every (c, a) with $0 < c < 1$ and $0 < a < w(1-c)$,*

$$a_1(c, a) \leq x = d - 1 - h < 1. \quad (7.16)$$

Proof. Fix an extra label depth $\sigma > 0$ with $c + \sigma < 1$. Lemma 6.3 bounds the normalized conditional X entropy at every sufficiently small fixed gap r by xr in the limit $N \rightarrow \infty$. Its base cell is $Q(c, a)$ in the e_c frame. Given $e_{c+\sigma}$, this full cell and the full cell Q_a in the finer frame determine one another up to bounded overlap, by (7.3). Replacing the base cell therefore changes the ordinary conditional entropy by $O(1)$, which vanishes in the N limit before division by r . The resulting entropy is exactly $K_{\{X\}}^\sigma(a, a+r)$. Dividing by r and taking $r \downarrow 0$ gives $\phi_{\{X\}}^\sigma \leq x$ at almost every point. Finally Lemma 7.2, with $\sigma \downarrow 0$, gives $a_1 = \phi_{\{X\}} \leq x$. The positive-offset range is the range required by Lemma 6.3. $\square$

Proposition 7.4 (Entropy demand). *At almost every point of $\mathcal{U}$,*

$$\partial_c \mathcal{H}(c, a) \geq s_0 \ell + wg - \mathcal{M}. \quad (7.17)$$

Proof. For $b > 0$, label nesting gives the exact normalized limiting identity

$$\begin{aligned} \mathcal{H}(c+b, a) - \mathcal{H}(c, a) &= H_\infty(e_{c+b} \mid e_c, Q(c, a)) \\ &\quad + H_\infty(Q(c+b, a) \mid e_{c+b}) - H_\infty(Q(c, a) \mid e_{c+b}). \end{aligned} \quad (7.18)$$

Condition the first entropy additionally on the index. The old matrix cell is then deterministic given that index and e_c . The time branching in (5.20) contributes at least $s_0 \ell b$. The normal range on an index in the old time bin has only subpower many coarse angular choices, so conditioning on the old joint label does not cost a positive exponent of that time branching. The finite regularization includes these conditional label tests. This proves the stated lower bound for the first term.

Fix now $\sigma > b$ and condition both remaining terms further on $e_{c+\sigma}$. Given e_{c+b} , the two full observations are sandwiched between full observations at offsets $a - Cb$ and $a + Cb$ in the c frame, up to bounded overlap; all four speeds are bounded. The change in their entropy difference is bounded by the extra-label information on this containing gap, as proved in Lemma 7.2. At a common differentiation point it is at most b times a quantity tending to zero as $\sigma \downarrow 0$, plus $o(b)$.

In the common finer frame, changing c to $c + b$ changes the four thresholds by $(w - k_i)b$. Use a common full starting offset below all the thresholds, and subtract the two rectangular layering formulas. The axes with smaller k_i extend farther. The contribution is therefore

$$b \sum_G (w - k_G) j_G^\sigma + o(b).$$

Let $b \downarrow 0$ with σ fixed, and then let $\sigma \downarrow 0$. Combining this expression with the label contribution gives (7.17). $\square$

7.4 A strong trace for the last coordinate

The projection constraints will distinguish the sets where v_1 is positive or zero. Weak convergence alone does not preserve that distinction as the offset tends to zero. We therefore prove strong compactness for v_1 , using the fact that V is unchanged by frame changes.

Keep its absolute threshold fixed. As c increases, the label and the observations of the other three coordinates become finer. We will show that these later data determine the earlier conditioning up to bounded overlap, while the V increment itself remains the same. Its conditional entropy rate is therefore nonincreasing along each fixed threshold, which gives the required compactness.

Lemma 7.5 (Velocity rate along a fixed threshold). *The function $v_1(c, a)$ has an essentially nonincreasing version as c increases on lines where*

$$\lambda = \rho + (w - 1)c + a$$

is fixed. If $w < 1$, it has strong one-sided traces in $L_{\text{loc}}^1(dc)$ at $a = 0$, through a full-measure set of offsets. Its one-sided stripe averages have the same strong limits. If $w = 1$, every sequence of such generic positive offsets tending to zero has a subsequence on which $v_1(\cdot, a)$ converges strongly in $L_{\text{loc}}^1(dc)$.

Proof. At fixed λ , the last conditional rate, before removing an extra label, is the derivative at $r = 0+$ of

$$H_\infty(V_{\lambda+r} \mid V_\lambda, X_{\lambda+c+r}, Y_{\lambda+\ell c+r}, U_{\lambda+\tau c+r}, e_{c+\sigma}). \quad (7.19)$$

This follows by subtracting the rectangular entropy for the first three axes from that for all four, using (7.9). Compare two values of c with the same λ , using extra precision σ at both locations. The later label refines the earlier one. The later other-coordinate thresholds are also finer. In converting from the later frame to the earlier frame, the unknown part of V beyond V_λ enters Y, U, X with factors bounded respectively by

$$O(N^{-\ell(c+\sigma)}), \quad O(N^{-\tau(c+\sigma)}), \quad O(N^{-(c+\sigma)}),$$

where c is the earlier location. These errors fit in the earlier end cells when $r \ll \min\{\ell, \tau\}\sigma$. For clarity, write the two locations as $c_1 < c_2$ and put $\Delta c = c_2 - c_1$. In the conversion

$$X_1 = X_2 - \Delta t U_2 - \Delta \vartheta Y_2 + \Delta t \Delta \vartheta V,$$

the four uncertainty terms, divided by the earlier X -cell width $N^{-\lambda-c_1-r}$, are bounded respectively by constants times

$$N^{-\Delta c}, \quad N^{-\tau\Delta c-\ell\sigma}, \quad N^{-\ell\Delta c-\tau\sigma}, \quad N^{r-\sigma}.$$

The unresolved V terms in Y_1 and U_1 , divided by their respective earlier widths, are bounded by $O(N^{r-\ell\sigma})$ and $O(N^{r-\tau\sigma})$. The remaining observed-coordinate errors are smaller. Hence the later conditioning data determine the earlier conditioning data up to bounded overlap. The variable V itself is identical in both frames. Conditional entropy decreases when information is added, proving the required inequality for (7.19). First differentiate in r and then remove the extra label by Lemma 7.2. Fubini in the common threshold and the two locations gives the essential monotonicity assertion. More explicitly, the change of variables $(c, a) \mapsto (c, \lambda)$ is an invertible shear, and its interior domain is $c > 0$, $\lambda > 0$, $c + \lambda < 1$. Rate-existence and label-removal null sets remain null in these variables. After intersecting over countably many extra precisions, Fubini gives, for almost every λ , the comparison for almost every ordered pair $c_1 < c_2$. Integrating over two ordered small neighborhoods of Lebesgue points and shrinking them gives the comparison at every pair of Lebesgue points on that line. Extend by one-sided limits to obtain a monotone version. Taking limits of one-sided local averages supplies a jointly measurable choice of these versions.

For the trace statement suppose $\alpha_0 = 1 - w > 0$, and put $f(c, a) = v_1(c, a)$. On a compact interior strip define

$$h(r, a) = f(r + a/\alpha_0, a).$$

Here the common threshold is $\rho - \alpha_0 r$. For almost every r , the function $a \mapsto h(r, a)$ has a bounded nonincreasing version, and hence has one-sided limits $h_+(r)$ and $h_-(r)$ at zero. Dominated convergence gives strong L^1 convergence on compact r intervals on either side. For $J \Subset J' \Subset (0, 1)$, undoing the shear yields, for small $|a|$ on the appropriate side,

$$\begin{aligned} \|f(\cdot, a) - h_{\pm}\|_{L^1(J)} &\leq \|h(\cdot, a) - h_{\pm}\|_{L^1(J')} \\ &\quad + \|h_{\pm}(\cdot - a/\alpha_0) - h_{\pm}\|_{L^1(J)} \longrightarrow 0. \end{aligned}$$

The last term tends to zero by continuity of translation in L^1 . Fubini gives one full-measure set of offsets whose original slices agree almost everywhere with these representatives. Thus convergence holds through every sequence in that set. Averaging the L^1 estimate over a one-sided stripe gives the same limit for stripe averages.

If $w = 1$, the fixed-threshold lines are constant-offset lines. Every generic slice is a bounded monotone function of c . Helly's selection theorem, followed by dominated convergence, gives strong L^1 compactness on each compact c interval. A compact exhaustion and subsequence diagonal give strong L^1_{loc} compactness. No choice of values on exceptional offset slices is needed in either case. $\square$

7.5 Demand on thin stripes

Lemma 7.6 (Stripe limits of the demand). *Fix $J \Subset (0, 1)$. For sufficiently small $\epsilon > 0$, the right stripe averages of g are at least d almost everywhere on J :*

$$\frac{1}{\epsilon} \int_0^\epsilon g(c, a) da \geq d. \tag{7.20}$$

If $w < 1$, the left stripe averages satisfy

$$\frac{1}{\epsilon} \int_{-\epsilon}^0 g(c, a) da \leq d. \tag{7.21}$$

Along any sequence of generic admissible offsets tending to zero,

$$\partial_c \mathcal{H}(\cdot, a) \xrightarrow{*} dw - B_* - \tau + s_0 \ell \quad \text{in } L^\infty(J). \quad (7.22)$$

Consequently every weak-star subsequential limit of

$$\mathcal{M} - w(g - d) \quad (7.23)$$

is at least $\tau + B_*$ almost everywhere on J . This conclusion holds both for generic offset slices and for one-sided stripe averages. All these limit inequalities may be tested against arbitrary nonnegative $L^1(J)$ functions.

Proof. The fundamental theorem of calculus in the offset variable and (7.6) give

$$\mathcal{H}(c, \epsilon) - \mathcal{H}(c, 0) = \int_0^\epsilon g(c, a) da \geq d\epsilon.$$

On the left, that same inequality at $a = -\epsilon$ gives $\mathcal{H}(c, 0) - \mathcal{H}(c, -\epsilon) \leq d\epsilon$. This proves the two stripe bounds; they are not assertions about individual offset slices of g .

Local Lipschitz continuity gives uniform convergence $\mathcal{H}(\cdot, a) \rightarrow \mathcal{H}(\cdot, 0)$ on J and a common bound for the c derivatives. Against a smooth compactly supported test function ψ , integration by parts therefore gives

$$\int_J \psi(c) \partial_c \mathcal{H}(c, a) dc = - \int_J \psi'(c) \mathcal{H}(c, a) dc \longrightarrow (dw - B_* - \tau + s_0 \ell) \int_J \psi(c) dc.$$

The common L^∞ bound and L^1 approximation extend this to all L^1 tests, proving (7.22). It also proves the corresponding statement for stripe averages. Finally Proposition 7.4 rearranges as

$$\mathcal{M} - w(g - d) \geq s_0 \ell + wd - \partial_c \mathcal{H}.$$

The right side converges weak-star to $\tau + B_*$. Nonnegative L^1 tests preserve the inequality under every weak-star subsequential limit, proving the claim. $\square$

The distinction between weak stripe bounds and the strong trace of v_1 will matter in the closing argument. Lemma 7.5 allows localization to the sets where the limiting velocity rate is positive or zero, while Lemma 7.6 supplies the same lower demand on those measurable sets.

8 Projection constraints from two frames

We retain the stationary data of Proposition 5.6 and the entropy rates of Section 7. Thus

$$0 < \tau < 1, \quad \ell = 1 - \tau, \quad s_0 = 1 - \beta > 0, \quad (k_X, k_Y, k_U, k_V) = (0, \tau, \ell, 1).$$

The angular exponent supplied by Lemma 6.2 is denoted by $\gamma > 0$. All conditional probabilities below refer to the indexed event probability, with its inherited weights.

Proposition 8.1 (Projection constraints). *At almost every interior point (c, a) where the full-cell thresholds lie in the permitted range, the following assertions hold. In every case write $a_1 = j_X$ and $v_1 = j_V$. If $\tau \neq \ell$, write $y_1 = j_Y$ and $u_1 = j_U$, where the conditional rates are ordered by increasing speed k_i . Each belongs to $[0, 1]$, and*

$$\begin{aligned} a_1 &\geq y_1, & u_1 &\geq v_1, \\ v_1 > 0 &\implies y_1 \geq \min\{1, s_0 + v_1\}, \\ a_1 < 1 &\implies u_1 \leq (a_1 - s_0)_+. \end{aligned} \quad (8.1)$$

If $\tau = \ell$, put $m_1 = j_{\{Y,U\}}$. Then $a_1, v_1 \in [0, 1]$, $m_1 \in [0, 2]$, and

$$a_1 < 1 \implies m_1 \leq 1 + (a_1 - s_0)_+, \quad v_1 \leq (a_1 - s_0)_+. \quad (8.2)$$

These conclusions apply on both signs of the offset whenever those offsets are in the permitted range.

The geometric tests compare two frames of the same matrix. Write Δt and $\Delta \vartheta$ for the changes in time and normal parameter between them. A time change adds velocity to position, and a normal change adds the transverse component to the normal component. Thus, for example, $Y' = Y + \Delta t V$ projects the pair (Y, V) onto a line. If the second-frame coordinate occupied too few bins, the planar projection bound would force the frame label to reveal a positive amount of matrix information. An averaged information estimate will rule this out.

We first put these tests in the units of a short scale interval. The coincidence of the two middle speeds requires one additional projection, whose slope is $\Delta t \Delta \vartheta$. We state the pinning input for that test before choosing the pair laws. Its proof, together with the planar projection argument used here, is given in Section 9.

8.1 The local matrix experiment

Fix an interior c for which the entropy conclusions hold at almost every offset, and work on a compact interval of admissible offsets. Our base cell will stagger the four coordinate-refinement intervals: coordinates with different speeds will be refined on disjoint depth intervals. This makes the ordered conditional rates of Section 7 the dimensions of the source pairs used below. We shall let $b \downarrow 0$ and write

$$E_0 = e_{c+b}, \quad D = N^b.$$

In the E_0 -frame, let J_0 be the full matrix cell at thresholds

$$r_i^0 = \rho + (w - k_i)c + a + (1 - k_i)b, \quad i \in \{X, Y, U, V\}. \quad (8.3)$$

Equivalently, this is the full cell based at $c + b$ and offset $a + (1 - w)b$. The cell partitions are nested as a increases.

Choose a fixed rational $\zeta_{\max} > 0$ smaller than every nonzero difference $|k_i - k_j|$. Choose a fixed $C_0 > 1$ sufficiently large, and put

$$L = e_{c+C_0b}. \quad (8.4)$$

For small b all labels remain in the original scale interval. Write $\mathfrak{s}_i = N^{-r_i^0}$ for the four physical cell widths. Subtract a fixed matrix center of J_0 and express each form in its corresponding units $\mathfrak{s}_i$. Here X, Y are normal and transverse position, and U, V are the corresponding velocity coordinates, as in (5.18). The centered coordinates stay in a fixed bounded set in every frame whose label lies within E_0 . A time change adds a multiple of the velocity row of $\begin{pmatrix} X & Y \\ U & V \end{pmatrix}$ to its position row; a normal change adds a multiple of its transverse column to its normal column. Their composition gives the mixed term in the following formula.

Lemma 8.2 (Change of frame in relative units). *If two frame labels in E_0 have differences Δt and $\Delta \vartheta$, measured respectively in units R_{c+b} and Θ_{c+b} , their centered relative coordinates are related by*

$$(X, Y, U, V) \mapsto (X + \Delta t U + \Delta \vartheta Y + \Delta t \Delta \vartheta V, Y + \Delta t V, U + \Delta \vartheta V, V). \quad (8.5)$$

All coefficients are bounded. Reading the frame with the label L introduces relative errors $O(D^{-\min(\ell, \tau)(C_0-1)})$ on bounded coordinates. Thus C_0 can be fixed so that these errors are smaller than $D^{-\zeta_{\max}}$ by a fixed positive power.

Proof. The widths in (8.3) satisfy

$$\frac{s_X}{s_U} = R_{c+b}, \quad \frac{s_X}{s_Y} = \Theta_{c+b}, \quad \frac{s_Y}{s_V} = R_{c+b}, \quad \frac{s_U}{s_V} = \Theta_{c+b}, \quad \frac{s_X}{s_V} = R_{c+b}\Theta_{c+b}. \quad (8.6)$$

The last equality uses $\tau + \ell = 1$. Substituting these ratios in the exact change of the four linear forms gives (8.5). Using uncentered forms adds only known translations. The time and angular errors of L relative to E_0 are respectively $D^{-\ell(C_0-1)}$ and $D^{-\tau(C_0-1)}$. The formula, on a bounded set of coordinates and coefficients, gives the asserted error bound. Actual dyadic widths may be used as the units, or bounded rounding factors may be retained throughout. $\square$

For $\mathbf{u} = (u_X, u_Y, u_U, u_V)$ with $0 \leq u_i \leq \zeta_{\max}$, let $\mathcal{O}_{\mathbf{u}}$ observe axis i in the L -frame to relative depth u_i , that is, to threshold $r_i^0 + bu_i$. An observation at depth zero carries at most bounded additional entropy after E_0, J_0, L are known. Figure 3 shows the refinement intervals when all four depths equal a fixed $\zeta \in (0, \zeta_{\max}]$.

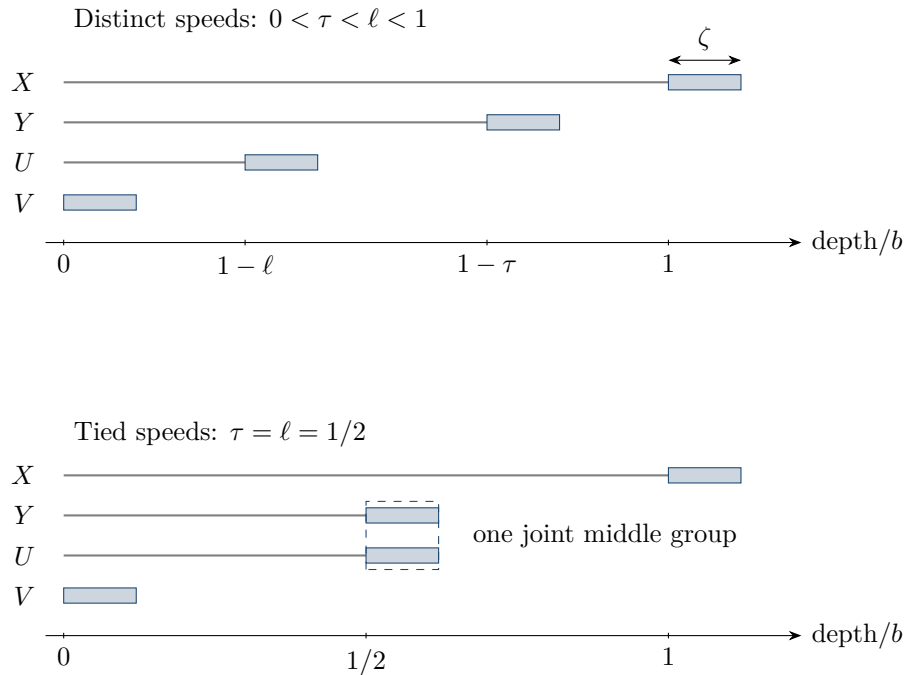


Figure 3: Schematic rebased coordinate windows for the local matrix experiment in Section 8. Gray segments show the base observations; shaded windows show the additional refinements. Relative to the original offset, the base of axis i is shifted by $(1 - k_i)b$, and its additional observation has depth ζb . When ζ is smaller than every nonzero speed separation, the windows of distinct groups are disjoint. At each window, all groups with smaller k_i are already observed in the full base cell. The other distinct-speed order exchanges Y and U . Tied coordinates overlap and must retain their joint rate.

Lemma 8.3 (Local counts and conditional Frostman laws). *There is a triangular choice $b_n \downarrow 0$, $N_n \rightarrow \infty$, with $D_n = N_n^{b_n} \rightarrow \infty$, and uniform refinements of the single-event graphs, chosen independently of the queried offset, with the following properties at almost every admissible offset a . At stage n , evaluate the finite experiment at a prescribed grid offset a_n with $|a_n - a| = o(b_n)$; all rates below are evaluated at the limiting point (c, a) . Conditional on*

$$\mathcal{B} = (E_0, J_0, L),$$

the entropy of $\mathcal{O}_{\mathbf{u}}$, divided by $\log D$, is

$$\mathcal{R}(\mathbf{u}) + o(1). \quad (8.7)$$

For distinct speeds,

$$\mathcal{R}(\mathbf{u}) = \sum_i j_i u_i.$$

For tied middle speeds, put

$$r_Y = \phi_{\{X,Y\}} - \phi_{\{X\}}, \quad r_U = \phi_{\{X,U\}} - \phi_{\{X\}}.$$

Then

$$\mathcal{R}(\mathbf{u}) = a_1 u_X + v_1 u_V + \begin{cases} r_Y(u_Y - u_U) + m_1 u_U, & u_Y \geq u_U, \\ r_U(u_U - u_Y) + m_1 u_Y, & u_U \geq u_Y. \end{cases} \quad (8.8)$$

These formulas hold for joint and unequal-depth observations. Their occupied conditional cells have probabilities

$$\mathbb{P}(\mathcal{O}_{\mathbf{u}} = o \mid \mathcal{B}) = D^{-\mathcal{R}(\mathbf{u}) + o(1)} \quad (8.9)$$

uniformly on the regularized graph at every tested depth. Ratios of joint and conditioning cell probabilities give the corresponding conditional formulas.

In particular, fix nuisance axes at relative depth ζ and observe a selected pair to any depth $0 \leq u \leq \zeta$. For distinct speeds its conditional point law is Frostman of exponent equal to the sum of the two selected rates, with subpower constant, through resolution $D^{-\zeta}$. For tied middle speeds, the additional U -rate after observing Y to depth ζ is $m_1 - r_Y$; the pair (X, V) after fixing Y, U has exponent $a_1 + v_1$.

The same triangular choice supplies, conditionally on the original index and E_0 , the time and normal-parameter Frostman bounds of exponents s_0 and γ in relative units. When $\tau = \ell$, the joint parameter square cells have masses $D^{-s_0 u + o(1)}$ for $0 \leq u \leq \zeta_{\max}$.

Proof. We prove the rectangular entropy formula first. We then realize its values as uniform finite cell masses and verify the time and normal laws needed to choose a second event.

Rectangular entropy increments. Fix $\sigma > 0$ and, temporarily, take b so small that $C_0 b < \sigma$ and $c + \sigma < 1$. Condition additionally on $e_{c+\sigma}$ and read all forms in that finer frame. Given L , Lemma 8.2 shows that the required observations in the two frames determine one another up to bounded overlaps. The full starting cell is included in this comparison; errors from unobserved components are below the finest tested width.

Apply the rectangular layering conclusion of Lemma 7.1. The starting offsets of the four increment intervals are

$$a + (1 - k_i)b.$$

Intervals belonging to different speed groups are disjoint, because $\zeta_{\max} < |k_i - k_j|$. During the interval for a group of speed k , every group of smaller speed has already been required to a depth beyond that interval, even in its *base* observation. Every group of larger speed ends below the interval. If S is the union of the groups of smaller speed and G is the current group, subtraction of the base rectangle from the refined rectangle leaves the density $\phi_{S \cup G}^\sigma - \phi_S^\sigma =: j_G^\sigma$. Thus the group's contribution is its ordered conditional rate times its refinement depth. This reasoning applies both to the order X, Y, U, V , when $\tau < \ell$, and to X, U, Y, V , when $\ell < \tau$. It gives (8.7) with the σ -conditioned rates, to error $o(b)$ in entropy divided by $\log N$.

For a tied group, first reveal Y and U together to the smaller of their two depths; then refine only the deeper coordinate. The common part costs m_1 times that depth. The remaining increment uses r_Y or r_U , according to the deeper coordinate. This gives (8.8), again first with σ -conditioned rates. Equivalently, one may subtract the layering formulas for a joint rectangle and its conditioning rectangle. All their thresholds lie within $a \pm Cb$ for a fixed C .

We next remove the extra label. Given L , the joint and conditioning observations in question are sandwiched between full e_c -frame observations at offsets $a - Cb$ and $a + Cb$, up to bounded overlaps. By Lemma 7.2, the change of a conditional increment upon adding $e_{c+\sigma}$ is bounded by the incremental label information on this full-cell interval. For almost every a , its upper first-order density tends to zero as $\sigma \downarrow 0$. Consequently the limsup of the error divided by b , as $b \rightarrow 0$ with σ fixed, is bounded by a constant times that density. Now let $\sigma \downarrow 0$ through a countable sequence. The rates converge to ϕ_S and j_G , and the error is $o(b)$. This proves the asserted formulas for the limiting normalized entropies without requiring a convergence speed uniform in a .

Uniform finite cell masses. Here is a simultaneous implementation on finite configurations. Fix a summable sequence of positive rational $b_n \downarrow 0$. At stage n , prescribe a finite offset grid of spacing $o(b_n)$ on the compact interval under consideration, and a finite grid of relative depths whose union is dense in $[0, \zeta_{\max}]$. Include the following in the countable family of tests: E_0, J_0, L ; every subset of the axis observations at equal and unequal depths; their joint and conditioning cells; the full-cell sandwiches just used; and the same observations with the fixed additional labels $e_{c+\sigma}$. Include also the within-index parameter-cell tests described below.

Use Lemma 2.6 on finitely many tests at a time, including the deficient-cell deletions relative to their original mass. This makes the joint and conditional cell masses uniform in exponent. Previously uniform tests keep their limits: surviving cells retain their original masses within the prescribed subpower factor, and the total restriction has the same order of loss. Thus the already defined stable entropy function and its rates do not change. Take subsequential limits on this countable family first. Their values satisfy the estimates established above with fixed σ and then $\sigma \downarrow 0$.

Finally choose N_n sufficiently large along the resulting configurations that the first n finite collections of tests have errors $o(b_n)$ in exponents normalized by $\log N_n$. The finite pigeonholings and cleaning can have total loss $N_n^{-o(b_n)}$: for a fixed finite number of tests their number of mass levels grows at most polynomially in $\log N_n$ and in the inverse requested accuracy, so increasing N_n makes their normalized logarithmic cost smaller than the chosen multiple of b_n . Choose the deletion cutoffs below that retention while still with exponent $o(b_n)$. Also require $b_n \log N_n \rightarrow \infty$. For each generic a , use its nearest offset-grid point. The stable full-cell and finer-label comparisons change the exponents by $o(b_n)$ at this replacement. The first-order density estimates hold at the actual generic point for every sequence $b_n \rightarrow 0$; no uniform differentiability rate over all such points was used.

Uniform joint cell masses now prove (8.9), rather than only its entropy average. For example, in the tied case and $u \leq \zeta$,

$$\begin{aligned} H(U_u, Y_\zeta \mid \mathcal{B}) - H(Y_\zeta \mid \mathcal{B}) \\ = (m_1 - r_Y)u \log D + o(\log D). \end{aligned}$$

The probability of an occupied U_u cell within an occupied Y_ζ cell is the ratio of their joint and marginal probabilities, hence $D^{-(m_1 - r_Y)u + o(1)}$. For distinct speeds the same subtraction removes all nuisance terms and leaves u times the sum of the selected rates. For (X, V) with Y, U fixed, the tied-group contribution cancels and leaves $(a_1 + v_1)u$. These statements hold at all the tested

intermediate depths. Bounded square coverings and increasingly fine depth meshes give the asserted Frostman upper bounds at every intermediate radius.

Time and normal parameters. It remains to verify the parameter hypotheses for these finite experiments. Lemma 6.2 and the uniform time branching give the two upper bounds conditional on the index and E_0 . Nondeficient E_0 cells have the same exponent as their time blocks, since each index uses only subpower many angular bins there. Add these conditional cells and their refinements to the tests before choosing the triangular sequence. When $\tau = \ell$, refining the label from $c + b$ to $c + b + ub/\ell$ multiplies both relative widths by D^{-u} . The time mass is $D^{-s_0 u + o(1)}$. Within each of its time blocks, the normal range occupies only subpower many corresponding angular bins. Its nondeficient occupied joint cells therefore have that same mass exponent. Cleaning them gives the lower as well as the upper bound, uniformly through the required depths. All of this cleaning precedes the choice of a pair law; a later bias is never applied to an uncontrolled exceptional set of single-event cells. $\square$

8.2 The product-slope pinning input

When $\tau = \ell$, the local count formula supplies the joint middle rate m_1 , rather than two separate ordered middle rates. To bound v_1 in this case, freeze both Y and U in the first formula of (8.5). The remaining projection is

$$X' = X + (\Delta t \Delta \vartheta) V + \Delta t U + \Delta \vartheta Y.$$

The last two terms are a translation once the nuisance coordinates and the two labels are fixed. The slope is the product of two parameter differences; their separate Frostman bounds do not assert that this product is nonconcentrated. The following lemma supplies that bound while controlling the change in each single-event law.

In its statement $D \rightarrow \infty$ is a resolution parameter and ζ is a positive depth. Neither denotes a critical parameter. A probability μ on a bounded square is called uniformly s -regular through depth $\zeta_{\max}$ if, for every occupied nested dyadic square Q of side comparable to D^{-u} ,

$$D^{-su - \epsilon_D} \leq \mu(Q) \leq D^{-su + \epsilon_D}, \quad 0 \leq u \leq \zeta_{\max}, \quad \epsilon_D \rightarrow 0. \quad (8.10)$$

Fixed factors from dyadic rounding can be included in D^{ϵ_D} . It is equivalent here to impose these estimates on depth meshes whose spacing tends to zero: monotonicity gives (8.10), with a slightly larger error. Uniformity for an ensemble means that the same error works for all its members after deleting a set of initial indices of probability tending to zero.

Lemma 8.4 (Pinning with controlled marginals). *Let $0 < s \leq 1$, $\gamma' > 0$, and $0 < \zeta_{\max} \leq 1$ be fixed. For each D , let I have a finite probability distribution π_D , and let μ_i be finitely supported probabilities on a fixed bounded square with coordinates (t, ϑ) . Suppose uniformly in i that (8.10) holds and that the two coordinate marginals satisfy*

$$\mu_i\{t \in J_r\} \leq D^{o(1)} r^s, \quad \mu_i\{\vartheta \in J_r\} \leq D^{o(1)} r^{\gamma'} \quad (D^{-\zeta_{\max}} \leq r \leq 1) \quad (8.11)$$

for all intervals J_r of radius r . For every $\epsilon > 0$, there are a subsequence, a fixed rational $\zeta \in (0, \zeta_{\max}]$, and a probability ν_D on triples (i, ξ, v) , with $\xi, v \in \text{supp } \mu_i$, such that:

- (i) *each marginal (i, ξ) and (i, v) is dominated by $D^{\epsilon \zeta} \pi_D(di) \mu_i$;*

(ii) for ν_D -almost every (i, ξ) , the conditional law of

$$\mathbf{p}_\xi(v) = (t_v - t_\xi)(\vartheta_v - \vartheta_\xi)$$

satisfies

$$\nu_D\{\mathbf{p}_\xi(v) \in J_r \mid i, \xi\} \leq CD^{\epsilon\zeta} r^{s-\epsilon}, \quad D^{-\zeta} \leq r \leq 1. \quad (8.12)$$

The pair law can be chosen using only the initial index and its parameter measure. In particular it need not depend on any later matrix-cell or offset test.

The proof of Lemma 8.4 is in Section 9.2.

8.3 Two events on one index

Let μ be the probability on the regularized single-event graph. First draw (I, E_0) from its marginal under μ . For the *ordinary pair law*, draw ξ_0, ξ_1 conditionally independently from the original law given (I, E_0) . Both events use the matrix attached to I . Their conditional independence does not make that matrix independent of their frame labels. Let L_0, L_1 be their labels at the precision in (8.4). Conditional on (I, E_0, ξ_0) , the difference of the second time from the first is s_0 -Frostman, and the corresponding normal difference is γ -Frostman, in relative units. Translation does not change either estimate. Reading the differences from L_0, L_1 changes them by less than the testing resolution, by Lemma 8.2. Bounded enlargement of intervals therefore preserves their Frostman bounds.

In the tied case we need one additional coupling. Apply Lemma 8.4 to the within-index parameter laws given (I, E_0) , with $s = s_0$, using the last assertion of Lemma 8.3. For any fixed sufficiently small $\epsilon > 0$, it gives, along a further subsequence, a fixed rational depth $\zeta > 0$. Both single-event marginals are bounded by $D^{\epsilon\zeta}$ times their original laws. Conditional on (I, E_0, ξ_0) the product slope

$$\Theta_* = (t_1 - t_0)(\vartheta_1 - \vartheta_0) \quad (8.13)$$

is $(s_0 - \epsilon)$ -Frostman with constant $CD^{\epsilon\zeta}$ through resolution $D^{-\zeta}$. All coordinates here are in the relative parameter units of E_0 . The product is Lipschitz on this fixed bounded square, so the fine-label replacement preserves this estimate as well.

The original graph and these pair distributions are chosen before querying an offset. We can prescribe countably many loss tolerances and projection tests. Each can use a further subsequence; intersecting their resulting full-measure sets of offsets still gives a full-measure set. No one choice of pair distribution is asserted to work for every tolerance.

The same-index relation forces the matrix, read in the second frame, to belong to the support of that frame's original coordinate law. We will compare this certain event with its probability under the product of the conditional matrix and label marginals. A small probability under that product would require positive mutual information. The next estimate shows that the short matrix-refinement gap contains asymptotically less information than such a comparison would require. It applies equally to the ordinary and pinned pair laws. The finite pair law may depend on b, N, I, E_0 , but it must be fixed while the offset is integrated.

Lemma 8.5 (Pair information on the refinement gap). *Let $K = 1 + \zeta_{\max}$, and let J_f be the full observation in the E_0 -frame obtained by increasing every threshold of J_0 by Kb . Along a triangular subsequence as in Lemma 8.3, for almost every admissible offset,*

$$I(J_f; L_0, L_1 \mid E_0, J_0) = o(\log D). \quad (8.14)$$

Here J_0, J_f are evaluated at the nearest prescribed offset-grid point. The assertion holds for every one of the countably many pair experiments just described, with ordinary Shannon units.

Proof. Given E_0 , each label L_i has at most CN^{C_1b} possible values, where C_1 depends on C_0 and the fixed speeds. This is an alphabet bound from the time and normal grids, independent of the probabilities of these labels. It remains true under either marginal bias. Consequently

$$H(L_0, L_1 \mid E_0) \leq C_2(1 + b \log N). \quad (8.15)$$

Temporarily write $J_0(a)$ for the rebase grid at offset a and set

$$f(a) = I(J_0(a); L_0, L_1 \mid E_0).$$

This is nondecreasing, because the matrix grids are nested as a increases. Its range is contained in $[0, C_2(1 + b \log N)]$. The chain rule gives the exact identity

$$f(a + Kb) - f(a) = I(J_0(a + Kb); L_0, L_1 \mid E_0, J_0(a)). \quad (8.16)$$

For a nondecreasing function with range of length H_* ,

$$\int_A^B [f(a + q) - f(a)] da \leq qH_* \quad (q > 0),$$

as follows either by changing variables in the two integrals or by integrating its nonnegative increment measure. For a grid replacement a_n with $|a_n - a| = o(b_n)$, monotonicity also bounds the left side of (8.16) by

$$f(a + (K + 1)b) - f(a - b)$$

for large n . Integrate on a compact interval whose slightly larger neighborhood stays admissible. Equations (8.15) and (8.16) give

$$\int \frac{I(J_f; L_0, L_1 \mid E_0, J_0)}{\log D} da \leq C_3 \left(b + \frac{1}{\log N} \right). \quad (8.17)$$

The left side uses the stepwise chosen grid offsets. The containing-gap estimate proves the bound regardless of the sizes of the grid's inverse images.

Choose b_n summable, and increase N_n so that $1/\log N_n$ is summable as well as satisfying all the finite tests of Lemma 8.3. Tonelli's theorem applied to the nonnegative functions in (8.17) shows that their sum is finite almost everywhere. In particular the summands tend to zero, which proves (8.14). A further subsequence, such as the one used for pinning, preserves this conclusion. For countably many pair experiments use the same integral argument for each and intersect the full-measure sets. Finally exhaust the admissible offset domain by compact intervals. $\square$

We next record explicitly how the marginal bias is used. Denote the first single-event marginal of a chosen pair law by ν . Suppose

$$\nu \leq D^{\epsilon\zeta} \mu. \quad (8.18)$$

Let $\mathcal{B}_0 = (E_0, J_0, L_0)$. Discard base values B for which $\nu(B) < D^{-\epsilon\zeta} \mu(B)$. Their total ν -probability is at most $D^{-\epsilon\zeta}$. For every other base, division of (8.18) by its base mass gives

$$\nu(\cdot \mid B) \leq D^{2\epsilon\zeta} \mu(\cdot \mid B). \quad (8.19)$$

This also holds for the conditional matrix marginals. For the ordinary pair law the first marginal equals μ , and no base deletion is needed.

8.4 Comparing the coupling with a product law

For a fixed base $B = (E_0, J_0, L_0)$, the actual pair experiment defines a joint law of the matrix and L_1 . The conditional slope law is obtained by mixing its laws given (I, E_0, ξ_0) . Since J_0 is determined by the matrix and E_0 , this conditioning and mixture preserve each of the uniform time, angular, or pinned-product Frostman bounds above. We compare the joint law with the product of its two conditional marginals. Independence is imposed only in this comparison law.

The planar estimate we use keeps the entire direction label: two labels with the same slope may prescribe different target intervals. Its point and label measures are independent in the product below. The proof is given in Section 9.1.

Lemma 8.6 (Robust planar projections). *Fix*

$$0 < s \leq 1, \quad 0 < t \leq 2, \quad 0 \leq u < h_{\text{pr}} := \min\{t, (s+t)/2, 1\}.$$

There is $\kappa_0 = \kappa_0(s, t, u) > 0$ with the following property. For every $R, C_0 \geq 1$ and $0 < \kappa \leq \kappa_0$, there is $\delta_0 > 0$ such that the following holds for $0 < \delta < \delta_0$. Let μ be a Borel subprobability measure on $[-R, R]^2$, let ν be a subprobability measure on an arbitrary finite label set Λ , and let $\vartheta : \Lambda \rightarrow [-R, R]$. Suppose

$$\mu(B(p, r)) \leq \delta^{-\kappa} r^t, \quad (\vartheta_* \nu)(B(a, r)) \leq \delta^{-\kappa} r^s \quad (\delta \leq r \leq 1). \quad (8.20)$$

For each $\lambda \in \Lambda$, let A_λ be a union of at most $C_0 \delta^{-u}$ intervals of length at most $C_0 \delta$. Then

$$\sum_{\lambda \in \Lambda} \nu(\{\lambda\}) \mu\{(x, y) : x + \vartheta(\lambda)y \in A_\lambda\} < \delta^{2\kappa}. \quad (8.21)$$

Consequently there cannot be a set $B \subset \Lambda$ with $\nu(B) \geq \delta^\kappa$ such that for every $\lambda \in B$ there is a μ -measurable set E_λ with

$$\mu(E_\lambda) \geq \delta^\kappa, \quad \{x + \vartheta(\lambda)y : (x, y) \in E_\lambda\} \subset A_\lambda. \quad (8.22)$$

Lemma 8.7 (Transfer of a projection bound to rates). *Fix the original single-event law μ and one of the same-index pair laws above, with the conditional marginal domination already established. Suppose that after freezing the nuisance axes at relative depth ζ , the original conditional law of two source coordinates is Frostman with exponent $t > 0$ and subpower constant through resolution $\delta = D^{-\zeta}$, uniformly in its occupied nuisance fibers. Suppose that, in the pair law conditional on B , the changing projection slope has an available positive Frostman exponent $s \leq 1$. In a pinned experiment, “available” means exponents tending to s with arbitrarily small fixed marginal and Frostman losses. Suppose further that the target coordinate in the L_1 -frame has, conditional on (E_0, J_0, L_1) , at most $D^{a_*\zeta + o(1)}$ occupied bins of width comparable to $D^{-\zeta}$. Then*

$$a_* \geq \min\{t, (t+s)/2, 1\}. \quad (8.23)$$

Proof. Suppose the inequality fails strictly. Choose fixed $0 < t_* < t$, $0 < s_* < s$, and $u > a_*$, still satisfying

$$u < \min\{t_*, (t_* + s_*)/2, 1\}. \quad (8.24)$$

The choices are possible by continuity, and are made before any pinning loss is chosen. Let $\kappa > 0$ be an allowance from Lemma 8.6 for these three fixed exponents. In the pinned case choose the loss tolerance ϵ so small that

$$\epsilon < s - s_*, \quad 2\epsilon < \kappa/2. \quad (8.25)$$

Choose the pinned coupling and its fixed positive depth ζ with this tolerance. For the ordinary pair law any fixed rational $0 < \zeta \leq \zeta_{\max}$ works. The depth does not subsequently shrink with D . Subpower errors in the original single-event estimates therefore tend to zero also after division by ζ .

The product probability bound. Fix a good base B , and initially use the *original* conditional matrix law $\mu(\cdot | B)$, independently of the pair experiment's conditional law of L_1 . Within this original matrix law, freeze the nuisance bins at depth ζ . Lemma 8.3 gives the point Frostman bounds, with exponent t_* , in every occupied fiber. For sufficiently large D their constants are smaller than $\delta^{-\kappa}$. The slope bounds, with exponent s_* , have the same allowed constant: in the pinned case they cost $C\delta^{-\epsilon}$, and for the ordinary pair law a subpower.

The frame formula (8.5) writes the target coordinate as a two-coordinate scalar projection plus a translation depending on the frozen nuisance bins and the full label L_1 . The error from a nuisance bin has size $O(\delta)$, since all coefficients are bounded. For each L_1 , enlarge all its allowable target bins by this error and subtract the known translation. Their number is at most $C\delta^{-u}$, because $u > a_*$. Their lengths are at most $C\delta$. The label may contain more information than the projection slope; Lemma 8.6 permits precisely this dependence of the target intervals on the entire label.

Apply that lemma in each original nuisance fiber, with fixed bounded coordinate charts. Its independent product hit probability is less than $\delta^{2\kappa}$, uniformly in the fiber. Integrating over the original nuisance distribution gives the same bound for the entire original conditional matrix law. Only now replace that whole matrix law by its biased marginal. Equation (8.19) bounds the new product hit probability by

$$\delta^{-2\epsilon}\delta^{2\kappa} \leq \delta^\kappa \quad (8.26)$$

for large D . For the ordinary pair law the domination factor is one. This argument does not assume that biased nuisance fibers have the original Frostman estimates, or that a slope remains independent after conditioning inside the actual coupling.

The support event for the same-index law. Under the true coupling, the target matrix coordinate always belongs to the allowed support for (E_0, J_0, L_1) . Both pair marginals are supported on the original single-event graph and both events have exactly the same index matrix. Thus this support test has probability one in the true coupling. It requires no probability lower bound for target cells after the change of measure.

To make it a finite information test, replace the matrix by a representative of its J_f cell. The relative coordinate error is $O(D^{-(1+\zeta_{\max})})$, which is smaller than $D^{-\zeta}$ by a fixed positive power. The bounded frame coefficients and the precision of L_0, L_1 leave an $O(\delta)$ error. Enlarge the target bins once more. The resulting event A_B depends only on (J_f, L_1) at fixed B , has conditional probability one in the true coupling, and still has probability at most δ^κ under the product of the conditional marginals of J_f and L_1 . For the last assertion, sample the exact matrix from its conditional marginal and then round it. A rounded hit implies an exact hit in one further bounded enlargement, already allowed in the preceding application of Lemma 8.6.

The information contradiction. Let P_B be the conditional joint law of (J_f, L_1) and Q_B the product of its marginals. Data processing of relative entropy by the indicator of A_B gives

$$I(J_f; L_1 | B) = D_{\text{KL}}(P_B \| Q_B) \geq \log \frac{1}{Q_B(A_B)} \geq \kappa\zeta \log D.$$

The good bases have pair probability tending to one by (8.19). Averaging the last inequality therefore gives

$$I(J_f; L_1 | E_0, J_0, L_0) \geq (1 - o(1))\kappa\zeta \log D. \quad (8.27)$$

On the other hand the chain rule and nonnegativity imply

$$I(J_f; L_1 | E_0, J_0, L_0) \leq I(J_f; L_0, L_1 | E_0, J_0) = o(\log D)$$

by Lemma 8.5. Since $\kappa\zeta > 0$ is fixed, this contradicts (8.27) and proves the lemma. $\square$

8.5 Deriving the individual constraints

Proof of Proposition 8.1. Fix a good interior c . Work at offsets where the entropy densities, the local counts, and the pair information conclusions all hold. Strict violations of any inequality can be witnessed by rational choices of the smaller exponents and loss tolerances in (8.24)–(8.25). There are countably many such choices, so the preceding constructions apply on a common full-measure set of offsets. Fubini then gives the asserted almost-everywhere statement in (c, a) .

For distinct speeds the four tests needed below are displayed in Table 1. Each uses the ordinary pair law. Source exponents are conditional on the listed nuisance bins in the original matrix law. They follow from the all-depth probability ratios in Lemma 8.3, in either of the two possible speed orders.

Target	Source pair	Frozen axes	Slope	Source exponent	Target exponent
X	(X, Y)	U, V	$\Delta\vartheta$	$a_1 + y_1$	a_1
X	(X, U)	Y, V	Δt	$a_1 + u_1$	a_1
U	(U, V)	X, Y	$\Delta\vartheta$	$u_1 + v_1$	u_1
Y	(Y, V)	X, U	Δt	$y_1 + v_1$	y_1

Table 1: The four ordinary-pair tests for distinct speeds. The angular and time slope exponents are respectively γ and s_0 .

For example the first formula of (8.5), with U, V frozen, is

$$X' = X + \Delta\vartheta Y + \Delta t U + \Delta t \Delta\vartheta V.$$

The last two terms are a full-label-dependent translation up to $O(D^{-\zeta})$. With Y, V frozen, the same expression is $X + \Delta t U$ plus such a translation. The second and third formulas give the other two rows. Thus every row has exactly the form required in Lemma 8.7.

It is useful to spell out the numerical implication. For $0 \leq a, b \leq 1$ and $s > 0$, suppose the test with source exponent $a + b > 0$ gives

$$a \geq \min\{a + b, (a + b + s)/2, 1\}. \quad (8.28)$$

If $a < 1$ and $b > 0$, both the first and the third entries of this minimum exceed a . The second must therefore be at most a , so $b \leq a - s$. If $b = 0$ there is no further restriction. In either case

$$a < 1 \implies b \leq (a - s)_+. \quad (8.29)$$

When $a + b = 0$, that conclusion is immediate without applying the projection lemma.

The first row gives $y_1 \leq (a_1 - \gamma)_+$ when $a_1 < 1$. Together with $y_1 \leq 1$, this implies $a_1 \geq y_1$ also when $a_1 = 1$. The second row gives $u_1 \leq (a_1 - s_0)_+$ whenever $a_1 < 1$. The third row analogously gives $u_1 \geq v_1$. For the last row, if $v_1 > 0$ and $y_1 < 1$, (8.29) gives $y_1 \geq s_0 + v_1$. If $y_1 = 1$, the required $y_1 \geq \min\{1, s_0 + v_1\}$ holds automatically. These are all assertions of (8.1).

Suppose now that $\tau = \ell$. Put

$$r_Y = \phi_{\{X, Y\}} - \phi_{\{X\}}.$$

Conditional entropy monotonicity and the one-dimensional alphabet bound give $0 \leq r_Y \leq 1$ and $0 \leq m_1 - r_Y \leq 1$. Use the second row's time projection with Y, V frozen. For every $0 \leq u \leq \zeta$, the tied rectangle formula gives the conditional source exponent

$$t = a_1 + (m_1 - r_Y).$$

The target exponent is a_1 and the slope exponent is s_0 . Equation (8.29), with $b = m_1 - r_Y$, yields

$$a_1 < 1 \implies m_1 - r_Y \leq (a_1 - s_0)_+.$$

Since $r_Y \leq 1$, this proves the bound for m_1 in (8.2).

Finally freeze Y, U and use the pinned pair distribution. The first formula in (8.5) becomes

$$X' = X + (\Delta t \Delta \vartheta) V + \Delta t U + \Delta \vartheta Y.$$

The source pair is (X, V) , of conditional exponent $a_1 + v_1$, by Lemma 8.3. Its target exponent is a_1 . The product slope has available exponent s_0 by Lemma 8.4 and the preceding coupling construction. Lemma 8.7 includes both its marginal bias and its small direction loss. Thus (8.29) gives $v_1 \leq (a_1 - s_0)_+$ when $a_1 < 1$, as required.

Every construction above takes place on a compact subset of $\mathcal{U}$ from (7.2). Indeed the smallest and largest of the four unshifted thresholds are respectively

$$(1 - w)(1 - c) + a, \quad 1 - w(1 - c) + a.$$

If $w < 1$, a compact interior c -interval admits sufficiently small offsets of either sign, and the $O(b)$ rebase and refinement shifts stay in range. If $w = 1$, only positive small offsets are interior. This proves the stated conclusion for every permitted sign and completes the proposition. $\square$

9 Planar projection and product-pinning proofs

This section proves the two planar statements used in Section 8. We first derive the robust projection bound from the quantitative Furstenberg theorem. We then use that bound to construct the pair law whose product slope is nonconcentrated.

9.1 The robust projection bound

We prove Lemma 8.6. Its parameters s, t, u are fixed, and $h_{\text{pr}} = \min\{t, (s + t)/2, 1\}$ is the planar projection exponent. A positive product hit probability would produce a Furstenberg configuration with too few tubes: comparable slope masses and incidence counts give the required quantitative point and direction bounds.

Proof. Put $\Delta = h_{\text{pr}} - u > 0$ and $\varepsilon = \Delta/4$. Let $\eta_{\text{pr}} = \eta_{\text{pr}}(\varepsilon, s, t) > 0$ be the Frostman tolerance in Ren and Wang [17, Theorem 4.1]. We may take

$$\kappa_0 = \min\{1/20, \Delta/16, \eta_{\text{pr}}/12\}. \tag{9.1}$$

All constants below may depend on R, C_0, s, t , but not on the number of labels or on their weights. We decrease δ_0 finitely many times.

First put the incidences in the bounded coordinates used for dyadic tubes. Set $B_0 = R + 1$ and

$$v = \frac{y + B_0}{2B_0}, \quad w = \frac{x + B_0}{2B_0^2}, \quad a_\lambda = -\frac{\vartheta(\lambda)}{B_0}.$$

Then $(v, w) \in (0, 1)^2$, $a_\lambda \in (-1, 1)$, and

$$w - a_\lambda v = \frac{x + \vartheta(\lambda)y + B_0(1 + \vartheta(\lambda))}{2B_0^2}.$$

Apply this affine transformation to A_λ , writing the result as J_λ . Each J_λ has at most $C\delta^{-u}$ constituent intervals of length at most $C\delta$. The transformed point and slope measures satisfy (8.20) with an additional fixed factor C . We keep the notation μ for the transformed point measure.

Choose a dyadic number ϱ with $\delta \leq \varrho < 2\delta$, and put

$$L = 8\lceil \log_2(1/\delta) \rceil + 8, \quad q_* = \delta^{2\kappa}.$$

Suppose that the left side of (8.21) is at least q_* . Write $f(\lambda) = \mu\{(v, w) : w - a_\lambda v \in J_\lambda\}$. Partition the slopes into dyadic ϱ -intervals I . Give I its total label weight $W_I = \nu\{\lambda : a_\lambda \in I\}$, and choose a label λ_I maximizing f in that interval. There are finitely many labels, so

$$\sum_I W_I f(\lambda_I) \geq q_*. \quad (9.2)$$

There are at most $C\delta^{-1}$ occupied slope intervals. Discarding those with $W_I < \delta^4$ loses at most $C\delta^3 \leq q_*/2$ from (9.2). The remaining weights have at most L dyadic levels. One level, consisting of n slope intervals, satisfies

$$H_* := \sum_{I \text{ in the level}} W_I f(\lambda_I) \geq \frac{q_*}{2L}.$$

Let $a_* = \sum_I W_I$ on this level. Its weights are within a factor of two, and $H_* \leq a_* \leq 1$. Consequently the uniform distribution of its slope intervals satisfies

$$\frac{\#\{I : I \cap B(b, r) \neq \emptyset\}}{n} \leq CL\delta^{-3\kappa} r^s \quad (\varrho \leq r \leq 1). \quad (9.3)$$

Indeed, relative counting costs at most $2/a_*$ times the original slope mass of a fixed enlargement of the ball, and $a_* \geq q_*/(2L)$. For enlarged radii exceeding one, the same bound follows from the total mass bound.

Let $d(p)$ be the number of selected representative tests satisfied by $p = (v, w)$. Comparability of the slope weights gives

$$\int \frac{d(p)}{n} d\mu(p) \geq \frac{H_*}{2a_*} \geq \frac{q_*}{4L}.$$

Since $0 \leq d(p)/n \leq 1$ and $\mu(\mathbb{R}^2) \leq 1$, the set $G_0 = \{p : d(p) \geq q_* n/(8L)\}$ has mass at least $q_*/(8L)$. Pigeonhole its positive integer degrees into at most L dyadic levels. This gives a Borel set $G \subset G_0$ and an integer $M \geq 1$ such that

$$\mu(G) \geq \frac{q_*}{8L^2}, \quad M \leq d(p) < 2M \quad (p \in G), \quad M \geq \frac{q_* n}{16L}. \quad (9.4)$$

Partition the point square into dyadic ϱ -squares and give Q the weight $\mu(G \cap Q)$. There are at most $C\delta^{-2}$ squares. Discard weights below δ^4 ; their total is at most $C\delta^2 \leq q_*/(16L^2)$. A dyadic weight pigeonhole gives a family $\mathcal{P}$ of squares with comparable weights and total retained mass

$$b_* \geq \frac{q_*}{16L^3}. \quad (9.5)$$

For each $Q \in \mathcal{P}$ choose $p_Q \in G \cap Q$. Comparison between counting and these comparable cell weights shows that $\mathcal{P}$ has relative t -Frostman constant

$$CL^3\delta^{-3\kappa} \quad (9.6)$$

down to ϱ : passage to relative counts costs at most $2/b_*$, and the original measure bounds the mass of a fixed enlargement of each testing ball.

Here a dyadic ϱ -tube is the union of lines $w = av + b$ with (a, b) in a fixed dyadic parameter square of side ϱ . For each selected slope representative a_I and each $b' \in J_{\lambda_I}$, include the tube whose parameter square contains (a_I, b') . Call the resulting family $\mathcal{T}$. Every J_{λ_I} meets at most $C\delta^{-u}$ intercept intervals, so

$$|\mathcal{T}| \leq Cn\delta^{-u}. \quad (9.7)$$

For $Q \in \mathcal{P}$ and each representative test satisfied by $p_Q = (v_Q, w_Q)$, select the tube with parameter cell containing $(a_I, w_Q - a_I v_Q)$. This tube intersects Q . Different slope cells give different tubes. Denote this fiber by $\mathcal{T}(Q)$; then $M \leq |\mathcal{T}(Q)| < 2M$.

A ball of radius r in tube-parameter space restricts its slope coordinate to an interval of comparable radius. By (9.3) and (9.4),

$$\frac{\#\{\text{parameter cells of } \mathcal{T}(Q) \text{ meeting } B(z, r)\}}{|\mathcal{T}(Q)|} \leq CL^2 \delta^{-5\kappa} r^s \quad (\varrho \leq r \leq 1). \quad (9.8)$$

Thus the base cells and selected incident fibers form exactly the quantitative configuration of Ren and Wang [17, Definition 3.1 and Theorem 4.1]. The selected fiber cardinalities lie between M and $2M$, as permitted by the cited theorem. For sufficiently small δ , the constants in (9.6) and (9.8) are at most $\varrho^{-\eta_{\text{pr}}}$. In detail, arrange $CL^3 \leq \delta^{-\kappa}$ and use $6\kappa \leq \eta_{\text{pr}}/2$ together with $\varrho < 2\delta$.

The cited theorem now gives

$$|\mathcal{T}| \geq c_\varepsilon M \varrho^{-h_{\text{pr}} + \varepsilon} \geq \frac{c'_\varepsilon}{16L} n \delta^{-h_{\text{pr}} + \varepsilon + 2\kappa}.$$

Combining this with (9.7) yields

$$1 \geq \frac{c''_\varepsilon}{L} \delta^{-(\Delta - \varepsilon - 2\kappa)}.$$

But (9.1) implies $\Delta - \varepsilon - 2\kappa \geq 5\Delta/8 > 0$, so the right side tends to infinity. This proves (8.21). Finally, (8.22) would make the product hit probability at least $\nu(B)\delta^\kappa \geq \delta^{2\kappa}$, a contradiction. $\square$

Remark 9.1 (Measures, labels, and fixed charts). The point measure need not be atomic: finite point families appear only after restricted cell masses are pigeonholed. The statement also holds for a general measurable label space if the slope map and the target incidence set in the product of label and point spaces are jointly measurable. In each of the finitely many slope cells, choose a label whose hit probability is at least half the conditional average. This replaces (9.2) by the same inequality with a factor $1/2$, absorbed by the constants in the proof.

A fixed finite collection of bounded projection charts is permitted: a violating product hit probability must have a fixed fraction in one chart, and restrictions are still subprobability measures. Fixed affine coordinate changes, label-dependent translations of the target, and scalar normalizations bounded above and away from zero also only alter fixed constants and the small-scale threshold.

At a local scale $\delta = (\delta')^{h_1}$, an inherited bound $(\delta')^{-a} r^t$ fits (8.20) when $a \leq \kappa h_1$. This explicitly determines how small losses must be chosen on each fixed finite depth ladder. If a measure to be tested is bounded above by δ^{-b} times the independent product, where $b < 2\kappa$, its hit probability is at most $\delta^{2\kappa - b}$. No independence of a preceding coupled pair measure is asserted.

9.2 Constructing the product-pinning law

We now prove Lemma 8.4. There are two possibilities. If a thin line neighborhood carries substantial mass, the coordinate Frostman bounds keep its line away from the coordinate directions; on that cluster the product map is a nondegenerate quadratic in time. Otherwise, avoidance of line

neighborhoods gives an initial positive ray exponent. Repeated planar projection estimates improve that exponent toward the dimension of the parameter measure.

This ray-improvement argument adapts the projection bootstrap of Shmerkin and Wang [18], developed further by Orponen et al. [16]. We prove the finite-scale form required here, including the product map and bounds for both single-event marginals.

We use unoriented directions, with their usual angular metric, and write

$$R_\xi(v) = [v - \xi] \in \mathbb{P}^1(\mathbb{R}) \quad (v \neq \xi).$$

The next lemma supplies the nonlinear part of the proof. Absolute section masses, rather than normalized conditional probabilities, make the symmetry and deletion estimates transparent. The ray improvement and ensuing pinning iteration are inspired by the thin-tube bootstrap of Orponen–Shmerkin–Wang [16, Section 1.3, Lemma 2.8 and Corollary 2.18]. The regular finite-scale and product-pinning arguments are proved here, using the separately credited Ren–Wang projection input; these interfaces are not invoked as consequences of the cited bootstrap.

Lemma 9.2 (Improving ray sections). *Let $0 < s \leq 1$ and $\delta \rightarrow 0$. For each δ , let μ be a finitely supported probability on a bounded square such that every occupied square in a fixed nested dyadic grid of side comparable to δ^u has mass $\delta^{su+o(1)}$, uniformly for $0 \leq u \leq 1$. Suppose $G \subset \text{supp } \mu \times \text{supp } \mu$ is symmetric, $\mu^2(G) = 1 - o(1)$, and, for some $0 < \sigma < s$,*

$$\mu\{v : (\xi, v) \in G, R_\xi(v) \in B_r\} \leq \delta^{-o(1)} r^\sigma \quad (\delta \leq r \leq 1) \quad (9.9)$$

for every anchor ξ and every direction ball B_r . For every

$$0 < \sigma_* < \min\{s, (s + \sigma)/2\},$$

one can restrict to a symmetric relation $G_ \subset G$ of probability $1 - o(1)$ on which (9.9) holds with σ_* in place of σ . One can instead obtain the same absolute section bound for $\mathbf{p}_\xi(v)$, with intervals in place of direction balls. All assertions are uniform for ensembles satisfying the hypotheses uniformly.*

Proof. We localize the second point to a small square and linearize the output map there. Symmetry supplies the reverse rays from the second point to the anchors; those rays provide the directions for the planar projection estimate. A joint pruning then makes the local estimates iterable over a finite ladder of depths.

Choose

$$\sigma_* < \sigma_m < \min\{s, (s + \sigma)/2\}. \quad (9.10)$$

We first prove the bound at a fixed depth $T \in (0, 1]$. Choose a small starting depth $d_0 > 0$ and a finite ladder

$$d_0 < d_1 < \cdots < d_n = T, \quad d_{j+1} < 2d_j. \quad (9.11)$$

The choices of d_0 and of a further small parameter $\lambda > 0$ will be specified below. Delete pairs of distance less than δ^λ . Their probability is at most $\delta^{s\lambda-o(1)}$, by the square-mass upper bounds.

A local projection estimate. Consider one step $d_1 \rightarrow d_2$ of the ladder, and put $h_1 = d_2 - d_1$. Choose λ small enough that

$$d_1 - \lambda > h_1, \quad 2d_1 - 2\lambda > d_2. \quad (9.12)$$

Let J be an occupied square of side δ^{d_1} and let $\mu_J = \mu|_J / \mu(J)$, rescaled to a bounded unit square. For $0 \leq u \leq h_1$, the ratio of child and parent masses is $\delta^{su+o(1)}$. A bounded square covering therefore makes μ_J an s -Frostman probability through resolution

$$\delta_J = \delta^{h_1}.$$

This uses the lower bound on $\mu(J)$ as well as the upper bound on its children.

For a fixed $v \in J$, consider the anchors ξ paired with v by the old relation and satisfying the separation condition. Symmetry and (9.9) give an absolute σ -Frostman bound on their reverse rays. Replacing v by the center of J changes the direction by $O(\delta^{d_1-\lambda})$. By (9.12), this is smaller than δ_J . The same absolute bound thus holds for the directions perpendicular to the rays from these anchors to the center of J .

Apply Lemma 8.6 at resolution δ_J , with point exponent s , direction exponent σ , and target exponent σ_m . Fix its positive power allowance κ . A reverse section of mass less than $\delta_J^{\kappa/4}$ may be discarded at total pair cost at most $\delta_J^{\kappa/4}$. For larger sections, normalization increases the Frostman constant by at most $\delta_J^{-\kappa/4}$. All inherited subpower constants are smaller than $\delta_J^{-\kappa/4}$ for sufficiently small δ , since h_1 is fixed. Decreasing κ if necessary leaves the loss required by the projection lemma.

For each direction, call a projection bin of width δ_J heavy if its μ_J -mass exceeds $\delta_J^{\sigma_m}$. The heavy bins number at most $O(\delta_J^{-\sigma_m})$. The robust projection lemma implies that the directions for which their union has mass at least δ_J^κ occupy at most a fixed positive power of δ_J in every normalized reverse section. One can use the product bound of that lemma and Markov's inequality; shrinking κ accommodates constants. Call these directions exceptional for J .

There are two different removals here. First remove pairs whose anchor direction is exceptional for the cell containing the other point. For each fixed v , its reverse-anchor mass is power-small; integration against $\mu(dv)$ gives a power-small total loss. Second, for each nonexceptional (ξ, J) , remove the points in J lying in heavy projection bins. Their *unconditional* μ -mass is at most $\delta_J^\kappa \mu(J)$. Integrating over ξ and summing over the disjoint cells J again gives a power-small loss. Exceptionality is a test of J and the anchor direction, independent of the point within J . Neither removal assumes independence on the old relation G .

For the ray map, its derivative on J is a perpendicular scalar projection multiplied by the inverse distance to the anchor. That multiplier is bounded below by a positive constant, since the ambient square is bounded. Its quadratic error is $O(\delta^{2d_1-2\lambda})$. Consequently (9.12) implies the following local bound on the surviving section:

$$\mu\{v \in J : (\xi, v) \text{ survives}, R_\xi(v) \in B_{\delta^{d_2}}\} \leq C\delta^{\sigma_m h_1} \mu(J). \quad (9.13)$$

Finite angular charts and overlapping projection grids only change C .

For the product map, the derivative is

$$\nabla_v \mathbf{p}_\xi(v) = (\vartheta_v - \vartheta_\xi, t_v - t_\xi). \quad (9.14)$$

Its direction is the swapped ray, whose direction law has the same Frostman exponent. Thus the product test still uses the old ray bound as its direction input; it does not require a previous product bound. Its size is at least δ^λ . An output interval of width δ^{d_2} consequently uses at most $\delta^{-O(\lambda)}$ projection bins at resolution δ_J . The quadratic error is $O(\delta^{2d_1})$. Thus the product version of (9.13) has right side $C\delta^{\sigma_m h_1 - C\lambda} \mu(J)$. In what follows this slightly weaker bound works for both maps.

Pruning and iteration. Perform these removals for every step in the finite union of tested ladders, also with the legs interchanged. After all these removals, perform one joint pruning process over every tested level and both legs: if, for an anchor and a tested cell J , the remaining nonempty section has mass less than $\delta^\lambda \mu(J)$, delete it. Continue with either leg until no deficient section remains at any tested level. Each anchor–cell combination is charged at most once, because it is empty after its deletion. Integrating over anchors and summing over J bounds the charge by δ^λ per tested level and per leg, even if the deletions cause a cascade. There are finitely many levels,

so the total loss tends to zero. All operations are finite on the finitely supported measures. Any subrelation satisfying every section threshold survives each deletion, so the terminal relation is the unique largest subrelation with those thresholds. Interchanging the legs preserves both the initial relation and the tests; hence the terminal relation is symmetric.

For the final relation, let $Q(d)$ be the largest absolute section mass in an output ball of radius δ^d . The retained mass in a nonempty anchor section of J is now at least $\delta^\lambda \mu(J)$. The local estimate divided by this lower bound is a factor

$$C\delta^{\sigma_m h_1 - C_1 \lambda}$$

times that section mass. If the section has an output in a chosen fine ball, all its outputs lie in an enlarged parent ball of radius $O(\delta^{d_1 - \lambda})$. That ball is coverable by $\delta^{-O(\lambda)}$ balls of radius δ^{d_1} . The cells are disjoint, so summing their section masses gives the recurrence

$$Q(d_2) \leq C\delta^{\sigma_m(d_2 - d_1) - C_2 \lambda} Q(d_1). \quad (9.15)$$

Restrictions made at other levels preserve every upper bound.

Starting with $Q(d_0) \leq 1$, the ladder gives

$$Q(T) \leq C^n \delta^{\sigma_m(T - d_0) - C_2 n \lambda - o(1)}.$$

First choose d_0 so that $\sigma_m d_0 < (\sigma_m - \sigma_*)T/4$. After fixing the ladder, choose λ so that $C_2 n \lambda < (\sigma_m - \sigma_*)T/4$ and all the strict inequalities (9.12) hold. The positive remaining margin absorbs C^n and the subpower errors. This proves $Q(T) \leq \delta^{\sigma_* T}$ for small δ .

The same construction works for finitely many target depths at once: take the union of their ladders and choose all loss parameters below the minimum of their finitely many positive allowances. To obtain all scales, take successively finer finite rational meshes of target depths. For each fixed mesh first make δ small enough that its finite collection of projection and deletion bounds holds and its total deleted mass is as small as prescribed. Then let the mesh refine sufficiently slowly. Between tested depths monotonicity costs at most $\delta^{-\sigma_* e}$, where e is the mesh spacing; at depths below the first mesh point the trivial mass bound has the same cost. Since $e \rightarrow 0$, this gives the claimed subpower constant at all radii. This order never compares an uncontrolled error with a shrinking ladder step. All bounds used above are uniform for the stipulated ensembles. $\square$

Proof of Lemma 8.4. It is enough to prove the assertion for small $\epsilon < s/10$; we reserve a fixed fraction of ϵ for each normalization. Choose a small constant $\eta_0 > 0$, depending on s, γ', ϵ , to be fixed in the first case below.

A line cluster. Suppose first that, for some fixed rational $\zeta \in (0, \zeta_{\max}]$, a set of initial indices of probability bounded below has a line neighborhood of width $\delta = D^{-\zeta}$ carrying mass at least δ^{η_0} . Pass to that subsequence, restrict the initial index there, and choose one such cluster C_i on each index. The normalization of the initial-index law costs a fixed factor.

Let (v_t, v_ϑ) be a unit vector parallel to its line. The time and normal-coordinate ranges of the cluster have lengths at most $C(|v_t| + \delta)$ and $C(|v_\vartheta| + \delta)$. The coordinate bounds and its mass lower bound imply

$$|v_t| \geq \delta^{C\eta_0}, \quad |v_\vartheta| \geq \delta^{C\eta_0}$$

for small δ , with C depending only on the fixed exponents. Thus its slope $k_i = v_\vartheta/v_t$ satisfies

$$\delta^{C\eta_0} \leq |k_i| \leq \delta^{-C\eta_0}, \quad \vartheta = k_i t + b_i + O(\delta^{1-C\eta_0}) \quad \text{on } C_i. \quad (9.16)$$

Here the subpower constants have been absorbed by slightly enlarging C . Choose K with $Ks > 3$, and require $K\eta_0 < 1$ in the standing choice of η_0 . Sample two points from the normalized cluster, restricting to

$$|t_v - t_\xi| \geq \delta^{K\eta_0}.$$

For every anchor the rejected conditional mass is at most $\delta^{(Ks-1)\eta_0-o(1)} = o(1)$. The pair restriction can be normalized symmetrically. Both marginals are bounded by $(1 - o(1))^{-1}\mu_i/\mu_i(C_i)$.

On these pairs,

$$\mathbf{p}_\xi(v) = k_i(t_v - t_\xi)^2 + O(\delta^{1-C\eta_0}).$$

The quadratic is monotone on each separated branch and its derivative there has magnitude at least $\delta^{C'\eta_0}$. For every $r \geq \delta$, the preimage of an output interval of radius r is contained in at most two time intervals of total length

$$C(r + \delta^{1-C\eta_0})\delta^{-C'\eta_0} \leq Cr\delta^{-C''\eta_0}.$$

Using the time Frostman bound before normalizing the cluster proves

$$\mathbb{P}\{\mathbf{p}_\xi(v) \in J_r \mid i, \xi\} \leq C\delta^{-C'''\eta_0-o(1)}r^s \quad (\delta \leq r \leq 1). \quad (9.17)$$

The estimate includes $r = \delta$: the larger approximation error costs the displayed small power, which has not been suppressed. Fix η_0 sufficiently small compared with ϵ and all the constants above, and then take D large. Equation (9.17) implies (8.12); the two marginal losses, including the initial-index restriction, are at most $D^{\epsilon\zeta}$.

No line cluster. In the other case, the proportion of indices with such a cluster tends to zero at every fixed rational depth. Testing finite depth meshes and discarding their exceptional indices, and only then refining the meshes, gives uniformly on the retained indices

$$\mu_i(\{\text{dist}(\cdot, L) \leq r\}) \leq D^{o(1)}r^{\eta_0} \quad (D^{-\zeta_{\max}} \leq r \leq 1) \quad (9.18)$$

for all lines L . To verify the interpolation, replace a radius by the next larger tested radius; if the depth mesh has spacing e , the loss is at most $D^{\eta_0 e}$. The discarded index probability can tend to zero while $e \rightarrow 0$, since every fixed finite test has exceptional probability tending to zero.

Fix any rational $0 < \zeta \leq \zeta_{\max}$ and put $\delta = D^{-\zeta}$. A ray-direction ball through an anchor is contained in a line neighborhood of comparable width. Equation (9.18) therefore supplies (9.9) with a positive exponent, decreasing η_0 if necessary to make it smaller than s . Start with all distinct pairs. Their probability tends to one, because the mass of any point is at most that of its square at resolution δ , which is $\delta^{s-o(1)}$.

Repeatedly apply Lemma 9.2. For example, choosing the next exponent halfway between the old one and $\min\{s, (s + \sigma)/2\}$ decreases its distance to s by a fixed factor. Finitely many iterations give a ray exponent as close to s as desired. Apply the product version once more to obtain an absolute product-section exponent larger than $s - \epsilon/2$, with subpower constant, on a symmetric relation of probability $1 - o(1)$.

Restrict first to anchors whose surviving section has mass at least $1/2$, and then normalize on their sections. The discarded anchor mass tends to zero. The first marginal is bounded by a constant times μ_i . The second marginal is also so bounded: its density is an integral of section densities bounded by 2, against a first-anchor law bounded by a constant times μ_i . The product-section bound survives these normalizations. Uniformity makes the same construction possible on every retained initial index; the initial-index normalization is $1 + o(1)$. All fixed and subpower constants fit inside the reserved $D^{\epsilon\zeta}$ allowance for sufficiently large D . This proves the lemma and its independence from subsequent offset tests. $\square$

The two auxiliary inputs are now proved. Together with the same-index information argument, they establish all the constraints in Proposition 8.1. We turn to their incompatibility with the stationary entropy demand.

10 Closing the contradiction and the maximal estimate

The stationary packets require the limiting adjusted rate $\mathcal{M} - w(g - d)$ to be at least $\tau + B_*$. We show that the projection constraints are incompatible with this demand and the bounds for g on thin offset stripes. The strong compactness of v_1 allows us to use the constraints on the sets where its limiting value is positive. This excludes a positive critical exponent. We then pass from the critical inequality to the maximal estimate, retaining the shading-density power through the reduction to arbitrary L^3 functions.

10.1 Excluding the stationary rates

Proposition 10.1 (Vanishing at the maximal endpoint). *There are values $p > 2$ arbitrarily close to 2 for which $h(p, 0) = 0$.*

Proof. Suppose otherwise. Choose an open interval I of p -values, with lower endpoint $p_{\min} > 2$, on which $h(p, 0) > 0$. Fix the smoothing parameter before making any of the subsequent selections, with

$$0 < \eta \leq \min \left\{ \frac{1}{32}, \frac{p_{\min} - 2}{16} \right\}. \quad (10.1)$$

This choice is legitimate because the critical inequality at $q = 0$ is independent of η . Lemma 2.10 supplies an interior differentiability point with $h > 0$ and $h_z > 0$. The construction in Proposition 5.4, Proposition 5.6, and Lemma 6.2 then supplies fixed parameters satisfying

$$p > 2, \quad 0 \leq q < 1, \quad d = 2 - q, \quad 0 < h \leq 1 - q, \quad x = 1 - q - h < 1, \quad (10.2)$$

$$0 < \tau < 1, \quad \ell = 1 - \tau, \quad 0 < \beta < 1, \quad s_0 = 1 - \beta, \quad (10.3)$$

$$B_* = h + \ell(p\beta - \mathcal{D}(q, \beta)), \quad w = h_z + \ell \mathcal{D}_q(q, \beta), \quad 0 < w \leq 1. \quad (10.4)$$

In particular, Lemma 6.2 supplies the strict inequality $s_0 > 0$.

Recall that g is the total matrix entropy rate and $\mathcal{M}$ its speed-weighted rate from (7.13). For brevity in this proof put

$$b_+ = (\beta - q)_+, \quad T_* = \tau + B_*, \quad \mathcal{Q} = \mathcal{M} - w(g - d), \quad (10.5)$$

where $r_+ = \max\{r, 0\}$. The smoothing definition in (2.4) gives

$$0 \leq \mathcal{D}(q, \beta) \leq \min\{q, \beta/2\}. \quad (10.6)$$

If $q < (1/2 - \eta)\beta$, then the integrand defining $\mathcal{D}$ equals q throughout its averaging interval. Consequently

$$\mathcal{D}(q, \beta) = q, \quad \mathcal{D}_q(q, \beta) = 1, \quad w > \ell. \quad (10.7)$$

We record precisely the limiting statements to be used. Fix a compact interval $J \subset (0, 1)$. Proposition 7.4 and Lemma 7.6 imply that every weak-star limit of $\mathcal{Q}(\cdot, a)$, along generic offsets $a \rightarrow 0$, is at least T_* almost everywhere on J . The same lower bound holds for every weak-star

limit of stripe averages. Moreover, the right stripe averages of g are at least d , and the left stripe averages are at most d , whenever the latter offsets are available. In formulas,

$$\frac{1}{\epsilon} \int_0^\epsilon g(c, a) da \geq d, \quad \frac{1}{\epsilon} \int_{-\epsilon}^0 g(c, a) da \leq d \quad (10.8)$$

for almost every $c \in J$ and sufficiently small $\epsilon > 0$, on the sides being used. These inequalities may be integrated against every nonnegative L^1 test function supported on J . The rates are uniformly bounded, so weak-limit inequalities initially established with smooth tests extend to such tests by L^1 approximation. We will use this extension to localize on measurable subsets of J .

For $w < 1$, Lemma 7.5 gives strong one-sided $L^1(J)$ traces for v_1 through a fixed full-measure set of generic offsets. The corresponding stripe averages converge to the same trace. For $w = 1$, that lemma gives strong subsequential $L^1(J)$ compactness along every sequence of generic offsets. We use this strong convergence to localize the constraints that hold where $v_1 > 0$; weak-star convergence suffices for the other rates.

By Corollary 7.3, on the right we also have

$$a_1 \leq x = 1 - q - h < 1. \quad (10.9)$$

It follows in particular that

$$(a_1 - s_0)_+ \leq (\beta - q - h)_+ \leq b_+. \quad (10.10)$$

The domain $\mathcal{U}$ in (7.2) contains small offsets of both signs uniformly on J when $w < 1$, and small positive offsets when $w = 1$. Thus Proposition 8.1 applies on every stripe used below. The bound (10.9) is available on the right.

The choice of stripe follows from the identity

$$\mathcal{Q} = (\mathcal{M} - rg) + rd + (r - w)(g - d), \quad r \in \mathbb{R}.$$

After averaging, an upper bound for $\mathcal{M} - rg$ gives an upper bound for $\mathcal{Q}$ on the right when $r \leq w$, and on the left when $r \geq w$. For distinct speeds the projection inequalities pair a_1 with y_1 and u_1 with v_1 , making $r = \tau/2$ useful. Direct bounds for $\mathcal{M}$, corresponding to $r = 0$, handle two further regimes. Table 2 records the five cases. In the last case, compactness first gives a gap common to all weak limits of generic slices; averaging then gives the stripe contradiction.

Speeds	Parameter regime	Stripe	Main comparison
Distinct	$w \geq \tau/2$	Right	$\mathcal{M} - \tau g/2$
Distinct	$w < \tau/2, \tau < 2\ell$	Right	$\mathcal{M}$
Distinct	$w < \tau/2, \tau \geq 2\ell$	Left	$\mathcal{M} - \tau g/2$, trace of v_1
Tied	$q \geq (1/2 - \eta)\beta$	Right	$\mathcal{M}$
Tied	$q < (1/2 - \eta)\beta$	Right	$\mathcal{M} - g/2$, compactness of v_1

Table 2: The comparisons used to exclude the stationary rates. The stripe bounds for g enter only after averaging.

Distinct speeds, $w \geq \tau/2$. Suppose first that $\tau \neq \ell$. In coordinate notation,

$$g = a_1 + y_1 + u_1 + v_1, \quad \mathcal{M} = \tau y_1 + \ell u_1 + v_1.$$

On the right, Proposition 8.1 and (10.10) give $a_1 \geq y_1$, $u_1 \geq v_1$, and $u_1, v_1 \leq b_+$. Thus

$$\mathcal{M} - \frac{\tau}{2}g = -\frac{\tau}{2}(a_1 - y_1) + \left(\ell - \frac{\tau}{2}\right)u_1 + \left(1 - \frac{\tau}{2}\right)v_1 \leq 2\ell b_+. \quad (10.11)$$

If $\ell - \tau/2 \geq 0$, bound both last rates by b_+ . If $\ell - \tau/2 < 0$, first use $u_1 \geq v_1$; the resulting coefficient of v_1 is $\ell + 1 - \tau = 2\ell > 0$. This verifies (10.11) for either ordering of the middle speeds. Since $\tau/2 - w \leq 0$, the right stripe inequality in (10.8) implies that every upper stripe limit of $\mathcal{Q}$ is at most

$$U_1 = \frac{\tau}{2}d + 2\ell b_+.$$

Its gap below the required lower limit is

$$T_* - U_1 = h + \frac{\tau q}{2} + \ell(p\beta - \mathcal{D}(q, \beta) - 2b_+).$$

For $q < \beta$, this is at least $h + \tau q/2 + \ell((p-2)\beta + q) > 0$. For $q \geq \beta$, it is at least $h + \tau q/2 + \ell(p-1/2)\beta > 0$. Either possibility contradicts the lower stripe limit T_* .

Distinct speeds, $w < \tau/2$ and $\tau < 2\ell$. Here $\ell > 1/3$. If $q < (1/2 - \eta)\beta$, (10.7) would give $w > \ell > \tau/2$. We must therefore have $q \geq (1/2 - \eta)\beta$. On the right, $y_1 \leq a_1 \leq 1 - q$ and $u_1, v_1 \leq b_+$, so

$$\mathcal{M} \leq U_2 := \tau(1 - q) + (1 + \ell)b_+.$$

The right stripe bound and $w > 0$ make U_2 an upper stripe limit for $\mathcal{Q}$ as well. If $q < \beta$, subtraction gives

$$\begin{aligned} T_* - U_2 &= h + (\ell(p-1) - 1)\beta - \ell\mathcal{D}(q, \beta) + 2q \\ &\geq h + (\ell(p-3/2) - 2\eta)\beta > 0. \end{aligned}$$

The last inequality follows from $\ell > 1/3$, $p > 2$, and $\eta \leq 1/32$; its coefficient is greater than $1/6 - 1/16 > 0$. If $q \geq \beta$, the gap is at least $h + \tau q + \ell(p-1/2)\beta > 0$. This again contradicts the demand.

Distinct speeds, $w < \tau/2$ and $\tau \geq 2\ell$. Use the left stripe. Since $w < \tau/2 < 1/2$, its offsets are available and v_1 has a strong left trace, denoted by v_* . Take a weak-star limit $\bar{g}$ of left stripe averages. The projection constraints imply, wherever $v_1 > 0$,

$$g \geq 2 \min\{1, s_0 + v_1\} + 2v_1. \quad (10.12)$$

We give the localization estimate, since convergence of the averaged rate alone would not suffice here. The full generic-slice trace implies

$$R_\epsilon := \frac{1}{\epsilon} \int_{-\epsilon}^0 \|v_1(\cdot, a) - v_*\|_{L^1(J)} da \longrightarrow 0. \quad (10.13)$$

Indeed the integrand tends to zero through a fixed full-measure set of offsets, so its essential supremum over $(-\epsilon, 0)$ tends to zero. For $E_\delta = \{c \in J : v_*(c) \geq \delta\}$, where $\delta > 0$, let $B_a = E_\delta \cap \{c : v_1(c, a) = 0\}$. Then

$$|B_a| \leq \delta^{-1} \|v_1(\cdot, a) - v_*\|_{L^1(J)}. \quad (10.14)$$

The function $f_0(t) = 2 \min\{1, s_0 + t\} + 2t$, for $0 \leq t \leq 1$, is bounded by 4 and has Lipschitz constant at most 4. For every bounded nonnegative test φ supported on E_δ , (10.12) and $g \geq 0$ consequently give

$$\int_J \varphi(c) \frac{1}{\epsilon} \int_{-\epsilon}^0 g(c, a) da dc \geq \int_J \varphi(c) f_0(v_*(c)) dc - 4\|\varphi\|_\infty (1 + \delta^{-1}) R_\epsilon.$$

Take the stripe weak-star limit, then let $\delta = 1/m \downarrow 0$. The resulting localized inequality is

$$\bar{g} \geq 2 \min\{1, s_0 + v_*\} + 2v_* \quad \text{almost everywhere on } \{v_* > 0\}.$$

On the other hand, $\bar{g} \leq d \leq 2$. If $s_0 + v_* \geq 1$, the displayed lower bound is greater than 2. Otherwise it is $2 - 2\beta + 4v_*$. It follows that

$$v_* \leq \beta/2 \quad \text{almost everywhere,} \tag{10.15}$$

including the set where the trace vanishes.

Because $\ell - \tau/2 \leq 0$, the inequalities $a_1 \geq y_1$ and $u_1 \geq v_1$ also imply

$$\mathcal{M} - \frac{\tau}{2}g \leq 2\ell v_1.$$

Now $\tau/2 - w > 0$. Combining the left stripe bound for g with (10.15), every upper stripe limit of $\mathcal{Q}$ is bounded by

$$U_3 = \frac{\tau}{2}d + \ell\beta.$$

But

$$T_* - U_3 = h + \frac{\tau q}{2} + \ell((p-1)\beta - \mathcal{D}(q, \beta)) \geq h + \frac{\tau q}{2} + \ell(p-3/2)\beta > 0.$$

This excludes the last distinct-speed case, including $\tau = 2\ell$.

Tied speeds with $q \geq (1/2 - \eta)\beta$. It remains to consider $\tau = \ell = 1/2$. Write m_1 for the middle-group rate. Then

$$g = a_1 + m_1 + v_1, \quad \mathcal{M} = m_1/2 + v_1.$$

On the right, Proposition 8.1 yields $m_1 \leq 1 + b_+$ and $v_1 \leq b_+$. Therefore

$$\mathcal{M} \leq U_4 := \frac{1}{2} + \frac{3}{2}b_+.$$

This is again an upper stripe limit for $\mathcal{Q}$. For $q < \beta$, its gap below demand satisfies

$$T_* - U_4 \geq h + \left(\frac{p-2}{2} - \frac{3\eta}{2}\right)\beta > 0.$$

Indeed (10.1) bounds the coefficient below by $13(p-2)/32$. For $q \geq \beta$, the gap is at least $h + (p-1/2)\beta/2 > 0$. This case is impossible as well. In particular, the same η fixed before the canonical construction works independently of the subsequently selected h, τ, ℓ, β .

Tied speeds with $q < (1/2 - \eta)\beta$. Here $\mathcal{D}(q, \beta) = q$ and $w > 1/2$. Put

$$U_0 = \frac{1}{2} + \frac{\beta - q}{2}, \quad K_0 = T_* - U_0 = h + \frac{p-1}{2}\beta > 0. \quad (10.16)$$

Take any sequence of generic positive offsets tending to zero. On a fixed compact interior interval J , extract a subsequence for which $v_1 \rightarrow v_*$ strongly in $L^1(J)$ and the other bounded rates converge weak-star. This is allowed also when $w = 1$. Denote their limits by bars. The demand gives $\bar{Q} \geq T_*$.

On $E_\delta = \{v_* \geq \delta\}$, use (10.14) along the selected sequence of slices: the zero-set measure is at most $\delta^{-1}\|v_{1,n} - v_*\|_1 \rightarrow 0$. Since $a_1 < 1$ on the right, positive v_1 and the projection constraint $v_1 \leq (a_1 - s_0)_+$ force $a_1 \geq s_0 + v_1$. Thus

$$\mathcal{M} - g/2 = (v_1 - a_1)/2 \leq -s_0/2$$

on the part with positive v_1 . Explicitly, for bounded $\varphi \geq 0$ supported on E_δ ,

$$\int_J \varphi(a_{1,n} - v_{1,n} - s_0) \geq -s_0\|\varphi\|_\infty \delta^{-1}\|v_{1,n} - v_*\|_1 \rightarrow 0 \quad \text{from below.}$$

Weak-star convergence of $a_{1,n}$, strong convergence of $v_{1,n}$, and then $\delta = 1/m \downarrow 0$ justify the inequality on $\{v_* > 0\}$. Passing to this localized limit gives

$$\begin{aligned} T_* \leq \bar{Q} &\leq \frac{d}{2} - \frac{s_0}{2} + (w - 1/2)(d - \bar{g}) \\ &= U_0 + (w - 1/2)(d - \bar{g}). \end{aligned}$$

Consequently $\bar{g} \leq d - K_0/(w - 1/2)$ on $\{v_* > 0\}$. On $\{v_* = 0\}$, the bounds $m_1 \leq 1 + \beta - q$ and strong convergence of v_1 give $\bar{\mathcal{M}} \leq U_0$. The same demand then gives $T_* \leq U_0 + w(d - \bar{g})$, and hence $\bar{g} \leq d - K_0/w$. Since $w - 1/2 < w$, the two parts imply the common bound

$$\bar{g} \leq d - \frac{K_0}{w} \quad \text{almost everywhere on } J. \quad (10.17)$$

Its positive gap depends only on the fixed parameters, not on the chosen sequence or the subsequential trace.

We now convert the common gap in (10.17), obtained from arbitrary offset sequences, to a stripe contradiction. Fix a nonnegative test $\varphi \in L^1(J)$ with positive integral. If arbitrarily small generic positive offsets satisfied

$$\int_J \varphi(c)g(c, a) dc > \left(d - \frac{K_0}{2w}\right) \int_J \varphi(c) dc,$$

choose a sequence of them tending to zero. The preceding compactness argument would give a weak-star limit violating (10.17). Thus all sufficiently small generic positive offsets satisfy the reverse inequality. Averaging it over $0 < a < \epsilon$ contradicts the first inequality of (10.8); the exceptional offsets form a null set.

Every possible choice of the stationary parameters has now been excluded. This contradicts the differentiability point obtained under the assumed failure of critical vanishing, and proves the proposition. $\square$

10.2 From the critical inequality to maximal functions

The following reduction uses only the indexed model, its regularization lemma, and the critical inequality at $q = 0$. In particular its proof is independent of the stationary construction and of Proposition 10.1.

Proposition 10.2 (Maximal reduction). *Let $p > 2$ be in the parameter range of Definition 2.3, and suppose that $h(p, 0) = 0$. For every $\kappa > 0$ there is a finite constant $C_{p,\kappa}$ such that, for all $0 < \delta < 1$ and all complex-valued $f \in L^3(\mathbb{R}^3)$,*

$$\|K_\delta f\|_{L^3(S^2, \sigma)} \leq C_{p,\kappa} \delta^{-(p-2)/3-\kappa} \|f\|_{L^3(\mathbb{R}^3)}. \quad (10.18)$$

Proof. We first establish a uniform discrete estimate, then an indicator estimate, and finally (10.18). All geometric constants in this proof are absolute unless their dependence is displayed.

A discrete estimate in one bounded chart. Let $N \geq 2$ be dyadic. Consider unit-weight lines in a bounded matrix chart, with pairwise c/N -separated spatial slopes, where $c > 0$ is a fixed constant. Let A be a finite collection of N^{-1} -cubes. Suppose each of n such lines has at least $c_1 \max\{1, \lambda N\}$ marked time bins whose centerline points lie within C_1/N of a cube in A . Here $0 < \lambda \leq 1$ and c_1, C_1 are fixed constants. Then, for each $\xi > 0$,

$$|A| \geq c_{p,\xi} N^{-\xi} n N \max\{N^{-1}, \lambda\}^{p+1}, \quad (10.19)$$

where $|A|$ is a cell count. Fixed bounded changes of the chart, of the resolution, and of the neighboring-cell allowance only change the constant.

Here is the uniformity argument. For a full-time plank of widths a, b in finest cell units, the slopes of lines contained in it lie in a rotated rectangle with side lengths $O(a/N)$ and $O(b/N)$. This follows by comparing the two endpoint cross-sections of the plank. Since $1 \leq a \leq b \leq N$, slope separation bounds their number by $O(ab)$. Consequently

$$\Delta_0 = \sup_B \frac{n_B}{ab} \leq C \quad (10.20)$$

for the original family and all its subfamilies.

If (10.19) failed uniformly for a fixed $\xi > 0$, choose a violating sequence with $N \rightarrow \infty$. Bounded N cannot cause this failure, because $n \leq CN^2$ and any nonempty A has at least one cell. On each line retain exactly $k \asymp N\theta$ of the allowed bins, where $\theta = \max\{N^{-1}, \lambda\}$; integer rounding is harmless. The union of their event cells lies in a fixed enlargement of A , of cell count at most $C|A|$. Along a subsequence write $\theta = N^{-\alpha+o(1)}$, with $0 \leq \alpha \leq 1$. Apply Lemma 2.6 to this indexed event graph. It retains $nkN^{-o(1)}$ events in weighted count and gives a limiting profile F . The number of events per active index is at most k , whereas retained index mass is at most n . Total incidence retention therefore forces this number to be $kN^{-o(1)}$ in exponent. It follows that $F(1) = \alpha$. The retained graph has raw multiplicity at least

$$\frac{nkN^{-o(1)}}{C|A|} \geq N^{p\alpha+\xi-o(1)}$$

along a sequence violating (10.19). At $q = 0$, the temporal cost of every profile is exactly $\Pi_F(1) = pF(1) = p\alpha$. The critical inequality with $h(p, 0) = 0$ and (10.20) bounds this same multiplicity by $N^{p\alpha+o(1)}$, a contradiction. This proves (10.19) with a constant uniform over all the finite configurations under discussion.

Cylinder geometry and translations. Use an auxiliary family of cylinders of length 4 and radius $4/N$, and let $\mathcal{K}_N$ denote the supremum of their normalized averages over all centers, as a function of direction. Thus $0 \leq \mathcal{K}_N \mathbf{1}_E \leq 1$ for every measurable set E . Let E be a finite union of

cubes of the fixed N^{-1} grid, and suppose one of these cylinders has average of $\mathbf{1}_E$ greater than $\lambda > 0$.

Cover S^2 by finitely many fixed small direction charts. After an axis permutation and, if needed, reversal, the time component in each chart is bounded away from zero. A cylinder in such a chart meets only $O(1)$ fine spatial cubes in each time bin of length comparable to N^{-1} . Its intersection with any one bin has volume $O(N^{-3})$, while its full volume is comparable to N^{-2} . An average greater than λ therefore supplies at least $c\lambda N$ occupied time bins, and at least one if this quantity is smaller than a constant. The centerline point in each such bin is within C/N of an occupied cube of E . These are the events needed in (10.19).

To allow arbitrary translations, group cylinder centers into unit grid cubes Q . Every cylinder in a group lies in a fixed enlargement Q^* , and the regions Q^* have bounded overlap as Q varies. For one group and one direction chart, all centerlines have bounded intercepts and slopes. Extend their lines across one fixed common time interval containing their cylinder traces, then translate and rescale that interval and the spatial chart by fixed factors to fit the indexed model. Only the marked events on the original cylinder are used. A comparable dyadic resolution and bounded neighboring-cell enlargements account for these fixed changes. Thus (10.19) applies to the cubes of E meeting Q^* , with constants independent of the location of Q . Pairwise angular separation in a fixed chart is equivalent, up to fixed constants, to separation of the spatial slopes.

A directional distribution bound. For $0 < \lambda < 1$, let

$$S_\lambda = \{\omega \in S^2 : \mathcal{K}_N \mathbf{1}_E(\omega) > \lambda\}.$$

In each fixed chart, choose a maximal c/N -separated subset of S_λ , of cardinality n , and choose for each selected direction a cylinder witnessing the strict inequality. Such a finite maximal subset exists by the packing bound on the sphere. The C/N -caps around its elements cover the high-value directions in that chart, so their measure is at most CnN^{-2} .

Apply (10.19) to the spatial groups just described and sum over the groups. Their ambient regions overlap a bounded number of times, whence

$$N^3|E| \geq c_{p,\xi} N^{-\xi} n N \max\{N^{-1}, \lambda\}^{p+1},$$

where $|E|$ now denotes Lebesgue volume. Sum over the finitely many direction charts. The cap measure bound gives

$$\sigma(S_\lambda) \leq C_{p,\xi} N^\xi |E| \max\{N^{-1}, \lambda\}^{-(p+1)}. \quad (10.21)$$

In particular, for $N^{-1} \leq \lambda < 1$,

$$\lambda^3 \sigma(S_\lambda) \leq C_{p,\xi} N^{p-2+\xi} |E|. \quad (10.22)$$

The factor N^{p-2} comes from $\lambda^{2-p} \leq N^{p-2}$. Thus the density cost has been retained throughout this reduction.

For a nonempty union of fine grid cubes, $|E| \geq N^{-3}$. The distribution formula, $\sigma(S^2) = 4\pi$, and (10.22) imply

$$\begin{aligned} \|\mathcal{K}_N \mathbf{1}_E\|_3^3 &= 3 \int_0^1 \lambda^2 \sigma(S_\lambda) d\lambda \\ &\leq 4\pi N^{-3} + C_{p,\xi} N^{p-2+\xi} |E| \int_{1/N}^1 \frac{d\lambda}{\lambda} \\ &\leq C_{p,\xi} N^{p-2+\xi} (1 + \log N) |E|. \end{aligned} \quad (10.23)$$

The empty set is immediate. This separately accounts for the entire range $\lambda < N^{-1}$.

Cell averages and general functions. Let $f \in L^3(\mathbb{R}^3)$ be complex-valued. The case $f = 0$ is immediate, and homogeneity reduces the proof to $\|f\|_3 = 1$. Choose a dyadic $N \geq 2$ with $N^{-1} \leq \delta < 2N^{-1}$, and let F_N be the function equal to the average of $|f|$ on each cube of the N^{-1} grid. Hölder's inequality on a cube and Jensen's inequality give

$$0 \leq F_N \leq N, \quad \|F_N\|_3 \leq 1. \quad (10.24)$$

Every grid cube meeting $T_\delta(a, \omega)$ lies in the cylinder with the same center and direction, length 4, and radius $4/N$. Indeed its points are within $\sqrt{3}/N$ of a point of the original cylinder, and $N \geq 2$. Summing the integrals of $|f|$ on all such cubes therefore gives

$$\int_{T_\delta(a, \omega)} |f| \leq \int_{\text{enlarged cylinder}} F_N.$$

The enlarged volume is $64\pi N^{-2}$, while $\pi\delta^2 \geq \pi N^{-2}$. Consequently

$$K_\delta f \leq 64\mathcal{K}_N F_N. \quad (10.25)$$

The part of F_N below N^{-1} contributes at most N^{-1} pointwise to $\mathcal{K}_N F_N$. For its remaining values use the disjoint level sets

$$E_j = \{2^j \leq F_N < 2^{j+1}\}, \quad -\log_2 N \leq j \leq \log_2 N.$$

There are $J_N = 2\log_2 N + 1$ such levels. Each is a finite union of fine grid cubes: its volume is finite by (10.24), and every occupied cube has volume N^{-3} . Sublinearity, the triangle inequality in $L^3(S^2)$, and (10.23) yield

$$\|\mathcal{K}_N F_N\|_3 \leq (4\pi)^{1/3} N^{-1} + 2C_{p, \xi} N^{(p-2+\xi)/3} (1 + \log N)^{1/3} \sum_j 2^j |E_j|^{1/3}.$$

Hölder's inequality for this finite sum gives

$$\sum_j 2^j |E_j|^{1/3} \leq J_N^{2/3} \left(\sum_j 2^{3j} |E_j| \right)^{1/3} \leq J_N^{2/3} \|F_N\|_3 \leq J_N^{2/3}.$$

Together with (10.25), this proves

$$\|K_\delta f\|_3 \leq C_{p, \xi} N^{(p-2+\xi)/3} (1 + \log N).$$

Given $\kappa > 0$, take $\xi = \kappa$ and absorb the logarithm into $C_\kappa N^{2\kappa/3}$. Since $N < 2/\delta$, this is (10.18) for normalized f . Homogeneity gives the general case. Fixed bounded resolutions are included by enlarging the constant; directly, Hölder gives $K_\delta f \leq (\pi\delta^2)^{-1/3} \|f\|_3$, uniformly in direction, when δ is bounded below.

Measurability. For clarity, all suprema used above define measurable functions of direction. If $(a_n, \omega_n) \rightarrow (a, \omega)$, the indicators of the corresponding cylinders converge in $L^{3/2}(\mathbb{R}^3)$: their boundaries have measure zero and their supports lie in one bounded set for a convergent sequence of parameters. Hölder's inequality therefore shows that the integral of $|f|$ over the cylinder is continuous in its center and direction. For each fixed direction, its supremum over all centers equals its supremum over the countable dense set $\mathbb{Q}^3$. This is a countable supremum of continuous functions of direction, hence is measurable. The same reasoning applies to $\mathcal{K}_N$ and to the finite-volume indicator functions used above. It also shows that changing f on a null set changes none of the tube integrals. Thus all distribution and L^3 estimates are valid for the stated L^3 equivalence classes. $\square$

Proof of Theorem 1.1. Fix $\varepsilon > 0$. Proposition 10.1 allows a choice of $p > 2$ with $h(p, 0) = 0$ and $(p - 2)/3 < \varepsilon/2$. Apply Proposition 10.2 with $\kappa = \varepsilon/2$. Since $0 < \delta < 1$,

$$\|K_\delta f\|_{L^3(S^2, \sigma)} \leq C_{p, \varepsilon/2} \delta^{-\varepsilon} \|f\|_{L^3(\mathbb{R}^3)}.$$

Choose this p once as a function of ε and rename the constant C_ε . It is independent of δ , of the complex-valued function f , and of all tube centers and directions, as required. $\square$

11 Nikodym and curved Kakeya consequences

We apply the transfer theorems of Gao, Liu, and Xi to the maximal estimate just proved. We state the local geometry explicitly so that the domains and constant dependence remain visible.

11.1 Nikodym maximal estimates on space forms

Theorem 1.1 also supplies the new input for the established Kakeya-to-Nikodym transfer of Gao, Liu, and Xi [9, Section 2.2, Proposition 2.3]. For a unit line segment $\gamma \subset \mathbb{R}^3$, write $T_\delta^{\text{Euc}}(\gamma) = \{y \in \mathbb{R}^3 : \text{dist}(y, \gamma) < \delta\}$, and define

$$N_\delta^{\text{Euc}} f(x) = \sup_{\substack{\gamma \ni x \\ \gamma \text{ a unit line segment}}} \frac{1}{\delta^2} \int_{T_\delta^{\text{Euc}}(\gamma)} |f(y)| dy.$$

This is the normalization in equation (5) of Gao et al. [9].

For the manifold statement, let (M, g) be a smooth three-dimensional Riemannian manifold of constant sectional curvature. Fix open sets $U \Subset V$ with $\bar{V}$ compact in a local chart that sends geodesics to straight lines. Work at a fixed segment length within this geometry, normalized to one by a fixed rescaling of g . Choose $0 < \delta_0 < 1$ small enough that the δ_0 -neighborhood of every segment contained in U lies in V . For $0 < \delta < \delta_0$, put

$$N_{\delta, U}^g f(x) = \sup_{\substack{\gamma \ni x \\ \gamma \subset U}} \frac{1}{\delta^2} \int_{T_\delta^g(\gamma)} |f(y)| dV_g(y), \quad x \in U,$$

where the supremum is over unit geodesic segments, and $T_\delta^g(\gamma) = \{y \in M : d_g(y, \gamma) < \delta\}$. An empty supremum is zero. This is a fixed local version of the geodesic Nikodym operator in equation (8) of Gao et al. [9].

Corollary 11.1 (Nikodym maximal estimates in dimension three). *For every $1 \leq p \leq 3$ and $\varepsilon > 0$, set $q = 2p'$, with $q = \infty$ at $p = 1$. Then, for $0 < \delta < 1$,*

$$\|N_\delta^{\text{Euc}} f\|_{L^q(\mathbb{R}^3)} \leq C_{p, \varepsilon} \delta^{1-3/p-\varepsilon} \|f\|_{L^p(\mathbb{R}^3)}.$$

For each fixed local geometry above and $0 < \delta < \delta_0$,

$$\|N_{\delta, U}^g f\|_{L^q(U, dV_g)} \leq C_{p, \varepsilon, U, V, g} \delta^{1-3/p-\varepsilon} \|f\|_{L^p(V, dV_g)}.$$

In particular, at $p = q = 3$ the operator norms are bounded by $C_\varepsilon \delta^{-\varepsilon}$ in Euclidean space and $C_{\varepsilon, U, V, g} \delta^{-\varepsilon}$ in the fixed local geometry.

Proof. Write $\widetilde{K}_\delta$ for the Keakeya operator of Gao et al. [9, Section 2.2], normalized by δ^{-2} over δ -neighborhoods of unit segments. For $0 < \delta < 1$, each such neighborhood is contained in the union of three of the cylinders $T_\delta(a, \omega)$ used here. Consequently $\widetilde{K}_\delta f \leq 3\pi K_\delta f$, so Theorem 1.1 gives its L^3 -to- L^3 bound with every positive power loss. Also

$$\|\widetilde{K}_\delta f\|_{L^\infty(S^2)} \leq \delta^{-2} \|f\|_{L^1(\mathbb{R}^3)}.$$

Interpolation for this sublinear operator, with $1/p = 1 - 2\theta/3$ and $1/q = \theta/3$, gives $q = 2p'$ and the power $-2(1 - \theta) - \eta\theta = 1 - 3/p - \eta\theta$ from an L^3 loss $\delta^{-\eta}$. Since $\eta > 0$ is arbitrary, including the two endpoint bounds gives $K(3)$ in the notation of Gao et al. [9, equation (6)].

Their Proposition 2.3 identifies $K(3)$ with the Euclidean Nikodym estimate $N(3)$, and with the geodesic Nikodym estimate on a constant-curvature manifold, in equations (7) and (9) respectively. The negligible exponent loss in that transfer is absorbed by starting with a smaller positive loss above. Its manifold comparison uses a geodesic-straightening diffeomorphism whose bi-Lipschitz and volume comparison constants are bounded on the fixed compact chart. Applying that comparison to $U \subseteq V$ gives the stated local estimate, with constants allowed to depend on this geometry. $\square$

The constants in the manifold estimate depend on the fixed compact straightening chart. The transfer uses the full maximal estimate of Theorem 1.1, including its dependence on arbitrary shading density.

11.2 A local curved Keakeya consequence

The same Euclidean maximal estimate also supplies the input for the curved Keakeya transfer of Gao, Liu, and Xi for the following fixed translation-invariant phase class. Fix a sufficiently small $\rho > 0$, put $\Omega = B_\rho^2 \times (-\rho, \rho)$ and $Y = B_\rho^2$, where B_ρ^2 is the open ball centered at the origin in $\mathbb{R}^2$, and fix a smooth real-valued phase on a neighborhood of $\overline{\Omega} \times \overline{Y}$ of the form

$$\phi(x, t; y) = x \cdot y + \psi(t; y), \quad \psi(0; y) = 0.$$

Assume the Hörmander mixed-rank and curvature nondegeneracy conditions (H1)–(H2) and Bourgain's condition there, in the senses of Gao et al. [9, Definitions 1.7, 1.10, and 1.11]. For this phase, the mixed-rank condition is automatic, while the other two conditions take the concrete form

$$\det \nabla_y^2 \partial_t \psi(t; y) \neq 0, \quad \nabla_y^2 \partial_t^2 \psi(t; y) = c(t; y) \nabla_y^2 \partial_t \psi(t; y)$$

for a scalar function c . The phase and chart are fixed after any shrinking needed for their local theorem. For $\omega, y \in B_\rho^2$ and $0 < \delta < \rho$, use the gradient-defined tubes and normalization of their Definitions 1.8 and 4.5:

$$\begin{aligned} T_{\delta, y}^\phi(\omega) &= \{(x, t) \in \Omega : |\nabla_y \phi(x, t; y) - \nabla_y \phi(\omega, 0; y)| < \delta\}, \\ \mathcal{K}_\delta^\phi f(y) &= \sup_{\omega \in B_\rho^2} \frac{1}{\delta^2} \int_{T_{\delta, y}^\phi(\omega)} |f(x, t)| \, dx \, dt. \end{aligned}$$

Corollary 11.2 (A local curved Keakeya maximal estimate). *For the phase and chart fixed above, there is $0 < \delta_{\text{curv}} \leq \rho$ such that, for every $1 \leq p \leq 3$ and $\varepsilon > 0$, there is a finite constant $C_{p, \varepsilon, \phi, \Omega, Y}$ for which*

$$\|\mathcal{K}_\delta^\phi f\|_{L^q(Y)} \leq C_{p, \varepsilon, \phi, \Omega, Y} \delta^{1-3/p-\varepsilon} \|f\|_{L^p(\Omega)}$$

for every $0 < \delta < \delta_{\text{curv}}$ and $f \in L^p(\Omega)$, where $q = 2p'$, with $q = \infty$ at $p = 1$. In particular, at $p = q = 3$,

$$\|\mathcal{K}_\delta^\phi f\|_{L^3(Y)} \leq C_{\varepsilon, \phi, \Omega, Y} \delta^{-\varepsilon} \|f\|_{L^3(\Omega)}.$$

Proof. The Euclidean comparison and interpolation in the preceding proof give $K(3)$ in the notation of Gao et al. [9, equation (6)]. Their Proposition 4.1 gives the equivalent assertion $\mathcal{K}(3)$ for the phase class above, which is exactly the displayed p, q range. Their straightening theorem is local [9, Theorem 1.12]; on the fixed chart its spatial and direction-parameter comparison constants may depend on the phase and the chart. This gives the stated estimates. $\square$

The constants are allowed to depend on the phase and localization, which are fixed independently of δ .

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